
LINC One

Comprehensive Product, Feature, Architecture, API, Data, Security, and
Source Reference

RELEASE 1.4

August 11, 2026

Author: Kirolos Sedra

Affiliation: T-TLabs / LInC Ministry

Document status: Comprehensive Source-Reviewed Release Baseline

Classification: Confidential – Internal Use Only

This release is intentionally a full reference manual rather than a short architecture assessment. It documents the reviewed repository from user-visible capability down to route, API, persistence, configuration, test, file, and named-function level.

Document Control

Field	Value
Title	LINC One: Comprehensive Product, Feature, Architecture, API, Data, Security, and Source Reference
Version	1.4
Date	August 11, 2026
Owner	Kirolos Sedra
Source package	LInC-Church-Management-main(2).zip
Review basis	Static source-code review of the complete supplied repository snapshot, including frontend source, Worker backend, automated tests, CI workflows, YAML assessment definitions, and repository ERD files.
Review scope	Current implementation and current risks. Recommendations are explicitly labeled and are not presented as implemented behavior.

Revision History

Version	Date	Change summary
1.4	Aug. 11, 2026	Rebuilt documentation as a comprehensive repository reference. Added complete feature-by-feature behavior, route/access matrix, backend endpoint reference, database/storage maps, assessment definition details, Bezalel behavior, NextGen portal and legacy paths, administrator capabilities, attendance, tutorials, CI/CD, source-file catalog, named-function catalog, test catalog, dependency catalog, and ERD-derived entity catalog.
1.3	Aug. 11, 2026	Re-baselined architecture around the expanded multi-domain backend, Bezalel, B2 storage, NextGen portal, security gaps, and CI/CD changes.
1.2	Jul. 24, 2026	Documented first Cloudflare Worker/Brevo backend slice and hybrid frontend/backend architecture.
1.1	Jul. 22, 2026	Documented landing-page floating action dock.
1.0	Jul. 22, 2026	Initial technical/modernization baseline.

Info – Coverage Statement

The document is designed to answer both “what can the product do?” and “where is that behavior implemented?”. The narrative chapters explain workflows and architectural intent. The appendices provide exhaustive inventories so a maintainer can locate every reviewed source module, named function, automated test, package dependency, and ERD entity without rediscovering the repository structure.

Contents

Document Control	1
1 Executive Summary	11
1.1 Current Product State	11
1.2 Major Functional Domains	11
1.3 Architecture Status	11
1.4 How This Release Differs from 1.3	12
2 Review Method and Source Coverage	13
2.1 Reviewed Artifacts	13
2.2 Evidence Classes	13
2.3 Build Verification Limitation	13
3 Product Map and User Roles	14
3.1 User Classes	14
3.2 Administrator Capability Model	14
3.3 Pastor Authorization	14
4 Frontend Route Reference	15
4.1 Route Matrix	15
4.2 Protected Route Behavior	16
4.3 Application Bootstrap Residue	16
5 Public Landing and Navigation	17
5.0.1 Hero and Top-Level Destinations	17
5.0.2 Spiritual Gifts Program Cluster	17
5.0.3 Community Carousel	17
6 Assessment Platform	18
6.0.1 Assessment Availability	18
6.0.2 Submission Trust Boundary	18
6.0.3 Administrator User Linkage	18
7 Public Booking	19
7.0.1 Public Schedule Projection	19
7.0.2 Request Contract	19

7.0.3	Conflict and Reservation Handling	19
8	Pastor Calendar and Meeting Operations	20
8.0.1	Calendar Records	20
8.0.2	Meeting Creation and Editing	20
8.0.3	Availability and Unavailability	20
8.0.4	Public Request Decisions	20
8.0.5	Calendar Export	20
9	Bezalel AI Assistant	21
9.0.1	Pastor Assistant	21
9.0.2	Public Booking Assistant	21
9.0.3	NextGen AI Roles	21
9.0.4	Provider Contract	21
10	People Development	22
10.0.1	Group Taxonomy	22
10.0.2	Member Assignment	22
10.0.3	Assignments and Combined Posts	22
10.0.4	Current Attachment Representation	22
10.0.5	Personal Notes	22
10.0.6	Recurring Group Meetings	22
11	Congregation Group Notes Portal	23
12	Confidential People Notes	24
13	NextGen Account Portal	25
13.0.1	QA Session Lifecycle	25
13.0.2	Question Submission	25
13.0.3	Voting and Privacy	25
13.0.4	Management	25
14	NextGen Files	26
15	Montreal Mission Trip Map	27
16	Legacy NextGen Activities Still in Source	28
17	Administrator Panel	29
17.0.1	Admin Areas	29
17.0.2	Chief Bootstrap	29

17.0.3 Authority Integrity	29
18 Administrator Attendance	30
18.0.1 People Records	30
18.0.2 Recording	30
18.0.3 Analytics	30
19 LInC Archives	31
20 Carousel Management	32
21 Tutorial Builder and Player	33
22 Email and Notification System	34
23 Localization, Motion, and Accessibility	35
24 Legal, About, and Guide Pages	36
25 Assessment Definition Reference	37
25.1 Configured Forms	37
25.1.1 five-service-pathways	37
25.1.2 spiritual-gifts-discovery	37
25.2 Five Service Pathways Detail	38
25.3 Spiritual Gifts Discovery Detail	38
26 Backend Architecture	39
26.1 Worker Composition	39
26.2 Authentication	39
26.3 Two Database Access Modes	39
26.4 Validation and Error Boundaries	39
26.5 Object Storage	39
27 Complete Backend API Reference	40
27.1 Endpoint Matrix	40
27.2 API Design Conventions	46
27.3 Cache-Control	46
28 Database and Persistence Reference	47
28.1 Observed Realtime Database Roots	47
28.2 Identity Fragmentation	47
28.3 Storage Classes	47

29 Security and Privacy Assessment	49
29.1 Implemented Controls	49
29.2 Current Risk Register	49
29.3 Recommended Closure Order	50
30 Testing, CI/CD, Deployment, and Operations	51
30.1 Frontend Workflow	51
30.2 Backend Verification and Deployment	51
30.3 Production Smoke	51
30.4 Hosting Surfaces	51
30.5 Netlify Secret Scanning	51
31 Configuration and Secret Catalog	52
31.1 Frontend/Browser Configuration	52
31.2 Backend and CI Configuration	52
31.3 Checked-In Worker Defaults	53
32 Technology and Dependency Catalog	54
32.1 Runtime Architecture	54
32.2 Exact Package Manifest	54
33 Complete End-to-End Workflow Reference	56
33.1 Application Startup, Authentication, and Route Protection	56
33.2 Public Assessment Submission	56
33.3 Assessment Administration and Member Linkage	57
33.4 Public Booking and Atomic Reservation	57
33.5 Pastor Meeting-Request Decision	57
33.6 Pastor Meeting Creation, Editing, and Participant Invitations	57
33.7 Pastor Availability and Unavailability	57
33.8 Bezalel Pastor Calendar Assistance	58
33.9 Bezalel Public Booking Assistance	58
33.10 People Development Pastor Management	58
33.11 Congregation Group Notes Portal	58
33.12 People Development Notification Delivery	58
33.13 Confidential People Notes	59
33.14 NextGen Account Session	59
33.15 NextGen QA Session and Question Submission	59
33.16 NextGen Voting	59
33.17 NextGen QA Management	59

33.18NextGen File Upload, Completion, Download, and Delete	59
33.19Montreal Mission Map	60
33.20Administrator Onboarding and Authority Allocation	60
33.21LInC Archives File Lifecycle	60
33.22Attendance Management	60
33.23Landing Carousel Management	60
33.24Tutorial Authoring, Simulation, Playback, and Progress	61
33.25Legacy Integrations and Residual Paths	61
34 Enterprise Target Architecture and Modernization Reference	62
34.1 Architecture Principles	62
34.2 Current Target Stack	62
34.3 Target Request Path	62
34.4 Capability and Role Model	63
34.5 Canonical Identity Target	63
34.6 Target Persistence Boundaries	63
34.7 API Standards	64
34.8 Transaction and Idempotency Standards	64
34.9 Security Architecture	64
34.10Privacy and Data Governance	65
34.11Audit Logging Target	65
34.12Observability and Incident Response	65
34.13Backup and Recovery	65
34.14Frontend Modernization Direction	65
34.15Backend Module Design	66
34.16Testing Strategy	66
34.17CI/CD and Environment Model	66
34.18Documentation and Governance Set	66
35 Current and Target Permission Catalog	67
35.1 Current Implemented Authorities	67
35.2 Recommended Capability Expansion	67
35.3 Non-Delegable or Highly Restricted Actions	68
36 Modernization Roadmap - Rebaselined to Release 1.4	69
36.1 Immediate Remediation Queue	70
36.2 Medium-Term Engineering Queue	70
37 Enterprise Readiness and Acceptance Criteria	71

37.1 Architecture Acceptance	71
37.2 Security Acceptance	71
37.3 Data Acceptance	71
37.4 Product Acceptance	71
37.5 Operational Acceptance	71
37.6 Commercial/Transfer Acceptance	72
38 Open Verification Items and Documentation Boundaries	73
39 Release Delta: Why the Old Documentation Was Insufficient	74
39.1 Major Changes Not Represented by the July Baseline	74
39.2 Documentation Model Change	74
39.3 Current Modernization Backlog	74
A ERD-Derived Entity Catalog	75
A.1 ADMIN_HIERARCHY	75
A.2 ADMIN_USER	75
A.3 ASSESSMENT_FORM	75
A.4 ASSESSMENT_FORM_CONTROL	76
A.5 ASSESSMENT_RESPONSE	76
A.6 ASSESSMENT_RESPONSE_IDENTITY	77
A.7 ATTENDANCE_PERSON	77
A.8 ATTENDANCE_PERSON_IDENTITY	78
A.9 AVAILABILITY	78
A.10 CAROUSEL_CONFIG	78
A.11 CAROUSEL_PHOTO	78
A.12 FIREBASE_AUTH_USER	79
A.13 MEETING	79
A.14 MEETING_PARTICIPANT	80
A.15 MEETING_REQUEST	80
A.16 MEETING_REQUEST_IDENTITY	81
A.17 NEXTGEN_PARTICIPATION_EVENT	81
A.18 NEXTGEN_QUESTION	81
A.19 NEXTGEN_REVIEW_LOG	82
A.20 NEXTGEN_SURVEY	82
A.21 NEXTGEN_SURVEY_RESPONSE	82
A.22 NEXTGEN_USER	83
A.23 NEXTGEN_USER_IDENTITY	83

A.24 NEXTGEN_VOTE	83
A.25 PD_ASSIGNMENT	84
A.26 PD_ASSIGNMENT_GROUP	84
A.27 PD_ATTACHMENT	84
A.28 PD_GROUP	85
A.29 PD_MEETING_SCHEDULE	85
A.30 PD_MEMBER	85
A.31 PD_MEMBER_IDENTITY	86
A.32 PD_MEMBER_RESPONSE	86
A.33 PD_NOTE	86
A.34 PD_PERSONAL_NOTE	87
A.35 PEOPLE_NOTES_PERSON	87
A.36 PEOPLE_NOTES_PERSON_IDENTITY	87
A.37 PEOPLE_NOTE_COMMENT	87
A.38 PEOPLE_NOTE_ITEM	88
A.39 RESPONSE_EMAIL_EVENT	88
A.40 RESPONSE_LINKAGE	88
A.41 TUTORIAL	89
A.42 TUTORIAL_PROGRESS	89
A.43 TUTORIAL_STEP	90
A.44 UNAVAILABILITY	90
B ERD Relationship Catalog	91
C Complete Source File Catalog	92
D Complete Named-Function Catalog	99
E Automated Test Catalog	119
F Residual Direct-Firebase Client Catalog	123
G Maintenance Checklist	124

List of Tables

List of Figures

Chapter 1

Executive Summary

1.1 Current Product State

LINC One is no longer accurately described as a spiritual-gifts form with a Pastor calendar. The supplied repository is a multi-workspace ministry application with public, participant, Pastor, and delegated-administrator experiences. The current implementation combines a React 19/TypeScript frontend on Netlify, Firebase Authentication, Firebase Realtime Database, a Cloudflare Worker/Hono backend, Brevo transactional email, Backblaze B2 private object storage, Google Maps, and Gemini-backed Bezalel AI workflows.

The source review counted **211 frontend TypeScript/TSX modules** (approximately **39,140 lines**), **46 backend source modules** (approximately **9,707 lines**), **16 backend test modules**, and **167 named automated test cases** across frontend and backend test files. The full named-function appendix records **981 named source functions and callbacks** identified in the current application code.

1.2 Major Functional Domains

- LINC One public landing and bilingual navigation.
- Two YAML-defined spiritual/ministry assessments.
- Public Pastor booking and privacy-safe schedule projection.
- Authenticated Pastor calendar, availability, unavailability, meeting requests, meeting creation, rescheduling, deletion, and ICS export.
- Bezalel AI for Pastor calendar assistance and public booking guidance.
- People Development member grouping, assignments, attachments, personal notes, and recurring group meetings.
- Identifier-based congregation group portal for assignments and meeting information.
- Confidential People Notes with strengths/weaknesses, comments, follow-up dates, and deletion workflows.
- Account-based NextGen QA sessions, question moderation, voting, participant verification, session management, files, and mission map.
- Residual legacy NextGen identifier/survey/certificate code still present in the repository.
- Administrator hierarchy, capability allocation, assessment management, carousel management, attendance, and LInC Archives.
- Tutorial authoring, playback, simulation overlays, target picking, and per-user progress.
- Brevo notifications, B2 signed upload/download workflows, Firebase server access, and CI/CD automation.

1.3 Architecture Status

The application has moved well beyond the v1.2 “single migrated email route” stage. Sensitive calendar, assessment, People Notes, People Development, administration, archives, NextGen QA/files, mission-map, and Bezalel operations now have dedicated Worker APIs. However, the migration is not

complete: attendance, landing carousel, tutorial data, participant aggregation, and legacy NextGen modules still include direct browser-side Realtime Database access.

Critical – Current High-Priority Security Findings

Three source-reviewed boundaries deserve immediate attention: (1) `/api/v1/people-development/notifications` has no authentication middleware; (2) NextGen file operations are currently available to any Firebase-authenticated account rather than a narrower NextGen/admin capability; and (3) the mission-map route checks only for an authenticated token carrying an email even though repository documentation describes a specific mission-login identity. The temporary public Brevo test endpoint is also still mounted.

1.4 How This Release Differs from 1.3

Release 1.3 was an architectural assessment. Release 1.4 is a **maintainer's reference manual**. It expands from major modules into UI behavior, route composition, backend contracts, validation rules, persistence paths, storage lifecycles, authorization boundaries, configuration, tests, and source-level catalogs. Page count is therefore a result of coverage rather than a target.

Chapter 2

Review Method and Source Coverage

2.1 Reviewed Artifacts

The review included the entire supplied project tree relevant to application behavior: `src/`, `backend/src/`, `backend/test/`, `.github/workflows/`, `src/components/forms/*.yaml`, `documentation/database/*.mmd`, package manifests, Worker configuration, Netlify configuration, and environment examples.

2.2 Evidence Classes

- **Implemented:** directly represented by active source code and route composition in the supplied snapshot.
- **Repository-documented:** described in README or ERD files and cross-checked where possible against source.
- **Legacy/residual:** source still exists but is not the primary route-level implementation.
- **Risk/inference:** a conclusion derived from the visible authorization, storage, or data-flow code; clearly labeled as such.

2.3 Build Verification Limitation

The supplied ZIP did not include installed `node_modules`. The documentation therefore uses source and test inspection as its primary evidence. The CI definitions themselves install dependencies with `npmci`, build the frontend, test/type-check the backend, deploy the Worker from main, and run the production NextGen contract smoke test. This release does not claim a fresh local TypeScript build result where dependencies were not installed in the review environment.

Chapter 3

Product Map and User Roles

3.1 User Classes

User class	Primary access mechanism	Current capabilities
Anonymous visitor	No account	Landing, About, privacy/terms, public assessments, booking schedule/request, public Bezalel booking assistant.
People Development participant	Personal identifier	Group assignment profile, group-targeted posts/attachments, recurring meeting schedule.
NextGen member	Firebase email/password account	NextGen QA sessions, question submission, voting, file workspace, mission map.
Pastor	Firebase account plus exact backend email allowlist	Calendar, requests, People Development, People Notes, NextGen management, tutorials, Bezalel Pastor agent, meeting invitations.
Chief administrator	Firebase account designated in admin hierarchy	All administrator authorities plus administrator allocation/suspension. Pastor can bootstrap as first Chief when no Chief exists.
Delegated administrator	Firebase account + active admin profile + allocated authorities	Only administrator capabilities enabled in stored authority record: assessment forms, carousel, attendance, archives, NextGen QA.

3.2 Administrator Capability Model

The current backend authority object contains five explicit booleans: `manageAssessmentForms`, `manageCarousel`, `manageAttendance`, `manageArchives`, and `manageNextGenQa`. Chief administrators receive the full authority set. A non-Chief administrator must be active and hold the specific capability for backend-protected operations that implement capability checks.

3.3 Pastor Authorization

Pastor access is not a database role lookup. The Worker currently uses a case-insensitive anchored email allowlist matching `rev.ibrahim@lincministry.com`. The browser calls the backend session/access endpoint through `usePastorAccess`; protected routes use the returned Pastor state. Sensitive Pastor APIs independently enforce the same server-side rule after Firebase ID-token verification.

Chapter 4

Frontend Route Reference

4.1 Route Matrix

Path	Access	Primary component	Purpose
/	Public	LandingPage	LINC One landing, locale switch, top-level navigation, carousel and program entry.
/about	Public	AboutUs	About/ministry information.
/calendar	Firebase login + Pastor allowlist	PastorDashboard	Pastor calendar, requests, People Development, NextGen management, tutorials, Bezalel.
/pastor/people-notes	Firebase login + Pastor allowlist	PeopleNotesPage	Confidential person-development notes and follow-up records.
/guide	Firebase login + Pastor allowlist	GuidePage	Protected operational guide.
/assessment	Public	AssessmentForm	Assessment selector, questionnaire, computed results and submission.
/booking	Public	BookingCalendar	Public Pastor booking schedule and request submission with optional Bezalel assistance.
/nextgen-activities	Firebase email/password account	NextGenActivities	NextGen QA sessions, private files and mission trip map.
/nextgen-activities/qa/:sessionId	Firebase email/password account	NextGenActivities	Specific QA session, question submission and voting.
/group-notes	Identifier access	CongregationGroupNotes	People Development assignments, attachments and recurring meeting schedule.
/administrator	Firebase login + Admin hierarchy	AdministratorPanel	Chief/admin command center, authorities, assessment controls, carousel, attendance and archives.
/privacy	Public	PrivacyPolicy	Privacy policy.
/tos	Public	TermsOfService	Terms of service.
*	Public redirect	Navigate	Unknown routes redirect to landing page.

4.2 Protected Route Behavior

The shared `ProtectedRoute` supports Google sign-in and email/password sign-in/sign-up. It presents a loading state while Firebase authentication and Pastor access are resolving, redirects unauthorized signed-in users when a fallback is supplied, and otherwise renders the protected child. The root `App` wraps all routes with the internationalization provider and tutorial provider.

4.3 Application Bootstrap Residue

At startup, `App.tsx` still processes the legacy Google OAuth callback and dynamically initializes EmailJS with a public key. This means the repository retains browser integration scaffolding even though many notification flows have moved to the Worker/Brevo boundary.

Chapter 5

Public Landing and Navigation

The landing route is a branded LINC One entry surface built from `LincOneHero`, `SpiritualProgramFeature`, `CommunityCarousel`, and `FloatingLandingNav`. It supports English/Arabic switching, responsive motion, reduced-motion behavior, footer links, and a scroll-triggered quick-access navigation.

5.0.1 Hero and Top-Level Destinations

The hero exposes the Spiritual Gifts Program as the dominant feature, plus NextGen, About Us, Pastor Login, and Administrator Panel. The floating navigation reproduces the five agreed top-level destinations after the hero action block passes above the viewport.

5.0.2 Spiritual Gifts Program Cluster

The program section groups the assessment, booking, and People Development group-note experiences instead of presenting them as unrelated products. This is reflected by route tab selection in `App.tsx`: assessment, booking, and group notes all map back to the spiritual-program navigation context.

5.0.3 Community Carousel

`CommunityCarousel.tsx` subscribes directly to `landingPage/carousel` in Realtime Database. It normalizes URL, data-URL, and legacy raw Base64 image values. Administrator carousel editing remains a direct-Firebase path in the current frontend.

Chapter 6

Assessment Platform

The assessment platform is a reusable YAML-driven engine rather than two hard-coded forms. **AssessmentForm** composes a selector, questionnaire, results view, and **useAssessmentForm**. Frontend calculations are implemented in deterministic utilities, while the backend independently validates answers and recomputes persisted results.

6.0.1 Assessment Availability

The public frontend obtains form state from `/api/v1/assessment/forms`. A form may be **active**, **disabled**, or **hidden**. Hidden forms are not intended for normal selection; disabled forms are visible but unavailable for response acceptance.

6.0.2 Submission Trust Boundary

The browser sends only form ID, locale, and answer values. The backend recognizes only the two configured form IDs, rejects unknown answer keys and invalid rating values, derives the stored fields/scores/results, owns timestamps, persists under **form**, and then attempts fixed-template Brevo notification. A successful database write is not rolled back merely because email delivery fails.

6.0.3 Administrator User Linkage

Response browsing, identifier linkage, response deletion, identifier email, and form-state changes have moved to administrator endpoints protected by Chief or **manageAssessmentForms** authority. The public assessment page no longer owns the legacy passcode-based linkage management.

Chapter 7

Public Booking

The public booking route renders a calendar/schedule experience and submits requests through the Hono booking API. The backend, not the browser, composes the authoritative schedule by combining availability, unavailability, meetings, pending requests, People Development recurring meetings, and active calendar reservations.

7.0.1 Public Schedule Projection

The schedule endpoint accepts **start** and **end** calendar dates and rejects ranges longer than 62 days. Returned data is privacy-safe: visitors receive availability and busy ranges rather than private meeting reasons or participant details.

7.0.2 Request Contract

A booking request requires a 2–100 character name, valid normalized email, date, reason up to 2,000 characters, locale, and an exact 30-minute slot. Slots must fall between 09:00 and 20:00 Toronto time. The schema also includes an empty **website** honeypot field.

7.0.3 Conflict and Reservation Handling

Before accepting a request, the backend checks that the slot is inside availability and does not overlap unavailability, confirmed meetings, pending requests, recurring group meetings, or another active reservation. A calendar-reservation key is claimed transactionally to reduce double-booking races. The request is persisted as pending and Pastor notification is attempted through Brevo.

Chapter 8

Pastor Calendar and Meeting Operations

The Pastor Dashboard consumes a shared authenticated calendar snapshot through `usePastorCalendarData`. The calendar feature refreshes after mutations and periodically while active. Calendar presentation is separated into hooks/utilities while persistence is provided by the Worker.

8.0.1 Calendar Records

The snapshot spans confirmed meetings, pending meeting requests, availability, unavailability, People Development recurring meeting schedules, and calendar reservation state. UI helpers derive month grids, selected-day content, display titles, acknowledged status, and occurrence labels.

8.0.2 Meeting Creation and Editing

Pastor can create a meeting/event with title, description, date, start/end time, location, meeting type (prayer, counseling, service, other), optional meeting link, and up to 500 participant IDs. Backend validation rejects end time not after start time and checks conflicts before storing.

8.0.3 Availability and Unavailability

Availability opens time for public booking. Unavailability closes time and is checked against meetings, requests, recurring group meetings, and reservations before creation/update. Both support create, edit, and delete paths.

8.0.4 Public Request Decisions

Accepting a request creates the confirmed meeting and advances the reservation/request state; rejecting releases the reservation. Requester notification is generated server-side. Meeting changes can also trigger localized reschedule/cancellation messages.

8.0.5 Calendar Export

The Pastor can export the selected month as an iCalendar `.ics` file for Apple, Google, Outlook, or another standards-compatible calendar client.

Chapter 9

Bezalel AI Assistant

Bezalel is a server-mediated Gemini assistant with two active roles: an authenticated Pastor calendar assistant and a public booking guide. The frontend never receives the Gemini API key.

9.0.1 Pastor Assistant

The Pastor route accepts at most 12 user/assistant messages, each capped at 2,000 characters. It builds a sanitized calendar context containing bounded fields from meetings, requests, availability, unavailability, and recurring group schedules. The Gemini response is constrained to JSON and one of the allowed actions: `none`, `open_availability`, `block_time`, `delete_availability`, `delete_unavailability`, `accept_request`, or `reject_request`. The frontend executes the chosen action through ordinary authenticated calendar APIs; Gemini does not write the database directly.

9.0.2 Public Booking Assistant

The public assistant receives only the privacy-safe booking projection. It may answer questions, collect visitor information, suggest available slots, and return a `ready_to_book` object. Final booking still passes through the ordinary booking request validation and conflict checks.

9.0.3 NextGen AI Roles

The same server-side Gemini service is used for bilingual QA-theme translation and on-theme question review. Question review accepts reasonable paraphrases and respectful disagreement but rejects spam, unrelated prompts, attacks, and instruction-override attempts. Rejected questions can receive a suggested on-theme rewrite.

9.0.4 Provider Contract

`generateGeminiJson` uses the configured Gemini model (default `gemini-3.6-flash`), low temperature, JSON MIME type, a supplied JSON schema, and Zod validation of the returned payload. Missing configuration, rate limits, upstream errors, empty responses, and invalid JSON are mapped into structured Bezalel errors.

Chapter 10

People Development

People Development is a Pastor-managed grouping and discipleship workspace backed by **peopleDevelopment**. It combines member assignment, group-targeted posts, attachments, private member notes, and recurring meeting schedules.

10.0.1 Group Taxonomy

The backend schema recognizes ten group codes: pastors, prophets, evangelists, teachers, apostles, helpers, mercy, facilitators, services, and giving. A member may also temporarily have no assigned group.

10.0.2 Member Assignment

Pastor assignment synchronizes a canonical People Development member record and writes group-linkage metadata back to one or more source assessment records. The source path is constrained and the update map is built by the backend rather than accepting arbitrary database paths from the browser.

10.0.3 Assignments and Combined Posts

Posts may target one or multiple groups, include text up to 20,000 characters, and include up to five attachments. The UI supports group-specific and combined posting, attachment replacement/removal, calendar-style assignment browsing, and deletion.

10.0.4 Current Attachment Representation

People Development attachments are still embedded in Realtime Database as Base64 objects. Each attachment is limited to 1 MiB and the schema caps a post at five attachments. This differs from LInC Archives and NextGen files, which use Backblaze B2 signed URLs.

10.0.5 Personal Notes

Pastor can record member-specific **strength** or **weakness** notes with group/member metadata. These are distinct from the separate **peopleNotes** subsystem.

10.0.6 Recurring Group Meetings

Schedules support group-specific or shared audiences, first/second/third/fourth/last ordinal, weekday 0–6, start time, 30–480 minute duration, start/end dates, and active status. The schedules feed both the Pastor calendar and participant group portal.

Chapter 11

Congregation Group Notes Portal

The `/group-notes` experience is an identifier-based People Development portal. The browser submits only the participant identifier to `/api/v1/people-development/portal`; the backend resolves the matching member and returns only that member's profile, assignments visible to the assigned group, and active matching/shared schedules.

The portal displays member name, identifier, email, primary gift, group identity/description, assignment count, latest assignment, searchable assignment history, attachment chips, a details dialog with open/download actions, the next group meeting, and the next shared meeting. A recognized member without a group receives an explicit unassigned state rather than another group's data.

The backend adds an in-memory attempt limiter to the identifier endpoint and returns HTTP 429 with **Retry-After** after the configured threshold. Because Worker instances are distributed and ephemeral, this is a lightweight abuse control rather than a globally durable rate-limit service.

Chapter 12

Confidential People Notes

People Notes is a dedicated Pastor-only development-record subsystem separate from People Development personal notes. The route `/pastor/people-notes` is protected in the frontend and every Worker route independently requires Firebase authentication plus the Pastor allowlist.

A person record may contain nested development items, comments, and follow-up dates. The UI supports creating people, adding development items, adding comments, setting/removing follow-up dates, deleting comments, deleting items, and deleting a person. Backend audit fields and timestamps are server-controlled. Mutations use narrow paths or atomic multipath patches so nested deletes and parent timestamp updates are coordinated.

Responses use private no-store caching. The backend differentiates unavailable configuration, Firebase Rules denial, and upstream database failures in its HTTP mapping.

Chapter 13

NextGen Account Portal

The active `/nextgen-activities` route is now an account-based portal. `NextGenAuthGateway` supports Firebase email/password account creation and login, with display-name update for new accounts. After authentication the portal exposes three primary sections: QA sessions, NextGen files, and the mission trip map.

13.0.1 QA Session Lifecycle

Managers create sessions with title, description, theme, and status **draft**, **open**, or **closed**. Theme text is translated into English and Arabic by Bezalel. Members can list non-draft sessions; only open sessions permit question submission/voting.

13.0.2 Question Submission

Each account is limited to two submitted questions per QA session. Before persistence, Bezalel compares the question with the bilingual session theme. Off-theme questions return HTTP 422 and are not written. Accepted records have fixed upvote/downvote choices and server-owned metadata.

13.0.3 Voting and Privacy

Votes are keyed so one current choice exists per normalized email identity; a member can change that choice while the session remains open. Public/member views do not expose aggregate vote totals in the ordinary voting UI. Manager views can include participant/voter detail, and aggregate ranking excludes participants marked discarded/unverified according to the backend logic.

13.0.4 Management

Pastor or an administrator with `manageNextGenQa` can create/update sessions, inspect session detail, verify/discard participants, and select or unselect questions for live discussion.

Chapter 14

NextGen Files

NextGen files use an authenticated folder/file workspace backed by Realtime Database metadata and Backblaze B2 object storage. The browser can create nested folders, drag/drop or choose multiple files, refresh state, navigate breadcrumbs, download ready files, delete files, and delete empty folders.

The upload lifecycle is two-phase: create a pending metadata record and signed upload URL, upload the browser file directly to B2, then call the completion endpoint. Completion performs a HEAD/inspection check and rejects a size mismatch before marking the record ready. Downloads are short-lived signed private URLs with a content-disposition filename override.

Warning – Current Authorization Scope

The route-level middleware verifies Firebase authentication globally, but the current file endpoints do not call the QA-manager capability check or another NextGen membership authorization helper. Therefore any authenticated Firebase account can currently list, create, upload, download, and delete NextGen files. This is an implementation boundary, not merely a documentation concern.

Chapter 15

Montreal Mission Trip Map

The NextGen mission map renders private locations and connection information using a browser Google Maps key and server-provided location data. The backend stores the map payload in the `NEXTGEN_MISSION_MAP_DATA` secret, validates location IDs, names, types, addresses, latitude/longitude, notes, parking, languages, and connection references, and returns the result with private/no-store caching.

The frontend supports mission-location cards, map markers, route visualization, Google/Apple Maps links, location-type labels, and travel-distance/time presentation. The code includes both Google-route behavior and fallback distance/duration logic.

Warning – Authorization Drift

Repository README text states that the mission map requires a Firebase password login for a specific NextGen account. The route implementation itself currently authorizes any verified Firebase user whose token contains a non-null email. The source implementation is authoritative for this release and the mismatch is recorded as documentation/security drift.

Chapter 16

Legacy NextGen Activities Still in Source

Although the active route uses the account-based portal, the repository still contains the earlier four-character-identifier NextGen activity subsystem under `src/components/nextgen/activities` and Pastor-side legacy Firebase modules. That subsystem includes registration/approval, four-character IDs, certificate generation, question categories, peer voting, survey responses, participation history, and review logging.

The legacy survey definition contains eight binary preference questions and three rating questions. Browser PDF certificate generation remains implemented in `nextGenCertificate.ts`. The backend new QA service also contains a one-time legacy migration path that can migrate old QA questions into a closed “QA Session 1” and records migration state below `migrations/legacyQaSession1`.

This code should be treated as compatibility/migration surface until an explicit deletion decision is made; it is not the primary NextGen route-level experience in the supplied snapshot.

Chapter 17

Administrator Panel

The Administrator Panel is a dedicated command center rather than a Pastor subpage. Its access hook coordinates Firebase authentication with the backend admin session, pending/suspended states, Chief initialization, authority drafts, activation, suspension, and logout.

17.0.1 Admin Areas

The current feature boundary includes hierarchy/authority management, assessment-form controls, assessment user linkage, carousel management, attendance management, and LInC Archives. An administrator without a required allocation receives a no-authority state rather than silently inheriting access.

17.0.2 Chief Bootstrap

When the admin hierarchy has no Chief and the authenticated user satisfies the exact Pastor allowlist, the backend sets that user's UID as Chief. A newly seen non-Chief account is created in pending status with no authorities. Chief can activate it by saving at least one authority or can suspend an existing administrator.

17.0.3 Authority Integrity

The authority schema requires all five capability booleans and refuses an all-false activation payload. Chief accounts cannot be edited/suspended through the ordinary target-administrator endpoints.

Chapter 18

Administrator Attendance

Attendance is currently a frontend-controlled Firebase feature gated by the administrator `manageAttendance` allocation. The feature is decomposed into people management, recording, analytics, personal analysis, copy/styles, deterministic utilities, and a controller hook.

18.0.1 People Records

Administrators can create/edit people records, search/select people, capture or choose profile photos, switch cameras, remove photos, and persist attendance person data under `attendance/people`. The normalizer supports URL/data-URL images and legacy raw Base64 photos.

18.0.2 Recording

The recording panel builds Sunday-oriented calendar views, selects an attendance date, and records attendance while avoiding duplicate date entries. Stored attendance is normalized from comma-separated or legacy representations into sorted date lists in the UI.

18.0.3 Analytics

Derived analytics include attendance-day ranges, Sunday series beginning with the first Sunday in May, summary counts, histograms/timelines, and a personal analysis modal.

Warning – Migration Status

The attendance controller still imports `firebase/database` directly. The administrator capability protects UI exposure, but the final security boundary therefore still depends on Firebase Security Rules rather than a dedicated Worker attendance API.

Chapter 19

LInC Archives

LInC Archives is a delegated-administrator private document workspace. It supports nested folder trees, breadcrumbs, location filtering, multi-file selection, upload-progress/pending UI, verification, download, and deletion.

Folder and file metadata live under **administration/archives**. Objects are stored in a configured Backblaze B2 S3-compatible bucket. Folder names reject slashes and dot/dot-dot names; file names reject slashes/control characters; declared file size is capped at 5 GB.

Archive upload follows pending-record -> signed upload -> object verification -> ready-record. If signed URL creation fails, the backend attempts to clean the pending metadata. Folder deletion refuses non-empty/nested structures according to archive-domain rules. Access requires Chief or **manageArchives**.

Chapter 20

Carousel Management

The landing carousel is managed in the Administrator Panel and displayed publicly by `CommunityCarousel`. The admin UI supports enabling/visibility behavior, adding selected images, local pending uploads, image conversion to data URLs, multilingual photo text, ordering/move operations, and deletion.

Unlike archives and NextGen files, carousel data is still represented directly in Realtime Database beneath `landingPage/carousel`. The public carousel subscribes to that path directly. This makes the carousel one of the remaining direct-Firebase content domains.

Chapter 21

Tutorial Builder and Player

The tutorial subsystem is globally available through **TutorialProvider** and includes a builder section, library, editor modal, player, simulation layer, and target picker. Tutorials have title, description, category, audience, publication state, and ordered steps. A step can carry title/explanation, route, DOM target, requested action, and simulated result.

The builder validates required title/category, requires at least one step, and checks title, explanation, target, and simulated result for each step. Published tutorials can be run using React Joyride-style guided behavior and route/target coordination. Per-user progress tracks last step, completion, completion time, and most recent start.

Tutorial definitions and progress still use direct Realtime Database operations under **tutorialBuilder/tutorials** and **tutorialBuilder/progress**.

Chapter 22

Email and Notification System

Brevo is the primary server-side transactional email provider. The backend contains specialized builders for assessment messages, identifier messages, public booking requests, calendar notifications, meeting invitations, and People Development notifications. Templates escape user-controlled HTML values and several provide bilingual English/Arabic content.

The shared Brevo service supports ordinary recipient delivery and BCC delivery. Meeting invitations are Pastor-authenticated. Public booking notification is server-generated to a configured fixed recipient. Assessment notifications use fixed configured/repository form recipients and optionally the submitter according to the assessment workflow.

Legacy Gmail/Google Calendar and EmailJS code remains in `src/services/gmail.ts`; `App.tsx` still initializes EmailJS and handles an OAuth callback. This code provides Google OAuth token storage, Gmail send, Calendar event/Meet link creation, placeholder link generation, and EmailJS helpers, but it is not the authoritative path for the newer backend-migrated workflows.

Chapter 23

Localization, Motion, and Accessibility

The application uses a custom internationalization context with English and Arabic locale switching and RTL/LTR direction. Assessment definitions mix translation keys and embedded bilingual text. Backend email builders and Bezalel prompts also contain bilingual behavior.

The landing page and other interfaces use Motion/AnimatePresence. Reduced-motion preference is respected in prominent landing animations. Branding components include accessibility tests for meaningful versus decorative logo usage. The codebase uses semantic buttons/links, aria labels in several interactive controls, and responsive layouts, although a formal WCAG conformance audit is not present in the supplied repository.

Chapter 24

Legal, About, and Guide Pages

Public informational routes include About Us, Privacy Policy, and Terms of Service. The Guide route is Pastor-protected. These are conventional React pages and participate in the shared layout where applicable.

Because the technical stack has changed materially—Gemini, Backblaze B2, server-side Brevo, new NextGen account/file behavior, and expanded Firebase server access—the legal copy should be periodically checked against actual provider/data flows. Release 1.3 already identified policy drift as a risk; the broader v1.4 source audit reinforces that requirement.

Chapter 25

Assessment Definition Reference

25.1 Configured Forms

25.1.1 five-service-pathways

Version: 1. YAML status: active. Approximate configured input/rating items: 56. Notification provider: brevo. Include submitter: True.

Section ID	Renderer type	Configured items
trainee	fields	10
faith	fields	5
gifts	groupedRating	25
ministry	ratingList	10
vision	fields	6

Calculation pipeline:

- giftTotals: sumGroups; source gifts.
- ministryTotals: fieldScores; source ministry.
- rankedGifts: rankGroups; source giftTotals.
- rankedMinistry: rankFields; source ministryTotals.
- primaryGift: topGroup; source rankedGifts.
- secondaryGift: topGroup; source rankedGifts.
- recommendedMinistry: topField; source rankedMinistry.

25.1.2 spiritual-gifts-discovery

Version: 1. YAML status: active. Approximate configured input/rating items: 46. Notification provider: brevo. Include submitter: True.

Section ID	Renderer type	Configured items
trainee	fields	4
spiritualGifts	groupedRating	42

Calculation pipeline:

- giftTotals: sumGroups; source spiritualGifts.
- rankedGifts: rankGroups; source giftTotals.

- primaryGift: topGroup; source rankedGifts.
- secondaryGift: topGroup; source rankedGifts.
- thirdGift: topGroup; source rankedGifts.

25.2 Five Service Pathways Detail

The Five Service Pathways definition collects ten participant-information fields, five faith/narrative responses, five grouped spiritual-gift dimensions with five ratings each, ten ministry ratings, and six vision/reflection fields. Its calculation pipeline sums gift groups, maps ministry fields to scores, ranks both dimensions, and derives a primary gift, secondary gift, and recommended ministry.

25.3 Spiritual Gifts Discovery Detail

The Spiritual Gifts Discovery definition contains participant information plus fourteen gift groups with three statements each, for a 42-item gift questionnaire. The gift groups are Prophecy, Serving/Helps, Teaching, Encouragement/Exhortation, Giving, Leadership, Mercy/Compassion, Wisdom, Knowledge, Faith, Administration, Evangelism, Pastoring/Shepherding, and Hospitality. The calculation pipeline sums each group, ranks groups, and returns the top three gifts.

Chapter 26

Backend Architecture

26.1 Worker Composition

`backend/src/index.ts` creates one Hono application and mounts domain routers below `/api/v1`. CORS allows the production site and local Vite origin, accepts `Content-Type` and `Authorization`, and permits GET, POST, PATCH, DELETE, and OPTIONS. The preflight max-age is 86,400 seconds.

26.2 Authentication

Firebase ID tokens are verified against Google's secure-token JWK endpoint using RS256, expected project audience, expected Firebase issuer, and time claims. The middleware additionally validates UID and `auth_time`, extracts email/email-verified/name/picture/provider fields, and stores both normalized user and raw ID token in Hono context.

26.3 Two Database Access Modes

The backend has two Realtime Database client modes. Some Pastor flows reuse the caller's Firebase ID token, allowing Firebase Security Rules to remain part of authorization. Other server-authoritative domains obtain a short-lived Google OAuth access token from configured Firebase service-account credentials and use an admin-style REST client. The service-account OAuth assertion requests Firebase Database and userinfo-email scopes and caches access tokens until close to expiry.

26.4 Validation and Error Boundaries

Request contracts are generally strict Zod objects. Unknown fields are rejected in security-sensitive routes. Provider/database errors are translated into structured JSON envelopes rather than returning raw credentials or upstream response bodies. Domain modules contain normalization functions so Realtime Database's schema-flexible records are converted before reaching UI/business logic.

26.5 Object Storage

Backblaze integration uses AWS Signature Version 4 through an S3-compatible endpoint. The service creates time-bounded upload/download URLs, inspects objects, deletes objects, validates required bucket/endpoint/region/application-key configuration, and retries inspection of newly uploaded objects to tolerate propagation delay.

Chapter 27

Complete Backend API Reference

27.1 Endpoint Matrix

Method	Endpoint	Access	Purpose	Primary data/provider
GET	/	Public	Worker health text route.	None
POST	/api/v1/email/test	Public (temporary)	Brevo sandbox/real test message to fixed configured recipient.	Brevo only
GET	/api/v1/auth/session	Firebase Bearer	Verify account and report Pastor authorization metadata.	Token only
GET	/api/v1/auth/pastor-access	Firebase Bearer + Pastor	Explicit Pastor authorization check.	Token only
GET	/api/v1/admin/session	Firebase Bearer	Create/update admin profile, bootstrap Pastor as Chief when hierarchy is empty, return Chief-visible account list.	administration/adminHierarchy
PATCH	/api/v1/admin/users/:uid/authority	Chief only	Activate administrator and allocate capability booleans.	administration/adminHierarchy/users
PATCH	/api/v1/admin/users/:uid/suspend	Chief only	Suspend administrator and clear authority set.	administration/adminHierarchy/users
GET	/api/v1/admin/archives/folders	Chief or manageArchives	List archive folder metadata.	administration/archives + Backblaze B2
POST	/api/v1/admin/archives/folders	Chief or manageArchives	Create nested archive folder.	administration/archives + Backblaze B2
DELETE	/api/v1/admin/archives/folders/:folderId	Chief or manageArchives	Delete an empty archive folder.	administration/archives + Backblaze B2
GET	/api/v1/admin/archives/files	Chief or manageArchives	List archive file metadata.	administration/archives + Backblaze B2
POST	/api/v1/admin/archives/files/upload-url	Chief or manageArchives	Create pending file record and signed Backblaze B2 upload URL.	administration/archives + Backblaze B2
POST	/api/v1/admin/archives/files/:fileId/complete	Chief or manageArchives	Verify uploaded object and mark file ready.	administration/archives + Backblaze B2
GET	/api/v1/admin/archives/files/:fileId/download-url	Chief or manageArchives	Create private signed download URL.	administration/archives + Backblaze B2
DELETE	/api/v1/admin/archives/files/:fileId	Chief or manageArchives	Delete B2 object and metadata.	administration/archives + Backblaze B2
GET	/api/v1/assessment/forms	Public	Return active/disabled/hidden state for known forms.	form + assessmentPage/-forms + Brevo
POST	/api/v1/assessment/submissions	Public	Validate answers, recompute server results, persist response, send assessment notification.	form + assessmentPage/-forms + Brevo
POST	/api/v1/assessment/direct-signups	Public	Create direct-signup identity record for the Five Service Pathways workflow.	form + assessmentPage/-forms + Brevo
GET	/api/v1/assessment/admin/responses	Chief or manageAssessment-Forms	List normalized responses for one assessment.	form + assessmentPage/-forms + Brevo
PATCH	/api/v1/assessment/admin/responses/:responseId/linkage	Chief or manageAssessment-Forms	Update user identifier/database-form linkage fields.	form + assessmentPage/-forms + Brevo
DELETE	/api/v1/assessment/admin/responses/:responseId	Chief or manageAssessment-Forms	Delete an assessment response.	form + assessmentPage/-forms + Brevo
POST	/api/v1/assessment/admin/responses/:responseId/identifier-email	Chief or manageAssessment-Forms	Send linked identifier to response email and record send metadata.	form + assessmentPage/-forms + Brevo
PATCH	/api/v1/assessment/admin/forms/:formId	Chief or manageAssessment-Forms	Set form state to active, disabled or hidden.	form + assessmentPage/-forms + Brevo

Method	Endpoint	Access	Purpose	Primary data/provider
GET	/api/v1/booking/schedule	Public	Return privacy-safe availability/busy schedule for at most 62 days.	availability, unavailability, meetings, meetingRequests, peopleDevelopment/meetingSchedules, calendarReservations
POST	/api/v1/booking/requests	Public	Validate exact 30-minute request, claim reservation atomically, persist pending request and notify Pastor.	meetingRequests + calendarReservations + Brevo
POST	/api/v1/bezalel/pastor/chat	Firebase Bearer + Pastor	Send sanitized calendar context and conversation to Gemini; return at most one validated calendar action.	Read-only calendar snapshot + Gemini
POST	/api/v1/bezalel/booking/chat	Public	Use public schedule projection with Gemini to answer availability questions and assemble a booking object.	Public schedule projection + Gemini
GET	/api/v1/pastor-calendar	Firebase Bearer + Pastor	Return complete normalized Pastor calendar snapshot.	meetings, meetingRequests, availability, unavailability, calendarReservations, People Development schedules, Brevo
GET	/api/v1/pastor-calendar/export.ics	Firebase Bearer + Pastor	Export a requested month as iCalendar data.	meetings, meetingRequests, availability, unavailability, calendarReservations, People Development schedules, Brevo
POST	/api/v1/pastor-calendar/meetings	Firebase Bearer + Pastor	Create confirmed meeting/event with conflict checks.	meetings, meetingRequests, availability, unavailability, calendarReservations, People Development schedules, Brevo
PATCH	/api/v1/pastor-calendar/meetings/:meetingId	Firebase Bearer + Pastor	Update meeting/event and notify requester when details change.	meetings, meetingRequests, availability, unavailability, calendarReservations, People Development schedules, Brevo
DELETE	/api/v1/pastor-calendar/meetings/:meetingId	Firebase Bearer + Pastor	Delete meeting/event and release associated reservation where applicable.	meetings, meetingRequests, availability, unavailability, calendarReservations, People Development schedules, Brevo

Method	Endpoint	Access	Purpose	Primary data/provider
POST	/api/v1/pastor-calendar/availability	Firebase Bearer + Pastor	Create public-booking availability block.	meetings, meetingRequests, availability, unavailability, calendar-Reservations, People Development schedules, Brevo
PATCH	/api/v1/pastor-calendar/availability/:blockId	Firebase Bearer + Pastor	Update an availability block.	meetings, meetingRequests, availability, unavailability, calendar-Reservations, People Development schedules, Brevo
DELETE	/api/v1/pastor-calendar/availability/:blockId	Firebase Bearer + Pastor	Delete an availability block.	meetings, meetingRequests, availability, unavailability, calendar-Reservations, People Development schedules, Brevo
POST	/api/v1/pastor-calendar/unavailability	Firebase Bearer + Pastor	Create blocking/unavailable calendar period with conflict validation.	meetings, meetingRequests, availability, unavailability, calendar-Reservations, People Development schedules, Brevo
PATCH	/api/v1/pastor-calendar/unavailability/:blockId	Firebase Bearer + Pastor	Update an unavailable block.	meetings, meetingRequests, availability, unavailability, calendar-Reservations, People Development schedules, Brevo
DELETE	/api/v1/pastor-calendar/unavailability/:blockId	Firebase Bearer + Pastor	Delete an unavailable block.	meetings, meetingRequests, availability, unavailability, calendar-Reservations, People Development schedules, Brevo
POST	/api/v1/pastor-calendar/meeting-requests/:requestId/decision	Firebase Bearer + Pastor	Accept or reject pending public request; accepting creates meeting and updates reservation; notify requester.	meetings, meetingRequests, availability, unavailability, calendar-Reservations, People Development schedules, Brevo
POST	/api/v1/meeting-invitations	Firebase Bearer + Pastor	Send localized participant meeting invitations via backend-controlled HTML/text templates.	Brevo
GET	/api/v1/people-notes	Firebase Bearer + Pastor	List normalized confidential People Notes records.	peopleNotes

Method	Endpoint	Access	Purpose	Primary data/provider
POST	/api/v1/people-notes	Firebase Bearer + Pastor	Create a person with server-owned audit fields.	peopleNotes
POST	/api/v1/people-notes/:personId/items	Firebase Bearer + Pastor	Create strength/weakness/development item under a person.	peopleNotes
POST	/api/v1/people-notes/:personId/items/:itemId/comments	Firebase Bearer + Pastor	Append development comment and update parent timestamps.	peopleNotes
PATCH	/api/v1/people-notes/:personId/items/:itemId/follow-up	Firebase Bearer + Pastor	Update follow-up date atomically with audit timestamps.	peopleNotes
DELETE	/api/v1/people-notes/:personId	Firebase Bearer + Pastor	Delete person and nested record.	peopleNotes
DELETE	/api/v1/people-notes/:personId/items/:itemId	Firebase Bearer + Pastor	Delete development item.	peopleNotes
DELETE	/api/v1/people-notes/:personId/items/:itemId/comments/:commentId	Firebase Bearer + Pastor	Delete one development comment.	peopleNotes
POST	/api/v1/people-development/portal	Public identifier + rate limit	Resolve participant profile, group-filtered assignments and active recurring meeting schedules.	form + peopleDevelopment
GET	/api/v1/people-development/pastor/snapshots	Firebase Bearer + Pastor	Return normalized members, assignments, personal notes and schedules.	peopleDevelopment
PATCH	/api/v1/people-development/pastor/members/:memberKey/group	Firebase Bearer + Pastor	Assign group and synchronize linkage fields back to source assessment records.	peopleDevelopment
POST	/api/v1/people-development/pastor/assignments	Firebase Bearer + Pastor	Create assignment/post targeted to one or more groups.	peopleDevelopment
PATCH	/api/v1/people-development/pastor/assignments/:assignmentId/attachments	Firebase Bearer + Pastor	Replace assignment attachment list.	peopleDevelopment
DELETE	/api/v1/people-development/pastor/assignments/:assignmentId	Firebase Bearer + Pastor	Delete assignment.	peopleDevelopment
POST	/api/v1/people-development/pastor/personal-notes	Firebase Bearer + Pastor	Create member-specific strength/weakness note.	peopleDevelopment
DELETE	/api/v1/people-development/pastor/personal-notes/:noteId	Firebase Bearer + Pastor	Delete member-specific personal note.	peopleDevelopment
POST	/api/v1/people-development/pastor/schedules	Firebase Bearer + Pastor	Create recurring group/shared meeting schedule.	peopleDevelopment
PATCH	/api/v1/people-development/pastor/schedules/:scheduleId	Firebase Bearer + Pastor	Update recurring group/shared meeting schedule.	peopleDevelopment
DELETE	/api/v1/people-development/pastor/schedules/:scheduleId	Firebase Bearer + Pastor	Delete recurring meeting schedule.	peopleDevelopment
POST	/api/v1/people-development/notifications	Public (security gap)	Send BCC People Development notification payload after schema validation. No authentication middleware is present in current route.	Brevo
GET	/api/v1/nextgen/mission-map	Firebase Bearer + any account email (current code)	Return secret-configured Montreal mission locations/connections after schema validation.	NEXTGEN_MISSION_MAP_DATA secret
GET	/api/v1/nextgen/health	Public	Contract health endpoint for production smoke test.	None
GET	/api/v1/nextgen/session	Firebase Bearer	Return current NextGen account identity and Pastor flag.	Token only
GET	/api/v1/nextgen/qa/sessions	Firebase Bearer	List non-draft QA sessions; ensure one-time legacy session migration.	nextGenPortal/qa + legacy nextGenActivities

Method	Endpoint	Access	Purpose	Primary data/provider
GET	/api/v1/nextgen/qa/sessions/:sessionId	Firebase Bearer + email + open session	Return member-safe session view with requested filter.	nextGenPortal/qa
POST	/api/v1/nextgen/qa/sessions/:sessionId/questions	Firebase Bearer + email + open session	Enforce two-question account limit, ask Bezalel to review theme relevance, then store accepted question.	nextGenPortal/qa + Gemini
POST	/api/v1/nextgen/qa/sessions/:sessionId/questions/:questionId/votes	Firebase Bearer + email + open session	Store/change one vote keyed by normalized email hash and account identity.	nextGenPortal/qa
GET	/api/v1/nextgen/pastor/qa/sessions	Pastor or manageNextGenQa administrator	List QA sessions with management counts.	nextGenPortal/qa + Gemini where applicable
POST	/api/v1/nextgen/pastor/qa/sessions	Pastor or manageNextGenQa administrator	Create QA session and translate theme with Bezalel.	nextGenPortal/qa + Gemini where applicable
PATCH	/api/v1/nextgen/pastor/qa/sessions/:sessionId	Pastor or manageNextGenQa administrator	Update title/description/status/theme; retranslate changed theme.	nextGenPortal/qa + Gemini where applicable
PATCH	/api/v1/nextgen/pastor/qa/sessions/:sessionId/questions/:questionId	Pastor or manageNextGenQa administrator	Select/unselect a question for discussion.	nextGenPortal/qa + Gemini where applicable
GET	/api/v1/nextgen/pastor/qa/sessions/:sessionId	Pastor or manageNextGenQa administrator	Return manager view including participants and voter identity detail.	nextGenPortal/qa + Gemini where applicable
PATCH	/api/v1/nextgen/pastor/qa/sessions/:sessionId/participants/:participantUid	Pastor or manageNextGenQa administrator	Verify or discard participant for vote aggregation.	nextGenPortal/qa + Gemini where applicable
GET	/api/v1/nextgen/files/folders	Any Firebase-authenticated account (current code)	List NextGen folder metadata.	nextGenPortal/files + Backblaze B2
POST	/api/v1/nextgen/files/folders	Any Firebase-authenticated account (current code)	Create NextGen folder.	nextGenPortal/files + Backblaze B2
DELETE	/api/v1/nextgen/files/folders/:folderId	Any Firebase-authenticated account (current code)	Delete empty NextGen folder.	nextGenPortal/files + Backblaze B2
GET	/api/v1/nextgen/files	Any Firebase-authenticated account (current code)	List NextGen file metadata.	nextGenPortal/files + Backblaze B2
POST	/api/v1/nextgen/files/upload-url	Any Firebase-authenticated account (current code)	Create pending file record and B2 upload URL.	nextGenPortal/files + Backblaze B2
POST	/api/v1/nextgen/files/:fileId/complete	Any Firebase-authenticated account (current code)	Inspect uploaded object, validate size and mark ready.	nextGenPortal/files + Backblaze B2
GET	/api/v1/nextgen/files/:fileId/download-url	Any Firebase-authenticated account (current code)	Create signed private download URL.	nextGenPortal/files + Backblaze B2
DELETE	/api/v1/nextgen/files/:fileId	Any Firebase-authenticated account (current code)	Delete B2 object and metadata.	nextGenPortal/files + Backblaze B2

27.2 API Design Conventions

Most application endpoints return `{ success: true, data: ... }` on success and `{ success: false, error: { code, message, ... } }` on failure. Create operations commonly return HTTP 201. Validation failures use 400; missing authentication 401; authorization 403; conflict/duplicate/state errors commonly 409; semantic AI rejection can use 422; missing records 404; provider/database failure commonly 502; missing server configuration commonly 503; and public portal rate limiting uses 429.

27.3 Cache-Control

Sensitive session, People Notes, mission-map, and other authenticated responses set private/no-store behavior. Public form-state and schedule data use short public or no-store behavior depending on endpoint. Maintaining this distinction is important because the Worker returns both public projections and highly confidential data.

Chapter 28

Database and Persistence Reference

28.1 Observed Realtime Database Roots

The following roots are directly evidenced by current backend or frontend code:

- `administration/adminHierarchy` – Chief UID, administrator profiles, roles/status/authority.
- `administration/archives/folders` and `administration/archives/files` – archive metadata.
- `assessmentPage/forms` – assessment state controls.
- `form` – assessment/direct-signup responses and identity/linkage data.
- `availability`, `unavailability`, `meetings`, `meetingRequests`, `calendarReservations` – `calendar/booking`.
- `attendance/people` – attendance people and attendance date history.
- `peopleDevelopment/members`, `peopleDevelopment/assignments`, `peopleDevelopment/personalNotes`, `peopleDevelopment/meetingSchedules` – People Development.
- `peopleNotes` – confidential People Notes hierarchy.
- `nextGenPortal/qa` – account-based QA sessions/questions/participants/votes.
- `nextGenPortal/files/folders` and `nextGenPortal/files/records` – NextGen object metadata.
- `nextGenActivities` and `nextGenUsers` – legacy identifier-based NextGen data.
- `migrations/legacyQaSession1` – QA legacy migration marker.
- `landingPage/carousel` – public carousel content.
- `tutorialBuilder/tutorials` and `tutorialBuilder/progress` – tutorial definitions/progress.

28.2 Identity Fragmentation

The repository still represents a person differently across Firebase Authentication, assessment responses, People Development member records, attendance people, People Notes people, booking request identities, and legacy NextGen identities. The ERD documentation explicitly models several of these as separate identity entities. Cross-domain linkage often uses email, a generated/personal identifier, source response keys, or Firebase UID rather than one canonical member ID.

28.3 Storage Classes

Data class	Store	Current implementation

Application records	Firebase Database	Realtime	JSON-like records accessed by browser SDK or Worker REST clients.
LInC Archives objects	Backblaze B2		Private signed URL upload/download; meta-data in Firebase.
NextGen file objects	Backblaze B2		Private signed URL upload/download; meta-data in Firebase.
People Development at- tachments	Firebase Database	Realtime	Embedded Base64, max 1 MiB each, max five per assignment.
Attendance/carousel pho- tos	Firebase data values		Data URL / URL / legacy Base64 normaliza- tion; no active Firebase Storage upload flow found.
Mission map locations	Cloudflare secret		JSON payload in NEXTGEN_MISSION_MAP_DATA.
AI prompts/results	Transient HTTP pay- loads		Gemini system instructions in backend source; conversations/context sent server-side per re- quest.

Security and Privacy Assessment

29.1 Implemented Controls

- Firebase Bearer-token verification on sensitive routers.
- Server-side exact Pastor authorization for Pastor APIs.
- Capability-based administrator checks for assessment, archives, and NextGen QA management.
- Strict Zod validation and rejection of unknown request fields on many write routes.
- Server-controlled timestamps/status fields and narrow database paths.
- Privacy-safe public booking projection rather than raw calendar reads.
- Calendar reservation claiming to reduce double-booking races.
- Server-held Brevo, service-account, B2, Gemini, and mission-map credentials/data.
- Backend-controlled email templates and HTML escaping.
- Signed, short-lived B2 URLs with object verification before ready state.
- Private/no-store cache policy on sensitive APIs.
- Question count limit and AI theme moderation in NextGen QA.
- Identifier portal attempt limiting.
- Automated backend tests, type-checking, deploy gating, and production contract smoke test.

29.2 Current Risk Register

ID	Severity	Area	Finding
R1	Critical	People Development notifications	Notification endpoint has no Firebase authentication or Pastor/admin authorization middleware. A caller that can reach it and satisfy schema validation can cause provider sends.
R2	High	NextGen files	All file/folder operations are protected only by Firebase authentication; no NextGen membership, Pastor, or administrator authority check is applied.
R3	High	Mission map	Current route accepts any authenticated Firebase user with an email; this is broader than repository README wording and may expose private addresses/location notes.

R4	Medium	Email test		Temporary <code>/api/v1/email/test</code> route remains publicly mounted and can produce provider traffic to a fixed recipient.
R5	High	Residual Firebase	direct	Attendance, carousel, tutorials, participant aggregation, and legacy NextGen paths still rely on browser Firebase access and therefore on Security Rules not included in the supplied snapshot.
R6	Medium	People Development attachments		Base64 files remain inside Realtime Database, increasing record size and making object lifecycle/authorization less clean than B2 domains.
R7	Medium	Legacy integrations		EmailJS and Google OAuth/Gmail/Calendar helpers remain initialized or reachable in source, increasing bundle/maintenance surface and documentation ambiguity.
R8	Medium	Identity model		Multiple domain-specific identities complicate access revocation, privacy requests, deduplication, and auditability.
R9	Medium	AI abuse/cost		Public Bezalel booking chat has no explicit durable global rate limiter in the reviewed route. Gemini quota errors are handled but abuse prevention is limited.
R10	Medium	Admin carousel		Carousel authority is represented in admin profile, but current carousel persistence remains direct from browser, so backend capability allocation is not the complete enforcement boundary.
R11	Medium	Attendance		Attendance authority is UI-gated but browser-controlled persistence makes Firebase Rules the decisive boundary.
R12	Low/Medium	Legal drift		Privacy/terms copy can become inaccurate as external providers and data categories evolve.
R13	Medium	Observability		Wrangler config shows observability commented out; no dedicated audit-log domain is visible in the supplied source.
R14	Medium	Revocation		ID-token verification validates JWT claims but does not implement an explicit Firebase revoked-token check in the reviewed middleware.
R15	Low/Medium	Repository drift		README and live source already disagree on at least mission-map authorization, so operational docs require release-level synchronization.

29.3 Recommended Closure Order

First protect or remove the unauthenticated notification/test routes; then tighten NextGen files and mission-map authorization; then complete backend migration for attendance/carousel/tutorials/legacy NextGen; then add durable abuse controls, audit events, security-rule documentation/tests, and a canonical identity strategy.

Chapter 30

Testing, CI/CD, Deployment, and Operations

30.1 Frontend Workflow

`.github/workflows/frontend-build.yml` runs on relevant pushes and pull requests, checks out the repository, installs Node.js 24, runs `npm ci`, and executes the production frontend build (TypeScript project build plus Vite build).

30.2 Backend Verification and Deployment

`backend-tests.yml` runs backend tests and TypeScript type checking. For main-branch push or manual dispatch, the production deploy job waits for tests, verifies Cloudflare credentials, runs the Wrangler deploy script, and then executes the production smoke test. Concurrency is serialized for the production backend environment.

30.3 Production Smoke

A separate scheduled workflow runs daily at cron `17 10 * * *` and can also be manually dispatched. It installs backend dependencies and runs the production NextGen API contract smoke script. The smoke script supports environment overrides for backend URL, attempt count, and retry delay.

30.4 Hosting Surfaces

The frontend is configured for Netlify SPA hosting with `npm run build` and `dist`. A wildcard redirect sends client routes to `/index.html`. The backend is an independently deployed Cloudflare Worker named `linc-backend`. Frontend and backend contract changes must remain compatible across these separate surfaces.

30.5 Netlify Secret Scanning

The current Netlify configuration excludes source-code paths from secret scanning through `SECRETS_SCAN_OMIT_PATHS`. This is broader than the earlier single-key omission approach and should be reviewed carefully because excluding source trees can reduce accidental-secret detection.

Chapter 31

Configuration and Secret Catalog

31.1 Frontend/Browser Configuration

Variable	Purpose	Classification
VITE_FIREBASE_API_KEY	Browser Firebase configuration	Public configuration
VITE_FIREBASE_AUTH_DOMAIN	Browser Firebase configuration	Public configuration
VITE_FIREBASE_PROJECT_ID	Browser Firebase configuration	Public configuration
VITE_FIREBASE_STORAGE_BUCKET	Browser Firebase configuration; storage is configured but no active upload API was found	Public configuration
VITE_FIREBASE_MESSAGING_SENDER_ID	Browser Firebase configuration	Public configuration
VITE_FIREBASE_APP_ID	Browser Firebase configuration	Public configuration
VITE_FIREBASE_MEASUREMENT_ID	Browser Firebase configuration	Public configuration
VITE_BACKEND_BASE_URL	Optional frontend override for Worker base URL	Public configuration
VITE_GOOGLE_MAPS_API_KEY	Restricted browser key used by Montreal mission map routing/map rendering	Browser-restricted secret-like key
VITE_GOOGLE_CLIENT_ID	Legacy Google OAuth client identifier used by Gmail/Calendar integration code	Public OAuth identifier

31.2 Backend and CI Configuration

Variable	Purpose	Classification
FIREBASE_PROJECT_ID	ID-token audience/issuer validation	Public server configuration
FIREBASE_DATABASE_URL	Realtime Database REST base URL	Public server configuration
FIREBASE_SERVICE_ACCOUNT_EMAIL	Google OAuth service-account identity for admin database access	Secret/protected
FIREBASE_SERVICE_ACCOUNT_PRIVATE_KEY	RSA private key used to mint Google OAuth assertion	Secret

Variable	Purpose	Classification
BOOKING_NOTIFICATION_EMAIL	Fixed Pastor notification recipient for public booking	Protected configuration
BREVO_API_KEY	Transactional email provider credential	Secret
BREVO_SENDER_EMAIL	Verified sender address	Protected configuration
BREVO_SENDER_NAME	Email sender display name	Protected configuration
BREVO_TEST_RECIPIENT	Fixed temporary email test recipient	Protected configuration
B2_BUCKET_NAME	Backblaze B2 archive bucket	Protected configuration
B2_S3_ENDPOINT	B2 S3-compatible endpoint	Public/protected configuration
B2_REGION	B2 region identifier	Public/protected configuration
B2_APPLICATION_KEY_ID	B2 application key ID	Secret
B2_APPLICATION_KEY	B2 application key	Secret
GEMINI_API_KEY	Gemini server-side API credential	Secret
GEMINI_MODEL	Gemini model name; checked-in default is gemini-3.6-flash	Public server configuration
NEXTGEN_MISSION_MAP_DATA	Private JSON location/connection payload	Secret
LINC_BACKEND_URL	Production smoke-test optional backend override	CI configuration
LINC_SMOKE_ATTEMPTS	Smoke retry count override	CI configuration
LINC_SMOKE_RETRY_DELAY_MS	Smoke retry delay override	CI configuration

31.3 Checked-In Worker Defaults

The supplied `wrangler.jsonc` checks in non-secret values for Firebase project/database, Pastor booking notification recipient, B2 bucket/endpoint/region, and Gemini model. Secret values are expected in Worker secret configuration or local `.dev.vars`; the repository's `.dev.vars.example` lists B2 and Gemini values but is not a complete inventory of all backend variables required by current routes.

Chapter 32

Technology and Dependency Catalog

32.1 Runtime Architecture

Frontend runtime is React 19.2.5 + React DOM, React Router 7.14.2, Firebase 12.12.1, Motion, Lucide, YAML, React Big Calendar, React Joyride, Leaflet/React Leaflet, Google Maps loader, and utility libraries. The package manifest also still includes EmailJS, OpenAI, and @google/genai; no active source import of OpenAI or @google/genai was found in the supplied snapshot because Gemini is called directly by HTTP from the Worker. The backend uses Hono, Zod, and aws4fetch, with Wrangler/Vitest tooling.

32.2 Exact Package Manifest

Scope	Package	Version constraint
Frontend runtime	@emailjs/browser	^4.4.1
Frontend runtime	@google/genai	^1.52.0
Frontend runtime	@googlemaps/js-api-loader	^2.1.1
Frontend runtime	clsx	^2.1.1
Frontend runtime	date-fns	^4.1.0
Frontend runtime	firebase	^12.12.1
Frontend runtime	leaflet	^1.9.4
Frontend runtime	lucide-react	^1.14.0
Frontend runtime	motion	^12.38.0
Frontend runtime	openai	^6.36.0
Frontend runtime	react	^19.2.5
Frontend runtime	react-big-calendar	^1.19.4
Frontend runtime	react-dom	^19.2.5
Frontend runtime	react-firebase-hooks	^5.1.1
Frontend runtime	react-joyride	^3.2.0
Frontend runtime	react-leaflet	^5.0.0
Frontend runtime	react-modal	^3.16.3
Frontend runtime	react-router-dom	^7.14.2
Frontend runtime	tailwind-merge	^3.5.0
Frontend runtime	yaml	^2.8.1
Frontend development	@eslint/js	^10.0.1
Frontend development	@testing-library/jest-dom	^7.0.0
Frontend development	@testing-library/react	^16.3.2
Frontend development	@types/google.maps	^3.65.4
Frontend development	@types/leaflet	^1.9.22

Scope	Package	Version constraint
Frontend development	@types/node	^24.12.2
Frontend development	@types/react	^19.2.14
Frontend development	@types/react-big-calendar	^1.16.3
Frontend development	@types/react-dom	^19.2.3
Frontend development	@types/react-modal	^3.16.3
Frontend development	@vitejs/plugin-react	^6.0.1
Frontend development	autoprefixer	^10.5.0
Frontend development	eslint	^10.2.1
Frontend development	eslint-plugin-react-hooks	^7.1.1
Frontend development	eslint-plugin-react-refresh	^0.5.2
Frontend development	globals	^17.5.0
Frontend development	jsdom	^30.0.1
Frontend development	nodemon	^3.1.14
Frontend development	postcss	^8.5.14
Frontend development	tailwindcss	^3.4.19
Frontend development	ts-node	^10.9.2
Frontend development	typescript	~6.0.2
Frontend development	typescript-eslint	^8.58.2
Frontend development	vite	^8.0.10
Frontend development	vitest	^4.1.10
Backend runtime	aws4fetch	^1.0.20
Backend runtime	hono	^4.12.32
Backend runtime	zod	^4.4.3
Backend development	@cloudflare/vitest-pool-workers	^0.18.8
Backend development	typescript	~6.0.3
Backend development	vitest	^4.1.10
Backend development	wrangler	^4.110.0

Chapter 33

Complete End-to-End Workflow Reference

This chapter complements the feature and API catalogs by tracing the principal application behaviors from user action through frontend orchestration, backend or Firebase access, persistence, external providers, and resulting state. It is intentionally redundant with lower-level catalogs where that redundancy helps a maintainer understand a complete transaction without jumping between chapters.

33.1 Application Startup, Authentication, and Route Protection

Actors: anonymous visitor, Firebase-authenticated account, Pastor, administrator. The browser initializes Firebase, internationalization, tutorial context, and React routing. Protected Pastor routes resolve Firebase authentication and then call the backend Pastor-access/session boundary. The backend verifies the Firebase ID token against Google's signing keys and then applies the exact Pastor email allowlist. Administrator access uses the separate administrator session endpoint and the hierarchy stored beneath `administration/adminHierarchy`. The application still processes legacy Gmail OAuth callback state and initializes EmailJS client configuration during startup, so authentication/bootstrap code contains both current and residual integration paths.

Info – Security Boundary

A successful Firebase login proves account identity only. Pastor authorization and administrator authority are separate decisions. Current NextGen account routes generally require a Firebase account plus an email, while selected management operations additionally require Pastor status or `manageNextGenQa`.

33.2 Public Assessment Submission

The visitor opens `/assessment`. The frontend obtains the current public assessment states, loads a selected YAML definition, renders fields and rating groups, validates required user input, and presents calculated results. On submission, the backend accepts only a known form identifier, validates the answer structure, recomputes the supported result calculations server-side, records timestamps and metadata, persists the response beneath `form`, and performs configured notification behavior. This prevents a browser from becoming authoritative for stored scores simply by submitting a precomputed result object.

The two supplied active definitions are `five-service-pathways` and `spiritual-gifts-discovery`. Their exact sections, configured item counts, scoring operations, and result outputs are documented in the Assessment Definition Reference chapter.

33.3 Assessment Administration and Member Linkage

An authorized administrator with assessment authority can inspect and change form publication/state controls through backend administration endpoints rather than editing the public definition from the browser. User-linkage operations connect submitted assessment records to People Development member identity. Current source retains a fragmented identity landscape, so the linkage layer remains important: an assessment response, a People Development member, a Firebase account, an attendance person, and a legacy NextGen user are not yet one canonical member record.

33.4 Public Booking and Atomic Reservation

The visitor opens `/booking`. The frontend requests a date-window projection from the booking API. The backend combines Pastor availability, unavailability, confirmed meetings, pending requests, active calendar reservations, and recurring People Development schedules and returns a sanitized public schedule rather than private meeting details. The visitor selects an exact 30-minute slot, supplies name, email, reason, locale, and the empty anti-bot honeypot, and submits a request.

Before persistence, the Worker revalidates date/time rules and conflicts. It transactionally claims a reservation key so concurrent requests cannot both own the same slot, stores the request as pending, and attempts Pastor notification through Brevo. The public request remains a request; it is not silently converted into a confirmed meeting.

33.5 Pastor Meeting-Request Decision

The Pastor Dashboard receives pending requests through the authenticated calendar snapshot. On accept, the backend verifies Pastor authorization, reloads current scheduling state, rechecks conflicts, creates the confirmed meeting, updates request state, reconciles reservation state, and sends requester notification. On reject, it updates request state and sends the corresponding communication. The public requester never receives the Pastor's private calendar contents.

33.6 Pastor Meeting Creation, Editing, and Participant Invitations

Pastor can create an internal meeting/event independently of a public request. The validated meeting contract includes title, description, date, start/end time, location, meeting type, optional meeting link, and participant identifiers. Calendar mutations are backend-owned and the client refreshes the shared snapshot after mutation. Participant invitation delivery is a separate backend Brevo workflow: the browser submits structured meeting/recipient data with a Firebase Bearer token, the Worker builds localized controlled HTML/plain-text email, and the provider credential remains server-side.

33.7 Pastor Availability and Unavailability

Availability and unavailability records are managed through dedicated authenticated calendar routes. The UI supports generated windows as well as record-level edits/deletions. These records feed both the Pastor calendar and the privacy-safe public schedule. Because booking conflict logic reads them again at request/decision time, the public schedule display is not the only defense against invalid slot claims.

33.8 Bezalel Pastor Calendar Assistance

Pastor opens Bezalel from the Pastor workspace and writes a natural-language calendar instruction. The Worker sends a controlled baseline prompt plus a sanitized calendar context to Gemini. The allowed structured action set is deliberately narrow: `none`, `open_availability`, `block_time`, `delete_availability`, `delete_unavailability`, `accept_request`, or `reject_request`. Bezalel does not directly mutate Firebase from the model call. The model returns a structured action proposal; the frontend executes the corresponding existing calendar API operation, preserving normal validation/authorization.

Warning – AI Control Model

The important control is not that the model is “trusted.” The model is constrained to a schema and the application owns execution. Calendar mutation APIs remain authoritative and must continue to validate identity, Pastor authority, input, and conflicts independently of the generated action.

33.9 Bezalel Public Booking Assistance

The public booking assistant operates on the sanitized public schedule rather than the Pastor’s private calendar. It can answer booking questions, collect missing booking information, and move the conversation toward a ready-to-book state. The final booking still goes through the normal booking request contract and conflict controls; the assistant is not a bypass around the booking API.

33.10 People Development Pastor Management

The Pastor snapshot normalizes members, assignments, personal notes, and recurring schedules from `peopleDevelopment`. Pastor can assign a member to a ministry group, create assignments/posts targeted to one or more groups, attach files, record member-specific strength/weakness notes, and create recurring group/shared meeting schedules. Group changes use backend-controlled multi-path updates so relevant linkage information in source assessment records stays synchronized with the People Development member.

Assignment attachments remain embedded as Base64 records in Realtime Database in the reviewed source, with explicit file-count/size validation. This differs from LInC Archives and NextGen Files, which use Backblaze B2 object storage.

33.11 Congregation Group Notes Portal

The participant opens `/group-notes` and supplies the ministry identifier expected by the public portal. The backend applies rate limiting, resolves the matching participant/member identity, determines group membership, and returns only assignments and active recurring schedules relevant to that profile/group. The portal is intentionally identifier-based rather than Firebase-account based in the current implementation, so identifier secrecy, rate limiting, and returned-data minimization matter.

33.12 People Development Notification Delivery

The current notification route accepts a validated People Development notification payload and sends BCC mail through Brevo. Unlike the Pastor management routes, the reviewed route is not wrapped

in Firebase authentication middleware. This is documented as a current implementation fact and a critical remediation item, not as an intended public feature.

33.13 Confidential People Notes

Pastor opens `/pastor/people-notes`. Every API operation requires Firebase authentication and Pastor authorization. Pastor can create a person record, add development items, append comments, update follow-up dates, and delete person/item/comment records. Server-owned timestamps are updated with mutations. Responses are treated as non-cacheable/private. This domain is separate from People Development personal notes and should remain a high-sensitivity authorization boundary.

33.14 NextGen Account Session

The active `/nextgen-activities` experience is Firebase email/password account based. After sign-in, the client obtains NextGen session identity and Pastor metadata and can load the account-visible QA sessions, files interface, and mission map. This differs materially from the legacy four-character identifier activities still present in source but no longer mounted as the primary NextGen route.

33.15 NextGen QA Session and Question Submission

An authenticated member lists non-draft QA sessions and opens an active session. Before accepting a question, the backend enforces session state and an account question limit. It then invokes Bezael/Gemini to compare the question against the session theme. The model response is schema-validated; an off-theme decision is returned as a semantic rejection, while an accepted question is stored in the session data. The one-time legacy migration creates a closed historical session for earlier QA data when required.

33.16 NextGen Voting

Voting is tied to the authenticated account/email rather than a freely supplied short identifier. The backend maintains one current vote per account/question and supports changing the vote. Member-facing data is intentionally narrower than management data; management counts and participant verification states are exposed only to the appropriate management boundary.

33.17 NextGen QA Management

A Pastor or administrator with `manageNextGenQa` can create/edit session title, description, status, and theme; select questions for discussion; and manage participant verification/discarded state. Theme translation can invoke Gemini when the source theme changes. Management endpoints use a shared authorization helper so Pastor authority and delegated QA authority remain separate from ordinary member access.

33.18 NextGen File Upload, Completion, Download, and Delete

The browser requests a signed upload URL, uploads the object directly to Backblaze B2, and then calls the completion endpoint. The backend inspects the object before marking metadata ready. Downloads

similarly use signed private URLs. Folder/file metadata is persisted beneath `nextGenPortal/files`. In the reviewed source, the route-level authorization requirement is only a Firebase-authenticated account; therefore upload, folder creation/deletion, file listing, signed download, and deletion are broader than a typical member-only/private-content policy would imply.

33.19 Montreal Mission Map

The frontend requests mission-map location/connection data from the Worker, then uses Google Maps/browser routing facilities for map display and route details. The sensitive location graph is stored in the `NEXTGEN_MISSION_MAP_DATA` Worker secret rather than compiled into the frontend. The current route checks for a valid Firebase user carrying an email. Repository prose that describes one specific mission account is therefore stricter than the code and should not be treated as the enforced rule.

33.20 Administrator Onboarding and Authority Allocation

A Firebase-authenticated account opening `/administrator` calls the administrator session API. When the hierarchy contains no Chief, the exact Pastor account can bootstrap the Chief. Other new accounts are represented as pending/non-authoritative until the Chief activates them. The Chief can allocate the five current authority booleans or suspend an administrator. Backend-protected domains check the relevant authority rather than trusting that a UI tab was hidden.

33.21 LInC Archives File Lifecycle

An authorized Chief or `manageArchives` administrator can create nested folder metadata, request a B2 upload URL, upload directly to B2, complete/verify the object, obtain a private signed download URL, and delete files. The backend prevents deletion of a non-empty folder and keeps Firebase metadata separate from B2 object bytes. File size is bounded by the archive contract and server inspection is used before a pending file is promoted to ready.

33.22 Attendance Management

Attendance remains a direct-client Firebase domain in the reviewed source. The administrator UI maintains people, optional photos/contact details, Sunday attendance records, analytics, and person-level analysis under `attendance/people`. Access is gated by the administrator UI/capability experience, but no equivalent Worker persistence boundary was found for attendance. Consequently, effective security still depends on Firebase database rules that were not supplied in this snapshot.

33.23 Landing Carousel Management

Public `CommunityCarousel` subscribes directly to `landingPage/carousel`. The administrator carousel UI also writes through client-side Firebase. Although `manageCarousel` exists in the administrator authority model, the persistence operation is not yet routed through a Worker endpoint that can independently enforce that capability. This makes Firebase Rules part of the actual enforcement boundary.

33.24 Tutorial Authoring, Simulation, Playback, and Progress

The tutorial subsystem supports builder/editor screens, ordered steps, route/target/action metadata, simulation overlays, target selection, tutorial playback, and per-user progress. Tutorial definitions and progress are stored directly through Firebase helper code beneath `tutorialBuilder/tutorials` and `tutorialBuilder/progress`. The tutorial context is mounted at application level so tutorials can span routes and identify UI targets.

33.25 Legacy Integrations and Residual Paths

The repository retains EmailJS startup/configuration, Gmail/Google OAuth code, legacy NextGen Firebase modules, and an older `BookMeeting` component in addition to the newer backend-centered paths. Package dependencies also include OpenAI and `@google/genai` without active imports in the reviewed source. These artifacts are not assumed to be active merely because they exist; they are cataloged because they affect maintenance, bundle surface, migration decisions, and future cleanup.

Chapter 34

Enterprise Target Architecture and Modernization Reference

This chapter preserves the architecture/governance purpose of the earlier documentation while reclassifying work that the current source has already implemented. It is a target-state reference: statements using “shall” or “should” describe desired controls, not necessarily current behavior.

34.1 Architecture Principles

1. **Server authority for sensitive mutations.** Confidential data, permissions, bookings, notifications, AI orchestration, and private files should cross an authenticated backend boundary.
2. **Least privilege.** Firebase authentication, Pastor authority, Chief status, and delegated capabilities are distinct concepts and should remain independently enforced.
3. **Public projections instead of public databases.** Anonymous features receive sanitized data shaped for the feature rather than broad database branches.
4. **Deterministic domain logic outside UI components.** Validation, scoring, conflict detection, authorization, and persistence rules belong in testable domain/server modules.
5. **Explicit data classification.** Public, internal, confidential, highly confidential, and secret data require different storage, caching, logging, and access rules.
6. **Incremental migration.** The existing product remains usable while direct-client Firebase domains are progressively moved behind APIs.
7. **Provider isolation.** Brevo, B2, Firebase Admin REST access, Google Maps, and Gemini should be hidden behind focused adapters rather than spread through page components.
8. **Operational evidence.** CI, smoke tests, backups, audit logs, monitoring, and runbooks are part of production completeness, not optional documentation afterthoughts.

34.2 Current Target Stack

The current code has validated a pragmatic stack: React/TypeScript/Vite on Netlify for the browser, Firebase Authentication for identity, Cloudflare Workers + Hono + Zod for API/orchestration, Firebase Realtime Database for current structured persistence, Backblaze B2 for private object storage, Brevo for transactional mail, and Gemini for bounded AI features. PostgreSQL remains an optional later migration rather than a prerequisite for finishing the server boundary. A database change should be driven by relational/transactional needs and migration benefit, not architecture aesthetics alone.

34.3 Target Request Path

For sensitive features, the preferred path is:

**Browser → Firebase identity token → Worker authorization/validation → domain service
→ database/object/provider**

Public endpoints are allowed when the feature itself is public, but they must expose feature-specific projections, validate input, apply abuse controls, and never rely on hidden UI state as authorization.

34.4 Capability and Role Model

The current source has implemented Chief plus five delegated authority booleans, while Pastor remains a separate exact-email authority. The longer-term capability model should become action-oriented rather than UI-oriented. Recommended capability namespaces include:

- `assessment.forms.manage`
- `assessment.responses.link`
- `carousel.manage`
- `attendance.manage`
- `archives.manage`
- `nextgen.qa.manage`
- `nextgen.files.manage`
- `booking.requests.decide`
- `calendar.availability.manage`
- `calendar.meetings.manage`
- `people-development.manage`
- `people-notes.manage`
- `tutorials.manage`
- `admin.authority.allocate`

Pastoral care decisions and highly confidential pastoral records should remain non-delegable by ordinary technical administrators unless an explicit ministry-governance decision creates a separate pastoral role.

34.5 Canonical Identity Target

The current repository contains several identities: Firebase account UID/email, administrator UID, assessment response identifier, People Development member key/identifier, attendance person key, and legacy NextGen IDs. The target model should introduce a stable member/person identity with explicit links to authentication accounts and feature records. Feature-specific confidential data should remain in separate domains rather than becoming one giant member document.

A canonical identity migration should preserve historical identifiers and support deterministic reconciliation. It should not rewrite old records in place without an audit/reversal plan.

34.6 Target Persistence Boundaries

Domain	Current boundary	Target / next control
Assessments	Worker + RTDB	Keep server calculations; add definition versioning/audit and canonical identity.
Booking/calendar	Worker + RTDB	Retain backend transactions; add durable audit/observability and richer recurrence if required.
People Notes	Worker + RTDB	Retain strict Pastor boundary; add audit access history/retention controls.

People Development	Worker + RTDB	Authenticate notification endpoint; move large attachments to object storage.
NextGen QA	Worker + RTDB	Keep account/session rules; formalize capability and retention policy.
NextGen files	Worker + B2	Add narrower read/write authorization and ownership/audience semantics.
Mission map	Worker secret	Enforce intended audience in code; define rotation/update procedure.
Archives	Worker + B2 + RTDB metadata	Add audit events, quotas, malware/content validation as risk warrants.
Attendance	Direct Firebase	Migrate CRUD/analytics mutations behind capability-enforced backend.
Carousel	Direct Firebase	Migrate publication/write path behind <code>manageCarousel</code> .
Tutorials	Direct Firebase	Migrate authoring/progress access or prove adequate Rules; define author capability.
Legacy NextGen	Direct Firebase modules retained	Retire or isolate after historical data migration/verification.

34.7 API Standards

Every production endpoint should document method/path, authentication, authorization/capability, request schema, response schema, error semantics, rate limits, cache policy, data classification, persistence side effects, provider side effects, and audit requirements. Zod strict objects should remain the default for mutation contracts. Error responses should not leak provider credentials, raw service-account assertions, secret configuration, or full upstream bodies.

An OpenAPI description is recommended for the growing API surface. With 81 cataloged endpoints, hand-maintained frontend types alone become increasingly difficult to keep synchronized.

34.8 Transaction and Idempotency Standards

Booking conflict/reservation logic already demonstrates transaction-aware design. Similar discipline should be applied to any workflow where duplicate submission matters: archive/file completion, email retry, invitation fan-out, attendance correction imports, NextGen votes, and administrator authority changes. Where external providers are involved, an idempotency key or durable job/event record should separate “domain transaction succeeded” from “notification/provider delivery succeeded.”

34.9 Security Architecture

The target security model includes Firebase identity verification, explicit authorization, server-side validation, provider-secret isolation, abuse/rate controls, private object URLs, least-privilege service credentials, secure cache headers, structured logs without sensitive payloads, and periodic access review. Firebase Rules remain defense-in-depth even after server migration and are the primary enforcement mechanism for any direct-client domain that remains.

Revoked-token checking should be considered for high-sensitivity operations. The current Worker validates signed token claims but the reviewed authentication implementation does not perform an

explicit Firebase revoked-token lookup.

34.10 Privacy and Data Governance

Data categories should have written retention and correction/deletion rules. Highly confidential pastoral notes require the strongest access controls and the least logging of content. Participant survey/QA claims must accurately describe whether data is anonymous, pseudonymous, or account-linked. Public legal text should name actual providers and data flows in the deployed architecture; current source and provider usage have evolved faster than the policy prose.

34.11 Audit Logging Target

A durable audit event should record security-relevant mutations without copying confidential content unnecessarily. Recommended fields include event ID, timestamp, actor UID, actor email or stable actor key, authorization source, capability, action, domain, target record ID, result, request correlation ID, and non-sensitive change metadata. Audit data should be append-only to ordinary application actors and covered by retention policy.

34.12 Observability and Incident Response

The Worker should emit structured request/error metrics and domain/provider failure counts. Monitoring should cover authentication failures, authorization denials, public endpoint rate-limit events, booking conflicts, Brevo failures, B2 completion failures, Gemini failures/429s, and smoke-test health. The runbook should define incident severity, containment, credential rotation, privacy assessment, recovery, and post-incident review.

34.13 Backup and Recovery

Realtime Database backups/exports and B2 object retention/versioning must be coordinated so metadata and objects can be restored consistently. Recovery documentation should specify responsible operator, backup schedule, retention, restore test cadence, and recovery-time/recovery-point objectives appropriate to ministry operations. Backups containing People Notes or pastoral data must be treated as highly confidential copies, not as ordinary developer artifacts.

34.14 Frontend Modernization Direction

The frontend should continue moving from page-owned persistence toward typed domain clients and hooks. Large components should compose smaller feature modules rather than owning authentication, database paths, business logic, provider calls, and presentation together. Direct Firebase imports should be treated as an explicit migration list; this release provides that list in Appendix F.

Localization should preserve English/Arabic directionality and keep feature-specific assessment text authoritative in versioned form definitions. Accessibility work should cover keyboard interaction, labels, focus state, reduced motion, semantic headings, readable contrast, and screen-reader behavior for dialogs/interactive maps.

34.15 Backend Module Design

The current Worker already demonstrates useful separation: routes, schemas, security/auth helpers, provider services, data access, AI prompt/service code, and domain-specific modules. Future work should preserve that separation and avoid turning `backend/src/index.ts` into a monolithic application service. Domain modules should own domain rules; infrastructure modules should own Firebase/B2/Brevo/Gemini mechanics.

34.16 Testing Strategy

The 167 cataloged named tests provide substantial regression coverage, especially for backend contracts. The target test strategy remains layered:

- pure unit tests for calculations, normalizers, date/slot logic, prompt schema handling, and authorization helpers;
- route tests for authentication, authorization, validation, success, conflict, not-found, provider failure, and configuration failure;
- integration tests around Firebase REST transactions and object-storage lifecycle where deterministic fixtures are practical;
- frontend behavior tests for complex administrator/Pastor workflows;
- production smoke tests limited to safe read/contract checks;
- migration reconciliation tests before deleting legacy/direct-client paths.

A feature should not be considered complete merely because the happy-path UI works.

34.17 CI/CD and Environment Model

The repository now builds the frontend in GitHub Actions, tests and type-checks the backend, deploys the Worker from the main branch after successful backend validation, runs a post-deploy smoke test, and runs a scheduled daily production NextGen smoke. This supersedes the v1.2 statement that Worker deployment was manual only.

A mature environment model should clearly separate local, test/preview, staging, and production credentials/data. Production service-account/Brevo/B2/Gemini secrets should never be available to frontend build contexts. Netlify secret-scan exclusions should be kept as narrow as practical; broad source exclusions reduce the value of automated secret detection.

34.18 Documentation and Governance Set

The repository should ultimately carry this reference plus operational documents: developer setup, environment-variable catalog, API/OpenAPI specification, data dictionary/ERDs, authorization matrix, deployment/runbook, backup/restore procedure, incident response, migration logs, architecture decision records, provider inventory, privacy data-flow register, and release notes. Each release should update the source counters, endpoints, residual Firebase list, risk register, and implementation-status matrix from the source tree rather than copying old assumptions.

Current and Target Permission Catalog

35.1 Current Implemented Authorities

Authority	Principal	Observed purpose
Pastor allowlist	Exact authorized Pastor Firebase account	Calendar, meeting decisions, People Development, People Notes, Bezalel Pastor functions, and other Pastor-only routes.
Chief	Administrator hierarchy Chief UID	Full current administrator authority and authority allocation/suspension.
<code>manageAssessmentForms</code>	Active delegated administrator	Assessment form state/control and related admin operations.
<code>manageCarousel</code>	Active delegated administrator	Administrator carousel UI authority; persistence still direct Firebase.
<code>manageAttendance</code>	Active delegated administrator	Administrator attendance UI authority; persistence still direct Firebase.
<code>manageArchives</code>	Active delegated administrator	LInC Archives folder/file operations behind Worker authorization.
<code>manageNextGenQa</code>	Active delegated administrator	NextGen QA management alongside Pastor authority.
Authenticated NextGen account	Any valid Firebase account meeting route requirements	Member QA/session access and, currently, broad NextGen file access.
Public identifier holder	Person who supplies matching People Development identifier	Group assignment/schedule projection through public portal.
Anonymous public user	No Firebase identity	Landing, assessment public state/-submission, public booking, selected health/test/public endpoints.

35.2 Recommended Capability Expansion

Capabilities should map to server actions, not page visibility. The capability namespace proposed earlier in this chapter can be extended with read/write distinctions when ministry governance needs them, for example `nextgen.files.read` versus `nextgen.files.manage`, or `attendance.read` versus `attendance.correct`. A technical administrator should not automatically gain confidential pastoral data merely because that administrator can configure public content.

35.3 Non-Delegable or Highly Restricted Actions

Unless ministry leadership explicitly decides otherwise, Pastor meeting-request decisions, highly confidential People Notes, pastoral/member care notes, and other ministry judgments should not be automatically inherited by Chief or general technical administration. Database/production operators may possess emergency infrastructure access, but that privilege should be documented, exceptional, and audited rather than represented as ordinary application authorization.

Modernization Roadmap - Rebaselined to Release 1.4

The earlier roadmap assumed only one backend slice existed. The supplied release has advanced several phases significantly. This rebaseline records what is demonstrably implemented and what remains.

Phase	Current status	Release 1.4 interpretation
0 - Immediate risk reduction	Partially complete	Major provider secrets moved server-side for migrated domains, but public notification/test endpoints and broad NextGen file/map authorization require immediate correction.
1 - Backend foundation	Substantially implemented	Worker/Hono, Firebase token verification, Pastor authorization, Firebase server REST access, Zod, provider adapters, tests, CI, deployment and smoke now exist. Audit persistence and mature rate limiting remain.
2 - Booking/no-tifications	Substantially implemented	Public schedule/request, conflict checks, transactional reservation, Pastor decisions, invitations and Brevo are backend-owned. Residual EmailJS/Gmail paths and public notification route remain cleanup work.
3 - Identity consolidation	Open/partial	Linkage exists, but canonical identity across assessment, People Development, attendance, NextGen and Firebase accounts is not complete.
4 - Admin/public content	Partially implemented	Chief/delegated capability model and archives/assessment backend exist. Carousel and attendance writes remain direct Firebase; tutorial administration also needs a formal capability boundary.
5 - NextGen/attendance/-groups	Mixed	NextGen QA/files/map and People Development have substantial backend coverage; attendance remains direct-client; file/map authorization needs tightening; legacy NextGen still remains in source.
6 - Pastoral high-sensitivity controls	Substantially implemented but incomplete	People Notes has a strict Pastor backend boundary. Durable audit logging, retention governance, access review, and formal backup handling remain.
7 - Database migration/de-commissioning	Not required yet / optional	Realtime Database remains active. Priority should be finishing backend boundaries and identity/data governance before a database technology migration unless relational needs justify it.

8 - Enterprise In progress
hardening

CI/deploy/smoke are materially stronger. Observability, audit, runbooks, backup/restore evidence, rules review, legal-provider alignment, and security review remain.

36.1 Immediate Remediation Queue

1. Require appropriate authentication/authorization for `/api/v1/people-development/notifications`.
2. Replace broad NextGen file access with an explicit audience/capability/ownership rule and tests.
3. Make mission-map code enforce the intended account/audience instead of relying on README prose.
4. Remove or lock the temporary public Brevo test endpoint in production.
5. Obtain and review production Firebase Security Rules because direct-client domains depend on them.
6. Move attendance and carousel writes behind backend capability checks or prove equivalent Rules enforcement.
7. Move People Development binary attachments from Base64 RTDB storage to private object storage when attachment volume warrants it.
8. Remove or isolate unused EmailJS/Gmail/legacy NextGen paths after confirming no production dependency.
9. Add durable audit events and structured observability for sensitive mutations/provider actions.
10. Bring privacy/terms/provider documentation into alignment with current Firebase, Cloudflare, Brevo, B2, Gemini, and Google Maps data flows.

36.2 Medium-Term Engineering Queue

Complete canonical identity design; formalize OpenAPI and generated/validated frontend contracts; implement robust per-principal/public rate limiting; establish environment/staging strategy; define storage/file scanning policy; automate database backup/restore verification; improve map/file authorization granularity; and continue reducing direct Firebase imports until every intentionally client-owned path has reviewed Rules and a documented reason to remain client-owned.

Chapter 37

Enterprise Readiness and Acceptance Criteria

37.1 Architecture Acceptance

The application is architecturally ready for controlled transfer when each sensitive domain has an explicit owner, a documented trust boundary, a tested persistence path, and no accidental provider secret in browser code. Direct Firebase access that remains must be intentional, listed, and backed by reviewed Rules.

37.2 Security Acceptance

Production security acceptance requires reviewed Firebase Rules; elimination/locking of unintended public mutation/provider endpoints; least-privilege NextGen file/map access; secret inventory/rotation procedure; audited high-sensitivity changes; rate/abuse controls for public AI and identifier portals; tested authorization denials; and a process for dependency/security updates.

37.3 Data Acceptance

Every active root/object namespace must have an owner, schema/normalization description, retention classification, backup plan, and migration policy. Identity links must be reconcilable. B2 metadata and object records must be restorable together. Pastoral/highly confidential records require explicit deletion/correction and access-governance procedures.

37.4 Product Acceptance

Public, participant, Pastor, and administrator workflows should have documented acceptance tests in both desktop and mobile layouts where applicable, including English/Arabic presentation, error states, empty states, loading states, and denied-access behavior. AI-assisted flows must degrade safely when Gemini is unavailable or rejects a request.

37.5 Operational Acceptance

A competent maintainer should be able to build frontend/backend, configure environments, run tests, deploy both hosting surfaces, verify smoke health, rotate provider credentials, restore backups, diagnose common provider/database failures, and identify the owner of each domain without relying on undocumented knowledge from the original developer.

37.6 Commercial/Transfer Acceptance

For external handoff or licensing, the repository should include clear IP ownership, dependency/license inventory, deployment costs/providers, data-processing/provider list, environment setup, architecture decisions, known-risk register, current roadmap, and support expectations. Estimates of monetary value should be maintained separately from engineering truth so commercial assumptions do not become architecture requirements.

Open Verification Items and Documentation Boundaries

The source audit is intentionally explicit about artifacts that were not available or facts that cannot be proven from static code alone.

1. Production Firebase Realtime Database Security Rules were not present in the supplied snapshot; therefore actual direct-client read/write enforcement for attendance, carousel, tutorials, and residual NextGen code cannot be certified from this repository alone.
2. Production Cloudflare secret values, Netlify environment values, Firebase console settings, Google Maps restrictions, Brevo account configuration, B2 bucket policy, and Gemini account/quota policy are external deployment state and were not inferred from checked-in names.
3. Static source proves that CI workflows are defined, not that every historical workflow run succeeded. The scheduled smoke workflow proves intended automation, not continuous production health.
4. The source tree contains legacy/unused-looking modules and dependencies. A module is classified as residual when active route/import tracing does not establish it as the current primary path; removal still requires production regression verification.
5. Database ERDs are repository documentation and may contain conceptual entities/fields broader than current raw Firebase records. The persistence chapter therefore distinguishes observed roots from ERD-derived conceptual structure.
6. The named-function catalog is static-source inventory. Anonymous inline closures and framework-generated functions are not represented as separate named entries unless they matched the catalog extraction rules.
7. Provider and legal/privacy obligations require organizational/legal review beyond source inspection. This document identifies technical data flows but is not legal advice.

Note – Documentation Maintenance Rule

For the next release, re-run the source inventory and compare route tables, endpoint mounts, environment names, Firebase imports, dependencies, test names, ERDs, and security middleware against this release. Do not carry a feature status forward merely because it was documented here; current source and deployment evidence must remain authoritative.

Release Delta: Why the Old Documentation Was Insufficient

39.1 Major Changes Not Represented by the July Baseline

The original v1.2 documentation correctly recorded only the first meeting-invitation backend slice. The supplied August repository now contains backend-owned assessment submission/administration, booking, Pastor calendar mutation, People Notes, People Development, administrator hierarchy and archives, NextGen QA/files/mission map, service-account database access, Backblaze B2, Gemini/Bezalel, automated backend deployment, and scheduled production contract verification.

39.2 Documentation Model Change

The earlier document mixed current state and future modernization heavily and summarized many features in one or two paragraphs. Release 1.4 changes the documentation model: current product behavior is documented first; architecture/security analysis is layered on top; and exhaustive source/test/function catalogs are included as maintenance references. Future work is intentionally shorter than implemented-current-state coverage.

39.3 Current Modernization Backlog

1. Close unauthenticated/broadly authenticated backend routes.
2. Move attendance, carousel, tutorials, and remaining legacy NextGen writes behind dedicated APIs.
3. Move People Development binary attachments from Realtime Database to object storage.
4. Remove or quarantine Gmail/EmailJS/Google OAuth legacy integration if no longer operationally required.
5. Add durable global rate limiting for public AI/identifier endpoints and provider-send controls.
6. Add audit-event records for sensitive administrative/Pastor mutations.
7. Document and test Firebase Security Rules as part of the repository.
8. Define a canonical member identity/linking model across assessment, attendance, People Development, People Notes, booking, and NextGen.
9. Synchronize privacy/terms/README documentation with the live provider and authorization model.

Appendix A

ERD-Derived Entity Catalog

The repository contains eight Mermaid ERD files. This appendix consolidates their entity declarations. ERD entities are documentation models: some correspond directly to Firebase records while others (for example conceptual email-event/linkage entities) may represent relationships or desired structure rather than a single raw database node.

A.1 ADMIN_HIERARCHY

- string hierarchy_id PK
- string chief_uid FK
- string hierarchy_id PK
- string chief_uid FK

A.2 ADMIN_USER

- string uid PK
- string role
- string status
- string uid PK
- string email
- string role
- string status
- string approved_by_uid FK
- string uid PK
- string email
- string role
- string status
- json authority
- string approved_by_uid FK

A.3 ASSESSMENT_FORM

- string form_id PK
- string state

A.4 ASSESSMENT_FORM_CONTROL

- string form_id PK
- string state
- timestamp updated_at
- string updated_by
- string form_id PK
- string state
- timestamp updated_at

A.5 ASSESSMENT_RESPONSE

- string response_id PK
- string form_id FK
- string user_identifier
- string full_name
- string email
- string response_id PK
- string form_id FK
- string response_kind
- string user_identifier
- string full_name
- string email
- string language
- json fields
- json scores
- json results
- timestamp created_at
- string response_id PK
- string user_identifier
- string full_name
- string email
- string response_id PK
- string user_identifier
- string people_development_group
- string response_id PK

- string form_id FK
- string response_kind
- string user_identifier
- string full_name
- string email
- string language
- json fields
- json scores
- json results
- string people_development_group
- timestamp created_at

A.6 ASSESSMENT_RESPONSE_IDENTITY

- string response_id PK
- string user_identifier
- string full_name
- string email

A.7 ATTENDANCE_PERSON

- string attendance_person_id PK
- string full_name
- string email
- string attendance_person_id PK
- string first_name
- string last_name
- string email
- string phone_number
- string attendance_dates_csv
- string attendance_person_id PK
- string first_name
- string last_name
- string email
- string phone_number
- string attendance_dates_csv

A.8 ATTENDANCE_PERSON_IDENTITY

- string attendance_person_id PK
- string full_name
- string email
- string phone_number

A.9 AVAILABILITY

- string availability_id PK
- date available_date
- string start_time
- string end_time
- boolean all_day
- string availability_id PK
- date available_date
- string start_time
- string end_time
- boolean all_day

A.10 CAROUSEL_CONFIG

- string carousel_id PK
- boolean enabled
- timestamp updated_at
- string carousel_id PK
- boolean enabled

A.11 CAROUSEL_PHOTO

- string photo_id PK
- string carousel_id FK
- string photo_url
- string alt_en
- string alt_ar
- int display_order
- string photo_id PK

- string carousel_id FK
- string photo_url
- int display_order

A.12 FIREBASE_AUTH_USER

- string uid PK
- string email
- string uid PK
- string email
- string uid PK
- string email
- string uid PK
- string email

A.13 MEETING

- string meeting_id PK
- string source_request_id FK
- date meeting_date
- string meeting_id PK
- date meeting_date
- string meeting_id PK
- string source_request_id FK
- string title
- date meeting_date
- string start_time
- string end_time
- string meeting_type
- boolean acknowledged
- string meeting_id PK
- string source_request_id FK
- string title
- date meeting_date
- string meeting_type
- boolean acknowledged

A.14 MEETING_PARTICIPANT

- string participant_link_id PK
- string meeting_id FK
- string response_id FK
- string participant_link_id PK
- string meeting_id FK
- string response_id FK
- string participant_link_id PK
- string meeting_id FK
- string response_id FK
- string participant_link_id PK
- string meeting_id FK
- string response_id FK

A.15 MEETING_REQUEST

- string request_id PK
- string status
- string created_meeting_id FK
- string request_id PK
- string name
- string email
- date requested_date
- string start_time
- string end_time
- string status
- string created_meeting_id FK
- timestamp created_at
- string request_id PK
- string name
- string email
- date requested_date
- string status
- string created_meeting_id FK

A.16 MEETING_REQUEST_IDENTITY

- string request_id PK
- string name
- string email

A.17 NEXTGEN_PARTICIPATION_EVENT

- string participation_id PK
- string user_id FK
- string activity_type
- string activity_id
- string filling_status
- timestamp completed_at
- string participation_id PK
- string user_id FK
- string activity_type
- string activity_id

A.18 NEXTGEN_QUESTION

- string question_id PK
- string submitted_by_user_id FK
- string question_text
- string question_id PK
- string submitted_by_user_id FK
- string question_text
- string category
- string status
- int total_upvotes
- int total_downvotes
- int net_votes
- string question_id PK
- string submitted_by_user_id FK
- string question_text
- string status

A.19 NEXTGEN_REVIEW_LOG

- string log_id PK
- string user_id FK
- string previous_status
- string status
- string reviewed_by
- timestamp reviewed_at
- string log_id PK
- string user_id FK
- string previous_status
- string status

A.20 NEXTGEN_SURVEY

- string survey_id PK
- int survey_version
- string survey_title
- string survey_id PK
- int survey_version
- string survey_title

A.21 NEXTGEN_SURVEY_RESPONSE

- string response_id PK
- string survey_id FK
- string user_id FK
- string response_id PK
- string survey_id FK
- string user_id FK
- string language
- string completion_status
- json answers
- timestamp completed_at
- string response_id PK
- string survey_id FK
- string user_id FK
- json answers

A.22 NEXTGEN_USER

- string user_id PK
- string full_name
- string status
- string user_id PK
- string normalized_user_id
- string full_name
- string email
- string status
- timestamp reviewed_at
- string user_id PK
- string full_name
- string email
- string status

A.23 NEXTGEN_USER_IDENTITY

- string user_id PK
- string normalized_user_id
- string full_name
- string email

A.24 NEXTGEN_VOTE

- string vote_id PK
- string question_id FK
- string user_id FK
- string vote_type
- string vote_id PK
- string question_id FK
- string user_id FK
- string vote_type
- timestamp completed_at
- string vote_id PK
- string question_id FK
- string user_id FK
- string vote_type

A.25 PD_ASSIGNMENT

- string assignment_id PK
- string assignment_text
- date assignment_date
- string assignment_id PK
- string assignment_text
- date assignment_date
- boolean has_attachments
- string assignment_id PK
- string assignment_text
- date assignment_date
- boolean has_attachments

A.26 PD_ASSIGNMENT_GROUP

- string assignment_group_id PK
- string assignment_id FK
- string group_code FK
- string assignment_group_id PK
- string assignment_id FK
- string group_code FK
- string assignment_group_id PK
- string assignment_id FK
- string group_code FK

A.27 PD_ATTACHMENT

- string attachment_id PK
- string assignment_id FK
- string file_name
- string mime_type
- int size_bytes
- string encoding
- string attachment_id PK
- string assignment_id FK
- string file_name

- string mime_type
- int size_bytes

A.28 PD_GROUP

- string group_code PK
- string group_label
- string group_code PK
- string group_label
- string group_code PK
- string group_label

A.29 PD_MEETING_SCHEDULE

- string schedule_id PK
- string group_code FK
- string audience
- string ordinal
- int weekday
- string start_time
- boolean active
- string schedule_id PK
- string group_code FK
- string audience
- string ordinal
- int weekday
- boolean active

A.30 PD_MEMBER

- string member_key PK
- string identifier
- string group_code FK
- string member_key PK
- string identifier
- string full_name
- string email

- string group__code FK
- string member__key PK
- string identifier
- string full__name
- string email
- string primary__gift
- string group__code FK
- string member__key PK
- string identifier
- string full__name
- string email
- string primary__gift
- string group__code FK

A.31 PD_MEMBER_IDENTITY

- string member__key PK
- string identifier
- string full__name
- string email

A.32 PD_MEMBER_RESPONSE

- string member__response_id PK
- string member__key FK
- string response_id FK
- string member__response_id PK
- string member__key FK
- string response_id FK
- string member__response_id PK
- string member__key FK
- string response_id FK

A.33 PD_NOTE

- string note_id PK
- string member__key FK
- string note_type

A.34 PD_PERSONAL_NOTE

- string note_id PK
- string member_key FK
- string group_code FK
- string note_type
- string note_text
- date note_date
- string note_id PK
- string member_key FK
- string group_code FK
- string note_type
- string note_text
- date note_date

A.35 PEOPLE_NOTES_PERSON

- string person_id PK
- string full_name
- string contact
- string person_id PK
- string full_name
- string contact

A.36 PEOPLE_NOTES_PERSON_IDENTITY

- string person_id PK
- string full_name
- string contact

A.37 PEOPLE_NOTE_COMMENT

- string comment_id PK
- string item_id FK
- string comment_text
- timestamp created_at
- string comment_id PK

- string item__id FK
- string comment__text

A.38 PEOPLE_NOTE_ITEM

- string item__id PK
- string person__id FK
- string item__type
- string title
- date date__added
- string item__id PK
- string person__id FK
- string item__type
- string title

A.39 RESPONSE_EMAIL_EVENT

- string email_event__id PK
- string response__id FK
- string language
- string provider
- string message__id
- timestamp sent__at
- string email_event__id PK
- string response__id FK
- string provider
- string language
- string message__id
- timestamp sent__at

A.40 RESPONSE_LINKAGE

- string linkage__id PK
- string response__id FK
- string user__identifier
- string database__form__id
- string linked__form__id

A.41 TUTORIAL

- string tutorial_id PK
- string title
- boolean published
- string tutorial_id PK
- string title
- string description
- string category
- string audience
- boolean published
- string created_by
- string tutorial_id PK
- string title
- string audience
- boolean published

A.42 TUTORIAL_PROGRESS

- string progress_id PK
- string uid FK
- string tutorial_id FK
- boolean completed
- string progress_id PK
- string uid FK
- string tutorial_id FK
- string progress_id PK
- string uid FK
- string tutorial_id FK
- boolean completed
- int last_step_index
- timestamp completed_at
- timestamp last_started_at
- string progress_id PK
- string uid FK
- string tutorial_id FK
- boolean completed

A.43 TUTORIAL_STEP

- string step_id PK
- string tutorial_id FK
- int step_order
- string title
- string route
- string target
- string action
- string step_id PK
- string tutorial_id FK
- int step_order
- string action

A.44 UNAVAILABILITY

- string unavailability_id PK
- date unavailable_date
- string start_time
- string end_time
- boolean all_day
- string unavailability_id PK
- date unavailable_date
- string start_time
- string end_time
- boolean all_day

Appendix B

ERD Relationship Catalog

- FIREBASE_AUTH_USER → ADMIN_USER: has. Mermaid cardinality `||-o|`.
- ADMIN_HIERARCHY → ADMIN_USER: contains. Mermaid cardinality `||-o{`.
- FIREBASE_AUTH_USER → ADMIN_HIERARCHY: may be chief. Mermaid cardinality `||-o|`.
- ADMIN_USER → ADMIN_USER: approves. Mermaid cardinality `||-o{`.
- ASSESSMENT_FORM_CONTROL → ASSESSMENT_RESPONSE: controls. Mermaid cardinality `||-o{`.
- ASSESSMENT_RESPONSE → RESPONSE_EMAIL_EVENT: records. Mermaid cardinality `||-o{`.
- MEETING_REQUEST → MEETING: creates. Mermaid cardinality `|o-o|`.
- MEETING → MEETING_PARTICIPANT: includes. Mermaid cardinality `||-o{`.
- ASSESSMENT_RESPONSE → MEETING_PARTICIPANT: identifies. Mermaid cardinality `||-o{`.
- PD_GROUP → PD_MEMBER: contains. Mermaid cardinality `||-o{`.
- PD_MEMBER → PD_MEMBER_RESPONSE: links. Mermaid cardinality `||-o{`.
- ASSESSMENT_RESPONSE → PD_MEMBER_RESPONSE: belongs to. Mermaid cardinality `||-o|`.
- PD_ASSIGNMENT → PD_ASSIGNMENT_GROUP: targets. Mermaid cardinality `||-o{`.
- PD_GROUP → PD_ASSIGNMENT_GROUP: receives. Mermaid cardinality `||-o{`.
- PD_ASSIGNMENT → PD_ATTACHMENT: contains. Mermaid cardinality `||-o{`.
- PD_MEMBER → PD_PERSONAL_NOTE: has. Mermaid cardinality `||-o{`.
- PD_GROUP → PD_PERSONAL_NOTE: classifies. Mermaid cardinality `||-o{`.
- PD_GROUP → PD_MEETING_SCHEDULE: schedules. Mermaid cardinality `||-o{`.
- PEOPLE_NOTES_PERSON → PEOPLE_NOTE_ITEM: contains. Mermaid cardinality `||-o{`.
- PEOPLE_NOTE_ITEM → PEOPLE_NOTE_COMMENT: contains. Mermaid cardinality `||-o{`.
- NEXTGEN_USER → NEXTGEN_QUESTION: submits. Mermaid cardinality `||-o{`.
- NEXTGEN_USER → NEXTGEN_VOTE: casts. Mermaid cardinality `||-o{`.
- NEXTGEN_QUESTION → NEXTGEN_VOTE: receives. Mermaid cardinality `||-o{`.
- NEXTGEN_USER → NEXTGEN_PARTICIPATION_EVENT: records. Mermaid cardinality `||-o{`.
- NEXTGEN_SURVEY → NEXTGEN_SURVEY_RESPONSE: receives. Mermaid cardinality `||-o{`.
- NEXTGEN_USER → NEXTGEN_SURVEY_RESPONSE: submits. Mermaid cardinality `||-o{`.
- NEXTGEN_USER → NEXTGEN_REVIEW_LOG: has. Mermaid cardinality `||-o{`.
- FIREBASE_AUTH_USER → TUTORIAL_PROGRESS: owns. Mermaid cardinality `||-o{`.
- TUTORIAL → TUTORIAL_STEP: contains. Mermaid cardinality `||-|{`.
- TUTORIAL → TUTORIAL_PROGRESS: tracks. Mermaid cardinality `||-o{`.
- CAROUSEL_CONFIG → CAROUSEL_PHOTO: contains. Mermaid cardinality `||-o{`.

Appendix C

Complete Source File Catalog

This appendix lists every reviewed TypeScript/TSX module in frontend and backend source directories. Total: 257 modules. Function lists are intentionally moved to the next appendix to keep this catalog readable.

Source module	Lines	Responsibility
src/App.tsx	394	Application router, protected-route authentication gateway, and top-level providers.
src/components/AssessmentForm.tsx	63	Source module for Assessment Form.
src/components/BookMeeting.tsx	465	Source module for Book Meeting.
src/components/Layout.tsx	160	Source module for Layout.
src/components/PageTitle.tsx	20	Source module for Page Title.
src/components/admin/AdminAccessScreens.tsx	192	Source module for Admin Access Screens.
src/components/admin/AdministratorPanel.tsx	720	Source module for Administrator Panel.
src/components/admin/admin.constants.ts	35	Source module for admin.constants.
src/components/admin/admin.types.ts	71	Source module for admin.types.
src/components/admin/admin.utils.test.ts	23	Automated test coverage for admin.utils.
src/components/admin/admin.utils.ts	247	Source module for admin.utils.
src/components/admin/archives/ArchiveFolderTree.tsx	90	LInC Archives feature module for Archive Folder Tree.
src/components/admin/archives/LincArchivesSection.tsx	341	LInC Archives feature module for Linc Archives Section.
src/components/admin/archives/archives.types.ts	24	LInC Archives feature module for archives.types.
src/components/admin/archives/archives.utils.test.ts	51	Automated test coverage for archives.utils.
src/components/admin/archives/archives.utils.ts	67	LInC Archives feature module for archives.utils.
src/components/admin/archives/index.ts	2	LInC Archives feature module for index.
src/components/admin/archives/useLincArchives.ts	232	React state/data hook for use Linc Archives.
src/components/admin/attendance/AttendanceAnalysisCharts.tsx	370	Administrator attendance feature module for Attendance Analysis Charts.
src/components/admin/attendance/AttendanceAnalysisPanel.tsx	383	Administrator attendance feature module for Attendance Analysis Panel.
src/components/admin/attendance/AttendanceManagement.tsx	158	Administrator attendance feature module for Attendance Management.
src/components/admin/attendance/AttendancePeoplePanel.tsx	314	Administrator attendance feature module for Attendance People Panel.
src/components/admin/attendance/AttendancePersonEditor.tsx	633	Administrator attendance feature module for Attendance Person Editor.
src/components/admin/attendance/AttendancePersonModal.tsx	138	Administrator attendance feature module for Attendance Person Modal.
src/components/admin/attendance/AttendancePersonalAnalysisModal.tsx	378	Administrator attendance feature module for Attendance Personal Analysis Modal.
src/components/admin/attendance/AttendanceRecordingPanel.tsx	504	Administrator attendance feature module for Attendance Recording Panel.
src/components/admin/attendance/attendance.copy.ts	175	Administrator attendance feature module for attendance.copy.
src/components/admin/attendance/attendance.styles.ts	214	Administrator attendance feature module for attendance.styles.
src/components/admin/attendance/attendance.types.ts	43	Administrator attendance feature module for attendance.types.
src/components/admin/attendance/attendance.utils.test.ts	51	Automated test coverage for attendance.utils.
src/components/admin/attendance/attendance.utils.ts	154	Administrator attendance feature module for attendance.utils.
src/components/admin/attendance/index.ts	1	Administrator attendance feature module for index.

Source module	Lines	Responsibility
src/components/admin/attendance/useAttendanceAnalytics.ts	200	React state/data hook for use Attendance Analytics.
src/components/admin/attendance/useAttendanceManagement.ts	636	React state/data hook for use Attendance Management.
src/components/admin/components/AdminCommandCenter.tsx	216	Source module for Admin Command Center.
src/components/admin/components/AdminHierarchySection.tsx	206	Source module for Admin Hierarchy Section.
src/components/admin/components/AdminUtilitySections.tsx	42	Source module for Admin Utility Sections.
src/components/admin/components/AssessmentFormsSection.tsx	201	Source module for Assessment Forms Section.
src/components/admin/components/AssessmentUserLinkageSection.tsx	193	Source module for Assessment User Linkage Section.
src/components/admin/components/CarouselManagementSection.tsx	413	Source module for Carousel Management Section.
src/components/admin/components/index.ts	12	Source module for index.
src/components/admin/hooks/index.ts	1	React state/data hook for index.
src/components/admin/hooks/useAdministratorAccess.ts	193	React state/data hook for use Administrator Access.
src/components/admin/index.ts	1	Source module for index.
src/components/assessment/AssessmentQuestionnaire.tsx	130	Assessment engine module for Assessment Questionnaire.
src/components/assessment/AssessmentResults.tsx	97	Assessment engine module for Assessment Results.
src/components/assessment/AssessmentSelector.tsx	164	Assessment engine module for Assessment Selector.
src/components/assessment/assessment.calculations.test.ts	101	Automated test coverage for assessment.calculations.
src/components/assessment/assessment.calculations.ts	130	Assessment engine module for assessment.calculations.
src/components/assessment/assessment.forms.ts	119	Assessment engine module for assessment.forms.
src/components/assessment/assessment.types.ts	109	Assessment engine module for assessment.types.
src/components/assessment/useAssessmentForm.ts	204	React state/data hook for use Assessment Form.
src/components/bezalel/BezalelChat.tsx	85	Bezalel AI feature module for Bezalel Chat.
src/components/bezalel/BookingBezalelAssistant.tsx	60	Bezalel AI feature module for Booking Bezalel Assistant.
src/components/bezalel/PastorBezalelAssistant.tsx	85	Bezalel AI feature module for Pastor Bezalel Assistant.
src/components/bezalel/index.ts	2	Bezalel AI feature module for index.
src/components/brand/LincLogo.test.tsx	21	Automated test coverage for linc logo.
src/components/brand/LincLogo.tsx	38	Source module for Linc Logo.
src/components/congregation-group-notes/AssignmentDetailsDialog.tsx	155	People Development congregation portal module for Assignment Details Dialog.
src/components/congregation-group-notes/AssignmentList.tsx	103	People Development congregation portal module for Assignment List.
src/components/congregation-group-notes/CongregationGroupDashboard.tsx	31	People Development congregation portal module for Congregation Group Dashboard.
src/components/congregation-group-notes/CongregationGroupHeader.tsx	23	People Development congregation portal module for Congregation Group Header.
src/components/congregation-group-notes/GroupSummary.tsx	35	People Development congregation portal module for Group Summary.
src/components/congregation-group-notes/IdentifierLogin.tsx	73	People Development congregation portal module for Identifier Login.
src/components/congregation-group-notes/LatestAssignment.tsx	45	People Development congregation portal module for Latest Assignment.
src/components/congregation-group-notes/MeetingCalendarPanel.tsx	77	People Development congregation portal module for Meeting Calendar Panel.
src/components/congregation-group-notes/MemberProfileHeader.tsx	49	People Development congregation portal module for Member Profile Header.
src/components/congregation-group-notes/UnassignedGroupNotice.tsx	24	People Development congregation portal module for Unassigned Group Notice.
src/components/congregation-group-notes/UpcomingMeetings.tsx	87	People Development congregation portal module for Upcoming Meetings.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	128	People Development congregation portal module for congregation Group Notes.config.

Source module	Lines	Responsibility
src/components/congregation-group-notes/congregationGroupNotes.types.ts	72	People Development congregation portal module for congregation Group Notes.types.
src/components/congregation-group-notes/congregationGroupNotes.utils.test.ts	74	Automated test coverage for congregation group notes.utils.
src/components/congregation-group-notes/congregationGroupNotes.utils.ts	335	People Development congregation portal module for congregation Group Notes.utils.
src/components/congregation-group-notes/useCongregationGroupNotes.ts	168	React state/data hook for use Congregation Group Notes.
src/components/landing/CommunityCarousel.tsx	127	Landing-page presentation/navigation module for Community Carousel.
src/components/landing/FloatingLandingNav.tsx	100	Landing-page presentation/navigation module for Floating Landing Nav.
src/components/landing/LincOneHero.tsx	162	Landing-page presentation/navigation module for Linc One Hero.
src/components/landing/SpiritualProgramFeature.tsx	118	Landing-page presentation/navigation module for Spiritual Program Feature.
src/components/landing/landingNavigation.test.tsx	63	Automated test coverage for landing navigation.
src/components/linc/LincPageHero.tsx	54	Source module for Linc Page Hero.
src/components/nextgen/GoogleMissionRouteMap.tsx	192	NextGen feature module for Google Mission Route Map.
src/components/nextgen/MontrealMissionTripMap.tsx	798	NextGen feature module for Montreal Mission Trip Map.
src/components/nextgen/activities/NextGenAccessSection.tsx	131	NextGen feature module for Next Gen Access Section.
src/components/nextgen/activities/NextGenActivityMenu.test.tsx	21	Automated test coverage for next gen activity menu.
src/components/nextgen/activities/NextGenActivityMenu.tsx	32	NextGen feature module for Next Gen Activity Menu.
src/components/nextgen/activities/NextGenParticipantHeader.tsx	11	NextGen feature module for Next Gen Participant Header.
src/components/nextgen/activities/NextGenPeerReviewPanel.tsx	21	NextGen feature module for Next Gen Peer Review Panel.
src/components/nextgen/activities/NextGenQuestionPanel.tsx	19	NextGen feature module for Next Gen Question Panel.
src/components/nextgen/activities/NextGenSurveyPanel.tsx	65	NextGen feature module for Next Gen Survey Panel.
src/components/nextgen/activities/index.ts	10	NextGen feature module for index.
src/components/nextgen/activities/nextGenActivities.constants.ts	115	NextGen feature module for next Gen Activities.constants.
src/components/nextgen/activities/nextGenActivities.firebase.ts	297	NextGen feature module for next Gen Activities.firebase.
src/components/nextgen/activities/nextGenActivities.types.ts	96	NextGen feature module for next Gen Activities.types.
src/components/nextgen/activities/nextGenActivities.utils.test.ts	68	Automated test coverage for next gen activities.utils.
src/components/nextgen/activities/nextGenActivities.utils.ts	155	NextGen feature module for next Gen Activities.utils.
src/components/nextgen/activities/nextGenCertificate.ts	147	NextGen feature module for next Gen Certificate.
src/components/nextgen/activities/useNextGenIdentity.ts	168	React state/data hook for use Next Gen Identity.
src/components/nextgen/activities/useNextGenQuestions.ts	154	React state/data hook for use Next Gen Questions.
src/components/nextgen/activities/useNextGenSurvey.ts	168	React state/data hook for use Next Gen Survey.
src/components/nextgen/portal/NextGenAuthGateway.tsx	81	NextGen feature module for Next Gen Auth Gateway.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	125	NextGen feature module for Next Gen Files Workspace.
src/components/nextgen/portal/NextGenQaSessions.test.tsx	118	Automated test coverage for next gen qa sessions.
src/components/nextgen/portal/NextGenQaSessions.tsx	204	NextGen feature module for Next Gen Qa Sessions.
src/components/nextgen/portal/index.ts	3	NextGen feature module for index.
src/components/pastor/PastorDashboard.tsx	1802	Pastor workspace module for Pastor Dashboard.
src/components/pastor/calendar/calendar.constants.ts	14	Pastor workspace module for calendar.constants.

Source module	Lines	Responsibility
src/components/pastor/calendar/calendar.email.ts	23	Pastor workspace module for calendar.email.
src/components/pastor/calendar/calendar.forms.ts	173	Pastor workspace module for calendar.forms.
src/components/pastor/calendar/calendar.slots.ts	279	Pastor workspace module for calendar.slots.
src/components/pastor/calendar/calendar.types.ts	38	Pastor workspace module for calendar.types.
src/components/pastor/calendar/calendar.utils.ts	92	Pastor workspace module for calendar.utils.
src/components/pastor/calendar/index.ts	6	Pastor workspace module for index.
src/components/pastor/hooks/index.ts	76	React state/data hook for index.
src/components/pastor/hooks/useAvailability.ts	688	React state/data hook for use Availability.
src/components/pastor/hooks/useCalendarMonth.ts	123	React state/data hook for use Calendar Month.
src/components/pastor/hooks/useMeetingRequests.ts	60	React state/data hook for use Meeting Requests.
src/components/pastor/hooks/useMeetings.ts	320	React state/data hook for use Meetings.
src/components/pastor/hooks/useNextGen.ts	240	React state/data hook for use Next Gen.
src/components/pastor/hooks/useParticipants.ts	265	React state/data hook for use Participants.
src/components/pastor/hooks/usePastorCalendarData.ts	76	React state/data hook for use Pastor Calendar Data.
src/components/pastor/hooks/usePeopleDevelopment.ts	1227	React state/data hook for use People Development.
src/components/pastor/hooks/usePeopleDevelopmentMeetingSchedules.ts	276	React state/data hook for use People Development Meeting Schedules.
src/components/pastor/meeting-requests/MeetingRequestsSection.tsx	194	Pastor workspace module for Meeting Requests Section.
src/components/pastor/meeting-requests/index.ts	4	Pastor workspace module for index.
src/components/pastor/meeting-requests/meetingRequests.types.ts	79	Pastor workspace module for meeting Requests.types.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	182	Pastor workspace module for meeting Requests.utils.
src/components/pastor/nextgen/NextGenQaSessionsAdmin.tsx	140	NextGen feature module for Next Gen Qa Sessions Admin.
src/components/pastor/nextgen/NextGenQuestionsSection.tsx	323	NextGen feature module for Next Gen Questions Section.
src/components/pastor/nextgen/NextGenRegistrationsSection.tsx	553	NextGen feature module for Next Gen Registrations Section.
src/components/pastor/nextgen/NextGenSurveyResultsSection.tsx	417	NextGen feature module for Next Gen Survey Results Section.
src/components/pastor/nextgen/index.ts	12	NextGen feature module for index.
src/components/pastor/nextgen/nextgen.actions.ts	147	NextGen feature module for nextgen.actions.
src/components/pastor/nextgen/nextgen.constants.ts	160	NextGen feature module for nextgen.constants.
src/components/pastor/nextgen/nextgen.firebase.ts	241	NextGen feature module for nextgen.firebase.
src/components/pastor/nextgen/nextgen.types.ts	187	NextGen feature module for nextgen.types.
src/components/pastor/nextgen/nextgen.utils.ts	884	NextGen feature module for nextgen.utils.
src/components/pastor/people-development/PeopleAssignmentsCalendarModal.tsx	586	Pastor workspace module for People Assignments Calendar Modal.
src/components/pastor/people-development/PeopleDevelopmentCombinedPostPanel.tsx	321	Pastor workspace module for People Development Combined Post Panel.
src/components/pastor/people-development/PeopleDevelopmentGroupPanel.tsx	592	Pastor workspace module for People Development Group Panel.
src/components/pastor/people-development/PeopleDevelopmentMeetingSchedulesSection.tsx	721	Pastor workspace module for People Development Meeting Schedules Section.
src/components/pastor/people-development/PeopleDevelopmentMeetingsCalendar.tsx	324	Pastor workspace module for People Development Meetings Calendar.
src/components/pastor/people-development/PeopleDevelopmentSection.tsx	1015	Pastor workspace module for People Development Section.

Source module	Lines	Responsibility
src/components/pastor/people-development/PeoplePersonalNoteModal.tsx	287	Pastor workspace module for People Personal Note Modal.
src/components/pastor/people-development/index.ts	39	Pastor workspace module for index.
src/components/pastor/people-development/peopleDevelopment.actions.ts	460	Pastor workspace module for people Development.actions.
src/components/pastor/people-development/peopleDevelopment.constants.ts	200	Pastor workspace module for people Development.constants.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	319	Pastor workspace module for people Development.selectors.
src/components/pastor/people-development/peopleDevelopment.types.ts	124	Pastor workspace module for people Development.types.
src/components/pastor/people-development/peopleDevelopment.utils.ts	585	Pastor workspace module for people Development.utils.
src/components/pastor/people-development/peopleDevelopmentEmail.ts	245	Pastor workspace module for people Development Email.
src/components/people-notes/AddDevelopmentItemForm.tsx	119	Confidential People Notes module for Add Development Item Form.
src/components/people-notes/AddPersonPanel.tsx	64	Confidential People Notes module for Add Person Panel.
src/components/people-notes/DevelopmentItemCard.tsx	164	Confidential People Notes module for Development Item Card.
src/components/people-notes/DevelopmentItemSection.tsx	41	Confidential People Notes module for Development Item Section.
src/components/people-notes/PeopleList.tsx	97	Confidential People Notes module for People List.
src/components/people-notes/PeopleNotesHeader.tsx	72	Confidential People Notes module for People Notes Header.
src/components/people-notes/SelectedPersonDetails.tsx	37	Confidential People Notes module for Selected Person Details.
src/components/people-notes/SelectedPersonHeader.tsx	44	Confidential People Notes module for Selected Person Header.
src/components/people-notes/peopleNotes.utils.test.ts	29	Automated test coverage for people notes.utils.
src/components/people-notes/peopleNotes.utils.ts	38	Confidential People Notes module for people Notes.utils.
src/components/people-notes/usePeopleNotes.ts	40	React state/data hook for use People Notes.
src/components/people-notes/usePeopleNotesData.ts	130	React state/data hook for use People Notes Data.
src/components/people-notes/usePeopleNotesMutations.ts	370	React state/data hook for use People Notes Mutations.
src/components/tutorial-builder/TutorialBuilderSection.tsx	392	Interactive tutorial builder/player module for Tutorial Builder Section.
src/components/tutorial-builder/TutorialContext.tsx	283	Interactive tutorial builder/player module for Tutorial Context.
src/components/tutorial-builder/TutorialEditorModal.tsx	773	Interactive tutorial builder/player module for Tutorial Editor Modal.
src/components/tutorial-builder/TutorialLibrary.tsx	231	Interactive tutorial builder/player module for Tutorial Library.
src/components/tutorial-builder/TutorialPlayer.tsx	258	Interactive tutorial builder/player module for Tutorial Player.
src/components/tutorial-builder/TutorialSimulationLayer.tsx	56	Interactive tutorial builder/player module for Tutorial Simulation Layer.
src/components/tutorial-builder/TutorialTargetPicker.tsx	180	Interactive tutorial builder/player module for Tutorial Target Picker.
src/components/tutorial-builder/index.ts	10	Interactive tutorial builder/player module for index.
src/components/tutorial-builder/tutorialBuilder.constants.ts	57	Interactive tutorial builder/player module for tutorial Builder.constants.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	223	Interactive tutorial builder/player module for tutorial Builder.firebase.
src/components/tutorial-builder/tutorialBuilder.types.ts	101	Interactive tutorial builder/player module for tutorial Builder.types.
src/components/tutorial-builder/tutorialBuilder.utils.ts	407	Interactive tutorial builder/player module for tutorial Builder.utils.
src/firebase.ts	24	Shared application infrastructure and contracts.
src/hooks/usePastorAccess.ts	84	React state/data hook for use Pastor Access.
src/i18n/index.tsx	49	Internationalization provider or translation catalog.
src/i18n/translations.ts	430	Internationalization provider or translation catalog.
src/lib/firebase.ts	51	Shared application infrastructure and contracts.

Source module	Lines	Responsibility
src/main.tsx	10	Source module for main.
src/pages/AboutUs.tsx	69	Route-level page and composition boundary for About Us.
src/pages/BookingCalendar.tsx	637	Route-level page and composition boundary for Booking Calendar.
src/pages/CongregationGroupNotes.tsx	23	Route-level page and composition boundary for Congregation Group Notes.
src/pages/GuidePage.tsx	104	Route-level page and composition boundary for Guide Page.
src/pages/LandingPage.tsx	66	Route-level page and composition boundary for Landing Page.
src/pages/NextGenActivities.tsx	70	Route-level page and composition boundary for Next Gen Activities.
src/pages/PeopleNotesPage.tsx	30	Route-level page and composition boundary for People Notes Page.
src/pages/PrivacyPolicy.tsx	106	Route-level page and composition boundary for Privacy Policy.
src/pages/TermsOfService.tsx	97	Route-level page and composition boundary for Terms Of Service.
src/services/administrator.test.ts	100	Automated test coverage for administrator.
src/services/administrator.ts	144	Frontend API/integration client for administrator.
src/services/assessment.ts	140	Frontend API/integration client for assessment.
src/services/bezalel.ts	87	Frontend API/integration client for bezalel.
src/services/booking.ts	128	Frontend API/integration client for booking.
src/services/congregationGroupNotes.test.ts	22	Automated test coverage for congregation group notes.
src/services/congregationGroupNotes.ts	28	Frontend API/integration client for congregation Group Notes.
src/services/gmail.ts	196	Frontend API/integration client for gmail.
src/services/googleMeet.ts	1	Frontend API/integration client for google Meet.
src/services/meetingInvitations.ts	101	Frontend API/integration client for meeting Invitations.
src/services/nextGenMissionMap.ts	103	Frontend API/integration client for next Gen Mission Map.
src/services/nextGenPortal.test.ts	29	Automated test coverage for next gen portal.
src/services/nextGenPortal.ts	207	Frontend API/integration client for next Gen Portal.
src/services/pastorAuthorization.ts	83	Frontend API/integration client for pastor Authorization.
src/services/pastorCalendar.ts	212	Frontend API/integration client for pastor Calendar.
src/services/peopleDevelopment.test.ts	66	Automated test coverage for people development.
src/services/peopleDevelopment.ts	187	Frontend API/integration client for people Development.
src/services/peopleDevelopmentNotifications.ts	160	Frontend API/integration client for people Development Notifications.
src/services/peopleNotes.ts	228	Frontend API/integration client for people Notes.
src/test/setup.ts	1	Automated test coverage for setup.
src/types.ts	58	Shared application infrastructure and contracts.
backend/src/admin/adminArchives.ts	336	Source module for admin Archives.
backend/src/admin/adminAuthorization.ts	99	Source module for admin Authorization.
backend/src/assessment/assessment.service.ts	467	Assessment engine module for assessment.service.
backend/src/bezalel/bezalel.ai.ts	159	Bezalel AI feature module for bezalel.ai.
backend/src/bezalel/bezalel.prompts.ts	39	Bezalel AI feature module for bezalel.prompts.
backend/src/booking/booking.service.ts	295	Source module for booking.service.
backend/src/calendar/calendar.ics.ts	56	Source module for calendar.ics.
backend/src/calendar/calendar.time.ts	49	Source module for calendar.time.
backend/src/emails/assessment.email.ts	54	Server-side email template/builder for assessment.
backend/src/emails/bookingRequest.email.ts	70	Server-side email template/builder for booking Request.
backend/src/emails/calendarNotification.email.ts	145	Server-side email template/builder for calendar Notification.
backend/src/emails/meetingInvitation.email.ts	284	Server-side email template/builder for meeting Invitation.
backend/src/emails/peopleDevelopmentNotification.email.ts	227	Server-side email template/builder for people Development Notification.
backend/src/index.ts	263	Source module for index.
backend/src/nextgen/nextGenFiles.ts	195	NextGen feature module for next Gen Files.
backend/src/nextgen/nextGenPortal.ts	660	NextGen feature module for next Gen Portal.
backend/src/pastorCalendar/pastorCalendar.normalize.ts	53	Source module for pastor Calendar.normalize.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	171	Source module for people Development.normalize.
backend/src/peopleNotes/peopleNotes.normalize.ts	109	Source module for people Notes.normalize.
backend/src/routes/admin.routes.ts	473	Cloudflare/Hono HTTP route module for admin.

Source module	Lines	Responsibility
backend/src/routes/assessment.routes.ts	319	Cloudflare/Hono HTTP route module for assessment.
backend/src/routes/auth.routes.ts	79	Cloudflare/Hono HTTP route module for auth.
backend/src/routes/bezalel.routes.ts	235	Cloudflare/Hono HTTP route module for bezalel.
backend/src/routes/booking.routes.ts	392	Cloudflare/Hono HTTP route module for booking.
backend/src/routes/meetingInvitations.routes.ts	173	Cloudflare/Hono HTTP route module for meeting Invitations.
backend/src/routes/nextGenMissionMap.routes.ts	169	Cloudflare/Hono HTTP route module for next Gen Mission Map.
backend/src/routes/nextGenPortal.routes.ts	555	Cloudflare/Hono HTTP route module for next Gen Portal.
backend/src/routes/pastorCalendar.routes.ts	893	Cloudflare/Hono HTTP route module for pastor Calendar.
backend/src/routes/peopleDevelopment.routes.ts	338	Cloudflare/Hono HTTP route module for people Development.
backend/src/routes/peopleDevelopmentNotifications.routes.ts	155	Cloudflare/Hono HTTP route module for people Development Notifications.
backend/src/routes/peopleNotes.routes.ts	470	Cloudflare/Hono HTTP route module for people Notes.
backend/src/schemas/assessment.schema.ts	61	Zod validation contracts for assessment requests and identifiers.
backend/src/schemas/booking.schema.ts	86	Zod validation contracts for booking requests and identifiers.
backend/src/schemas/meetingInvitation.schema.ts	103	Zod validation contracts for meeting Invitation requests and identifiers.
backend/src/schemas/pastorCalendar.schema.ts	89	Zod validation contracts for pastor Calendar requests and identifiers.
backend/src/schemas/peopleDevelopment.schema.ts	91	Zod validation contracts for people Development requests and identifiers.
backend/src/schemas/peopleDevelopmentNotification.schema.ts	108	Zod validation contracts for people Development Notification requests and identifiers.
backend/src/schemas/peopleNotes.schema.ts	46	Zod validation contracts for people Notes requests and identifiers.
backend/src/security/firebaseAuth.ts	147	Source module for firebase Auth.
backend/src/security/firebaseServiceAccount.ts	202	Source module for firebase Service Account.
backend/src/security/pastorAuthorization.ts	39	Source module for pastor Authorization.
backend/src/services/backblazeArchive.service.ts	182	Backend infrastructure/provider service for backblaze Archive.
backend/src/services/brevo.service.ts	151	Backend infrastructure/provider service for brevo.
backend/src/services/firebaseRealtimeDatabase.service.ts	266	Backend infrastructure/provider service for firebase Realtime Database.
backend/src/services/gemini.service.ts	111	Backend infrastructure/provider service for gemini.
backend/src/types/app.ts	43	Source module for app.

Appendix D

Complete Named-Function Catalog

The static review identified 981 named functions/callbacks in the current frontend and backend source modules. This is a location/reference catalog, not a replacement for source code; descriptions are concise statements of each function's role inferred from its name and owning module, while domain behavior is documented in the main chapters.

Source module	Function	Reference role
src/App.tsx	App	Named module function supporting app.
src/App.tsx	ProtectedRoute	Named module function supporting protected route.
src/App.tsx	AppRoutes	Named module function supporting app routes.
src/App.tsx	handleEmailAuth	UI/event handler for email auth.
src/App.tsx	getActiveTab	Retrieves or derives active tab.
src/components/AssessmentForm.tsx	AssessmentForm	Named module function supporting assessment form.
src/components/BookMeeting.tsx	BookMeeting	Named module function supporting book meeting.
src/components/BookMeeting.tsx	hourToLabel	Named module function supporting hour to label.
src/components/BookMeeting.tsx	hourToTime	Named module function supporting hour to time.
src/components/BookMeeting.tsx	timeToHour	Named module function supporting time to hour.
src/components/BookMeeting.tsx	slotOverlaps	Named module function supporting slot overlaps.
src/components/BookMeeting.tsx	handleDayClick	UI/event handler for day click.
src/components/BookMeeting.tsx	handleSlotClick	UI/event handler for slot click.
src/components/BookMeeting.tsx	handleSubmit	UI/event handler for submit.
src/components/Layout.tsx	Layout	Named module function supporting layout.
src/components/Layout.tsx	navItemClass	Named module function supporting nav item class.
src/components/PageTitle.tsx	PageTitle	Named module function supporting page title.
src/components/admin/AdminAccessScreens.tsx	AdminLoadingScreen	Named module function supporting admin loading screen.
src/components/admin/AdminAccessScreens.tsx	AdminLoginScreen	Named module function supporting admin login screen.
src/components/admin/AdminAccessScreens.tsx	AdminApprovalScreen	Named module function supporting admin approval screen.
src/components/admin/AdminAccessScreens.tsx	AdminAccessStage	Named module function supporting admin access stage.
src/components/admin/AdministratorPanel.tsx	AdministratorPanel	Named module function supporting administrator panel.
src/components/admin/AdministratorPanel.tsx	clearMessages	Named module function supporting clear messages.
src/components/admin/AdministratorPanel.tsx	handleVisibilityChange	UI/event handler for visibility change.
src/components/admin/AdministratorPanel.tsx	removePendingUpload	Removes pending upload from UI state or persistent storage.
src/components/admin/AdministratorPanel.tsx	uploadPendingPhotos	Uploads pending photos through the configured storage workflow.
src/components/admin/AdministratorPanel.tsx	saveStoredPhotoText	Saves stored photo text state or persistence changes.
src/components/admin/AdministratorPanel.tsx	savePhotoOrder	Saves photo order state or persistence changes.
src/components/admin/AdministratorPanel.tsx	movePhoto	Named module function supporting move photo.
src/components/admin/AdministratorPanel.tsx	deletePhoto	Deletes photo using the owning feature boundary.
src/components/admin/admin.utils.ts	normalizeAdminAuthority	Normalizes admin authority into the module's canonical representation.
src/components/admin/admin.utils.ts	normalizeAdminAccount	Normalizes admin account into the module's canonical representation.
src/components/admin/admin.utils.ts	firebaseAuthenticationErrorMessage	Named module function supporting firebase authentication error message.
src/components/admin/admin.utils.ts	humanizeIdentifier	Named module function supporting humanize identifier.
src/components/admin/admin.utils.ts	assessmentFormTitle	Named module function supporting assessment form title.
src/components/admin/admin.utils.ts	normalizeAssessmentFormState	Normalizes assessment form state into the module's canonical representation.
src/components/admin/admin.utils.ts	parsePhotos	Parses and validates photos.
src/components/admin/admin.utils.ts	fileToDataUrl	Named module function supporting file to data url.
src/components/admin/admin.utils.ts	createPhotoId	Creates photo id and/or its persisted representation.
src/components/admin/admin.utils.ts	normalizeNumber	Normalizes number into the module's canonical representation.
src/components/admin/admin.utils.ts	normalizeStoredPhoto	Normalizes stored photo into the module's canonical representation.
src/components/admin/archives/ArchiveFolderTree.tsx	ArchiveFolderTree	Named module function supporting archive folder tree.
src/components/admin/archives/ArchiveFolderTree.tsx	FolderTreeNode	Named module function supporting folder tree node.
src/components/admin/archives/LincArchivesSection.tsx	LincArchivesSection	Named module function supporting linc archives section.
src/components/admin/archives/LincArchivesSection.tsx	ArchiveMetric	Named module function supporting archive metric.
src/components/admin/archives/LincArchivesSection.tsx	ArchiveFileRow	Named module function supporting archive file row.
src/components/admin/archives/LincArchivesSection.tsx	submitFolder	Named module function supporting submit folder.
src/components/admin/archives/LincArchivesSection.tsx	confirmDeleteFolder	Named module function supporting confirm delete folder.
src/components/admin/archives/LincArchivesSection.tsx	acceptFiles	Named module function supporting accept files.
src/components/admin/archives/LincArchivesSection.tsx	handleDrop	UI/event handler for drop.
src/components/admin/archives/archives.utils.ts	buildArchiveTree	Builds archive tree from validated or normalized inputs.
src/components/admin/archives/archives.utils.ts	archiveBreadcrumbs	Named module function supporting archive breadcrumbs.
src/components/admin/archives/archives.utils.ts	foldersInArchiveLocation	Named module function supporting folders in archive location.
src/components/admin/archives/archives.utils.ts	filesInArchiveLocation	Named module function supporting files in archive location.
src/components/admin/archives/archives.utils.ts	formatArchiveBytes	Formats archive bytes for display or transport.
src/components/admin/archives/archives.utils.ts	sortNodes	Named module function supporting sort nodes.

Source module	Function	Reference role
src/components/admin/archives/useLincArchives.ts	useLincArchives	React hook coordinating linc archives state, data access, and UI actions.
src/components/admin/archives/useLincArchives.ts	addFolder	Named module function supporting add folder.
src/components/admin/archives/useLincArchives.ts	removeSelectedFolder	Removes selected folder from UI state or persistent storage.
src/components/admin/archives/useLincArchives.ts	addFiles	Named module function supporting add files.
src/components/admin/archives/useLincArchives.ts	removeFile	Removes file from UI state or persistent storage.
src/components/admin/archives/useLincArchives.ts	verifyFile	Named module function supporting verify file.
src/components/admin/archives/useLincArchives.ts	downloadFile	Downloads or prepares file for the user.
src/components/admin/attendance/AttendanceAnalysisCharts.tsx	AttendanceAnalysisCharts	Named module function supporting attendance analysis charts.
src/components/admin/attendance/AttendanceAnalysisPanel.tsx	AttendanceAnalysisPanel	Named module function supporting attendance analysis panel.
src/components/admin/attendance/AttendanceManagement.tsx	AttendanceManagement	Named module function supporting attendance management.
src/components/admin/attendance/AttendancePeoplePanel.tsx	AttendancePeoplePanel	Named module function supporting attendance people panel.
src/components/admin/attendance/AttendancePersonEditor.tsx	AttendancePersonEditor	Named module function supporting attendance person editor.
src/components/admin/attendance/AttendancePersonModal.tsx	AttendancePersonModal	Named module function supporting attendance person modal.
src/components/admin/attendance/AttendancePersonalAnalysisModal.tsx	AttendancePersonalAnalysisModal	Named module function supporting attendance personal analysis modal.
src/components/admin/attendance/AttendanceRecordingPanel.tsx	AttendanceRecordingPanel	Named module function supporting attendance recording panel.
src/components/admin/attendance/attendance.copy.ts	getAttendanceText	Retrieves or derives attendance text.
src/components/admin/attendance/attendance.utils.ts	normalizeNumber	Normalizes number into the module's canonical representation.
src/components/admin/attendance/attendance.utils.ts	normalizePerson	Normalizes person into the module's canonical representation.
src/components/admin/attendance/attendance.utils.ts	formatDateKey	Formats date key for display or transport.
src/components/admin/attendance/attendance.utils.ts	getAttendanceDays	Retrieves or derives attendance days.
src/components/admin/attendance/attendance.utils.ts	buildDaysOfAttendance	Builds days of attendance from validated or normalized inputs.
src/components/admin/attendance/attendance.utils.ts	buildCalendarDays	Builds calendar days from validated or normalized inputs.
src/components/admin/attendance/attendance.utils.ts	getFirstSundayInMay	Retrieves or derives first sunday in may.
src/components/admin/attendance/attendance.utils.ts	buildSundayDateKeysFromStart	Builds sunday date keys from start from validated or normalized inputs.
src/components/admin/attendance/attendance.utils.ts	parseDateKeyToTime	Parses and validates date key to time.
src/components/admin/attendance/attendance.utils.ts	normalizePersonPhotoSource	Normalizes person photo source into the module's canonical representation.
src/components/admin/attendance/useAttendanceAnalytics.ts	useAttendanceAnalytics	React hook coordinating attendance analytics state, data access, and UI actions.
src/components/admin/attendance/useAttendanceManagement.ts	useAttendanceManagement	React hook coordinating attendance management state, data access, and UI actions.
src/components/admin/attendance/useAttendanceManagement.ts	closePersonEditor	Closes or resets person editor UI state.
src/components/admin/attendance/useAttendanceManagement.ts	openNewPersonEditor	Opens or initializes new person editor UI state.
src/components/admin/attendance/useAttendanceManagement.ts	handleSelectPerson	UI/event handler for select person.
src/components/admin/attendance/useAttendanceManagement.ts	openPersonCamera	Opens or initializes person camera UI state.
src/components/admin/attendance/useAttendanceManagement.ts	closePersonCamera	Closes or resets person camera UI state.
src/components/admin/attendance/useAttendanceManagement.ts	switchPersonCamera	Named module function supporting switch person camera.
src/components/admin/attendance/useAttendanceManagement.ts	capturePersonPhotoFromLiveCamera	Named module function supporting capture person photo from live camera.
src/components/admin/attendance/useAttendanceManagement.ts	removePersonPhoto	Removes person photo from UI state or persistent storage.
src/components/admin/attendance/useAttendanceManagement.ts	handleSavePerson	UI/event handler for save person.
src/components/admin/attendance/useAttendanceManagement.ts	moveCalendarMonth	Named module function supporting move calendar month.
src/components/admin/attendance/useAttendanceManagement.ts	handleSelectAttendanceDate	UI/event handler for select attendance date.
src/components/admin/attendance/useAttendanceManagement.ts	handleMarkAttendance	UI/event handler for mark attendance.
src/components/admin/attendance/useAttendanceManagement.ts	handleEscape	UI/event handler for escape.
src/components/admin/attendance/useAttendanceManagement.ts	startCamera	Starts camera workflow.
src/components/admin/components/AdminCommandCenter.tsx	AdminCommandHeader	Named module function supporting admin command header.
src/components/admin/components/AdminCommandCenter.tsx	AdminAreaNavigation	Named module function supporting admin area navigation.
src/components/admin/components/AdminCommandCenter.tsx	AdminOverview	Named module function supporting admin overview.
src/components/admin/components/AdminHierarchySection.tsx	AdminHierarchySection	Named module function supporting admin hierarchy section.
src/components/admin/components/AdminUtilitySections.tsx	NoAuthorityCard	Named module function supporting no authority card.
src/components/admin/components/AdminUtilitySections.tsx	AttendanceAdminSection	Named module function supporting attendance admin section.
src/components/admin/components/AdminUtilitySections.tsx	AssessmentFormsSection	Named module function supporting assessment forms section.
src/components/admin/components/AssessmentUserLinkageSection.tsx	AssessmentUserLinkageSection	Named module function supporting assessment user linkage section.
src/components/admin/components/AssessmentUserLinkageSection.tsx	load	Loads into feature state.
src/components/admin/components/AssessmentUserLinkageSection.tsx	changeForm	Named module function supporting change form.
src/components/admin/components/AssessmentUserLinkageSection.tsx	save	Saves state or persistence changes.
src/components/admin/components/AssessmentUserLinkageSection.tsx	remove	Removes from UI state or persistent storage.
src/components/admin/components/AssessmentUserLinkageSection.tsx	sendIdentifier	Sends identifier through the configured communication or API boundary.
src/components/admin/components/AssessmentUserLinkageSection.tsx	sendAll	Sends all through the configured communication or API boundary.

Source module	Function	Reference role
src/components/admin/components/CarouselManagementSection.tsx	CarouselManagementSection	Named module function supporting carousel management section.
src/components/admin/hooks/useAdministratorAccess.ts	useAdministratorAccess	React hook coordinating administrator access state, data access, and UI actions.
src/components/admin/hooks/useAdministratorAccess.ts	handleLogin	UI/event handler for login.
src/components/admin/hooks/useAdministratorAccess.ts	handleLogout	UI/event handler for logout.
src/components/admin/hooks/useAdministratorAccess.ts	handleSaveAdminAuthority	UI/event handler for save admin authority.
src/components/admin/hooks/useAdministratorAccess.ts	handleSuspendAdmin	UI/event handler for suspend admin.
src/components/assessment/AssessmentQuestionnaire.tsx	AssessmentQuestionnaire	Named module function supporting assessment questionnaire.
src/components/assessment/AssessmentQuestionnaire.tsx	renderField	Named module function supporting render field.
src/components/assessment/AssessmentQuestionnaire.tsx	renderSection	Named module function supporting render section.
src/components/assessment/AssessmentResults.tsx	AssessmentResults	Named module function supporting assessment results.
src/components/assessment/AssessmentSelector.tsx	AssessmentSelector	Named module function supporting assessment selector.
src/components/assessment/assessment.calculations.test.ts	translate	Named module function supporting translate.
src/components/assessment/assessment.calculations.ts	calculationSourceSection	Named module function supporting calculation source section.
src/components/assessment/assessment.calculations.ts	buildResult	Builds result from validated or normalized inputs.
src/components/assessment/assessment.calculations.ts	scoreMap	Computes or returns score map for the domain.
src/components/assessment/assessment.calculations.ts	rankValues	Ranks values according to feature scoring rules.
src/components/assessment/assessment.calculations.ts	runCalculations	Executes the calculations workflow.
src/components/assessment/assessment.calculations.ts	isResultItem	Predicate evaluating whether result item.
src/components/assessment/assessment.calculations.ts	calculationDisplayValue	Named module function supporting calculation display value.
src/components/assessment/assessment.forms.ts	isUsableEmail	Predicate evaluating whether usable email.
src/components/assessment/assessment.forms.ts	localText	Named module function supporting local text.
src/components/assessment/assessment.forms.ts	humanizeId	Named module function supporting humanize id.
src/components/assessment/assessment.forms.ts	translateOrText	Named module function supporting translate or text.
src/components/assessment/assessment.forms.ts	template	Named module function supporting template.
src/components/assessment/assessment.forms.ts	ratingRange	Named module function supporting rating range.
src/components/assessment/assessment.forms.ts	allFields	Named module function supporting all fields.
src/components/assessment/assessment.forms.ts	getSection	Retrieves or derives section.
src/components/assessment/assessment.forms.ts	getGroup	Retrieves or derives group.
src/components/assessment/assessment.forms.ts	getField	Retrieves or derives field.
src/components/assessment/assessment.forms.ts	fieldLabel	Named module function supporting field label.
src/components/assessment/assessment.forms.ts	sectionTitle	Named module function supporting section title.
src/components/assessment/assessment.forms.ts	groupTitle	Named module function supporting group title.
src/components/assessment/assessment.forms.ts	cardTitle	Named module function supporting card title.
src/components/assessment/assessment.forms.ts	cardDescription	Named module function supporting card description.
src/components/assessment/assessment.forms.ts	pageTitle	Named module function supporting page title.
src/components/assessment/assessment.forms.ts	pageSubtitle	Named module function supporting page subtitle.
src/components/assessment/assessment.forms.ts	resultTitle	Named module function supporting result title.
src/components/assessment/assessment.forms.ts	resultCardLabel	Named module function supporting result card label.
src/components/assessment/assessment.forms.ts	initialAnswers	Named module function supporting initial answers.
src/components/assessment/assessment.forms.ts	isFilled	Predicate evaluating whether filled.
src/components/assessment/assessment.forms.ts	numberValue	Named module function supporting number value.
src/components/assessment/assessment.forms.ts	validate	Validates against feature rules.
src/components/assessment/assessment.useAssessmentForm.ts	useAssessmentForm	React hook coordinating assessment form state, data access, and UI actions.
src/components/assessment/useAssessmentForm.ts	setAnswer	Named module function supporting set answer.
src/components/assessment/useAssessmentForm.ts	selectForm	Named module function supporting select form.
src/components/assessment/useAssessmentForm.ts	resetForRetake	Named module function supporting reset for retake.
src/components/assessment/useAssessmentForm.ts	handleBackToAssessmentChoices	UI/event handler for back to assessment choices.
src/components/assessment/useAssessmentForm.ts	handleDirectSignup	UI/event handler for direct signup.
src/components/assessment/useAssessmentForm.ts	handleSubmit	UI/event handler for submit.
src/components/bezalel/BezalelChat.tsx	BezalelChat	Named module function supporting bezalel chat.
src/components/bezalel/BezalelChat.tsx	submit	Named module function supporting submit.
src/components/bezalel/BookingBezalelAssistant.tsx	BookingBezalelAssistant	Named module function supporting booking bezalel assistant.
src/components/bezalel/BookingBezalelAssistant.tsx	send	Sends through the configured communication or API boundary.
src/components/bezalel/PastorBezalelAssistant.tsx	PastorBezalelAssistant	Named module function supporting pastor bezalel assistant.
src/components/bezalel/PastorBezalelAssistant.tsx	executeAction	Named module function supporting execute action.
src/components/bezalel/PastorBezalelAssistant.tsx	send	Sends through the configured communication or API boundary.

Source module	Function	Reference role
src/components/brand/LincLogo.tsx	LincLogo	Named module function supporting linc logo.
src/components/congregation-group-notes/AssignmentDetailsDialog.tsx	AssignmentDetailsDialog	Named module function supporting assignment details dialog.
src/components/congregation-group-notes/AssignmentList.tsx	AssignmentList	Named module function supporting assignment list.
src/components/congregation-group-notes/CongregationGroupDashboard.tsx	CongregationGroupDashboard	Named module function supporting congregation group dashboard.
src/components/congregation-group-notes/CongregationGroupHeader.tsx	CongregationGroupHeader	Named module function supporting congregation group header.
src/components/congregation-group-notes/GroupSummary.tsx	GroupSummary	Named module function supporting group summary.
src/components/congregation-group-notes/IdentifierLogin.tsx	IdentifierLogin	Named module function supporting identifier login.
src/components/congregation-group-notes/LatestAssignment.tsx	LatestAssignment	Named module function supporting latest assignment.
src/components/congregation-group-notes/MeetingCalendarPanel.tsx	MeetingCalendarPanel	Named module function supporting meeting calendar panel.
src/components/congregation-group-notes/MemberProfileHeader.tsx	MemberProfileHeader	Named module function supporting member profile header.
src/components/congregation-group-notes/UnassignedGroupNotice.tsx	UnassignedGroupNotice	Named module function supporting unassigned group notice.
src/components/congregation-group-notes/UpcomingMeetings.tsx	UpcomingMeetings	Named module function supporting upcoming meetings.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	getGroupConfig	Retrieves or derives group config.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	getGroupLabel	Retrieves or derives group label.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	getGroupDescription	Retrieves or derives group description.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	getMeetingOccurrenceTimestamp	Retrieves or derives meeting occurrence timestamp.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	formatMeetingOccurrenceDate	Formats meeting occurrence date for display or transport.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	normalizeLookupKey	Normalizes lookup key into the module's canonical representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	unwrapStoredValue	Named module function supporting unwrap stored value.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	extractResponseValue	Extracts response value from a broader record.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	safeFirebaseKey	Named module function supporting safe firebase key.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	normalizeIdentifier	Normalizes identifier into the module's canonical representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	normalizeNumber	Normalizes number into the module's canonical representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	normalizePeopleDevelopmentGroup	Normalizes people development group into the module's canonical representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	extractPeopleDevelopmentGroup	Extracts people development group from a broader record.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	normalizeAssignmentGroups	Normalizes assignment groups into the module's canonical representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	getAssignmentDisplayGroupLabel	Retrieves or derives assignment display group label.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	formatDateLabel	Formats date label for display or transport.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	normalizeAssignmentAttachments	Normalizes assignment attachments into the module's canonical representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	formatAttachmentSize	Formats attachment size for display or transport.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	cleanBase64Payload	Named module function supporting clean base64 payload.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	createDecodedAttachmentUrl	Creates decoded attachment url and/or its persisted representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	buildProfileFromMemberRecord	Builds profile from member record from validated or normalized inputs.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	buildProfileFromFormRecord	Builds profile from form record from validated or normalized inputs.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	normalizeAssignment	Normalizes assignment into the module's canonical representation.
src/components/congregation-group-notes/congregationGroupNotes.config.ts	asRecord	Named module function supporting as record.

Source module	Function	Reference role
src/components/congregation-group-notes/useCongregationGroupNotes.ts	useCongregationGroupNotes	React hook coordinating congregation group notes state, data access, and UI actions.
src/components/congregation-group-notes/useCongregationGroupNotes.ts	openAttachment	Opens or initializes attachment UI state.
src/components/congregation-group-notes/useCongregationGroupNotes.ts	downloadAttachment	Downloads or prepares attachment for the user.
src/components/congregation-group-notes/useCongregationGroupNotes.ts	handleLogin	UI/event handler for login.
src/components/congregation-group-notes/useCongregationGroupNotes.ts	handleLogout	UI/event handler for logout.
src/components/landing/CommunityCarousel.tsx	CommunityCarousel	Named module function supporting community carousel.
src/components/landing/CommunityCarousel.tsx	normalizeImageSource	Normalizes image source into the module's canonical representation.
src/components/landing/CommunityCarousel.tsx	normalizePhotos	Normalizes photos into the module's canonical representation.
src/components/landing/CommunityCarousel.tsx	move	Named module function supporting move.
src/components/landing/FloatingLandingNav.tsx	FloatingLandingNav	Named module function supporting floating landing nav.
src/components/landing/LincOneHero.tsx	LincOneHero	Named module function supporting linc one hero.
src/components/landing/SpiritualProgramFeature.tsx	SpiritualProgramFeature	Named module function supporting spiritual program feature.
src/components/linc/LincPageHero.tsx	LincPageHero	Named module function supporting linc page hero.
src/components/nextgen/GoogleMissionRouteMap.tsx	GoogleMissionRouteMap	Named module function supporting google mission route map.
src/components/nextgen/GoogleMissionRouteMap.tsx	configureGoogleMaps	Named module function supporting configure google maps.
src/components/nextgen/GoogleMissionRouteMap.tsx	fallbackDistance	Named module function supporting fallback distance.
src/components/nextgen/GoogleMissionRouteMap.tsx	fallbackDuration	Named module function supporting fallback duration.
src/components/nextgen/GoogleMissionRouteMap.tsx	loadRoute	Loads route into feature state.
src/components/nextgen/MontrealMissionTripMap.tsx	MontrealMissionTripMap	Named module function supporting montreal mission trip map.
src/components/nextgen/MontrealMissionTripMap.tsx	googleMapsUrl	Named module function supporting google maps url.
src/components/nextgen/MontrealMissionTripMap.tsx	appleMapsUrl	Named module function supporting apple maps url.
src/components/nextgen/MontrealMissionTripMap.tsx	locationTypeLabel	Named module function supporting location type label.
src/components/nextgen/MontrealMissionTripMap.tsx	createLocationMarkerIcon	Creates location marker icon and/or its persisted representation.
src/components/nextgen/MontrealMissionTripMap.tsx	LocationTypeBadge	Named module function supporting location type badge.
src/components/nextgen/MontrealMissionTripMap.tsx	googleDirectionsUrl	Named module function supporting google directions url.
src/components/nextgen/activities/NextGenAccessSection.tsx	NextGenAccessSection	Named module function supporting next gen access section.
src/components/nextgen/activities/NextGenActivityMenu.tsx	NextGenActivityMenu	Named module function supporting next gen activity menu.
src/components/nextgen/activities/NextGenParticipantHeader.tsx	NextGenParticipantHeader	Named module function supporting next gen participant header.
src/components/nextgen/activities/NextGenPeerReviewPanel.tsx	NextGenPeerReviewPanel	Named module function supporting next gen peer review panel.
src/components/nextgen/activities/NextGenQuestionPanel.tsx	NextGenQuestionPanel	Named module function supporting next gen question panel.
src/components/nextgen/activities/NextGenQuestionPanel.tsx	saveAndClose	Saves and close state or persistence changes.
src/components/nextgen/activities/NextGenSurveyPanel.tsx	NextGenSurveyPanel	Named module function supporting next gen survey panel.
src/components/nextgen/activities/nextGenActivities.firebase.ts	createNextGenRegistration	Creates next gen registration and/or its persisted representation.
src/components/nextgen/activities/nextGenActivities.firebase.ts	getNextGenUser	Retrieves or derives next gen user.
src/components/nextgen/activities/nextGenActivities.firebase.ts	loadReviewedSessionIds	Loads reviewed session ids into feature state.
src/components/nextgen/activities/nextGenActivities.firebase.ts	subscribeToQuestions	Subscribes to to questions updates and exposes normalized changes.
src/components/nextgen/activities/nextGenActivities.firebase.ts	saveNextGenQuestion	Saves next gen question state or persistence changes.
src/components/nextgen/activities/nextGenActivities.firebase.ts	submitPeerVote	Named module function supporting submit peer vote.
src/components/nextgen/activities/nextGenActivities.firebase.ts	subscribeToSurveyCompletion	Subscribes to to survey completion updates and exposes normalized changes.
src/components/nextgen/activities/nextGenActivities.firebase.ts	saveSurveyResponse	Saves survey response state or persistence changes.
src/components/nextgen/activities/nextGenActivities.utils.ts	toRecord	Named module function supporting to record.
src/components/nextgen/activities/nextGenActivities.utils.ts	createInitialSurveyAnswers	Creates initial survey answers and/or its persisted representation.
src/components/nextgen/activities/nextGenActivities.utils.ts	normalizeNumber	Normalizes number into the module's canonical representation.
src/components/nextgen/activities/nextGenActivities.utils.ts	normalizeUserId	Normalizes user id into the module's canonical representation.
src/components/nextgen/activities/nextGenActivities.utils.ts	isUsableEmail	Predicate evaluating whether usable email.
src/components/nextgen/activities/nextGenActivities.utils.ts	getEasternTime	Retrieves or derives eastern time.
src/components/nextgen/activities/nextGenActivities.utils.ts	normalizeUserStatus	Normalizes user status into the module's canonical representation.
src/components/nextgen/activities/nextGenActivities.utils.ts	normalizeUserRecord	Normalizes user record into the module's canonical representation.
src/components/nextgen/activities/nextGenActivities.utils.ts	extractQuestionVoterIdentifiers	Extracts question voter identifiers from a broader record.
src/components/nextgen/activities/nextGenActivities.utils.ts	questionHasIdentifierVote	Named module function supporting question has identifier vote.
src/components/nextgen/activities/nextGenActivities.utils.ts	normalizeSavedSession	Normalizes saved session into the module's canonical representation.
src/components/nextgen/activities/nextGenActivities.utils.ts	getReviewableSessions	Retrieves or derives reviewable sessions.
src/components/nextgen/activities/nextGenActivities.utils.ts	countSurveyAnswers	Named module function supporting count survey answers.
src/components/nextgen/activities/nextGenActivities.utils.ts	addIdentifier	Named module function supporting add identifier.
src/components/nextgen/activities/nextGenActivities.utils.ts	addKeys	Named module function supporting add keys.
src/components/nextgen/activities/nextGenCertificate.ts	downloadNextGenCertificate	Downloads or prepares next gen certificate for the user.

Source module	Function	Reference role
src/components/nextgen/activities/nextGenCertificate.ts	encodeText	Named module function supporting encode text.
src/components/nextgen/activities/nextGenCertificate.ts	mergeByteArrays	Named module function supporting merge byte arrays.
src/components/nextgen/activities/nextGenCertificate.ts	createImagePdf	Creates image pdf and/or its persisted representation.
src/components/nextgen/activities/nextGenCertificate.ts	canvasToJpegBytes	Named module function supporting canvas to jpeg bytes.
src/components/nextgen/activities/nextGenCertificate.ts	appendBytes	Named module function supporting append bytes.
src/components/nextgen/activities/nextGenCertificate.ts	appendText	Named module function supporting append text.
src/components/nextgen/activities/nextGenCertificate.ts	markObject	Named module function supporting mark object.
src/components/nextgen/activities/useNextGenIdentity.ts	useNextGenIdentity	React hook coordinating next gen identity state, data access, and UI actions.
src/components/nextgen/activities/useNextGenIdentity.ts	chooseEntryMode	Named module function supporting choose entry mode.
src/components/nextgen/activities/useNextGenIdentity.ts	closeEntryMode	Closes or resets entry mode UI state.
src/components/nextgen/activities/useNextGenIdentity.ts	handleSignupSubmit	UI/event handler for signup submit.
src/components/nextgen/activities/useNextGenIdentity.ts	handleCertificateDownload	UI/event handler for certificate download.
src/components/nextgen/activities/useNextGenIdentity.ts	handleExistingUserSubmit	UI/event handler for existing user submit.
src/components/nextgen/activities/useNextGenIdentity.ts	handleLogout	UI/event handler for logout.
src/components/nextgen/activities/useNextGenQuestions.ts	useNextGenQuestions	React hook coordinating next gen questions state, data access, and UI actions.
src/components/nextgen/activities/useNextGenQuestions.ts	resetForm	Named module function supporting reset form.
src/components/nextgen/activities/useNextGenQuestions.ts	resetQuestionSession	Named module function supporting reset question session.
src/components/nextgen/activities/useNextGenQuestions.ts	handlePeerVote	UI/event handler for peer vote.
src/components/nextgen/activities/useNextGenSurvey.ts	useNextGenSurvey	React hook coordinating next gen survey state, data access, and UI actions.
src/components/nextgen/activities/useNextGenSurvey.ts	initialState	Named module function supporting initial state.
src/components/nextgen/activities/useNextGenSurvey.ts	updateState	Updates state while applying module validation and state rules.
src/components/nextgen/activities/useNextGenSurvey.ts	clearMessages	Named module function supporting clear messages.
src/components/nextgen/activities/useNextGenSurvey.ts	resetSurveySession	Named module function supporting reset survey session.
src/components/nextgen/activities/useNextGenSurvey.ts	setBinarySurveyAnswer	Named module function supporting set binary survey answer.
src/components/nextgen/activities/useNextGenSurvey.ts	setRatingSurveyAnswer	Named module function supporting set rating survey answer.
src/components/nextgen/activities/useNextGenSurvey.ts	handleSurveySubmit	UI/event handler for survey submit.
src/components/nextgen/portal/NextGenAuthGateway.tsx	NextGenAuthGateway	Named module function supporting next gen auth gateway.
src/components/nextgen/portal/NextGenAuthGateway.tsx	submit	Named module function supporting submit.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	NextGenFilesWorkspace	Named module function supporting next gen files workspace.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	load	Loads into feature state.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	addFolder	Named module function supporting add folder.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	upload	Uploads through the configured storage workflow.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	removeFolder	Removes folder from UI state or persistent storage.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	download	Downloads or prepares for the user.
src/components/nextgen/portal/NextGenFilesWorkspace.tsx	removeFile	Removes file from UI state or persistent storage.
src/components/nextgen/portal/NextGenQaSessions.tsx	NextGenQaSessionList	Named module function supporting next gen qa session list.
src/components/nextgen/portal/NextGenQaSessions.tsx	NextGenQaSessionPage	Named module function supporting next gen qa session page.
src/components/nextgen/portal/NextGenQaSessions.tsx	addQuestion	Named module function supporting add question.
src/components/nextgen/portal/NextGenQaSessions.tsx	vote	Named module function supporting vote.
src/components/pastor/PastorDashboard.tsx	PastorDashboard	Named module function supporting pastor dashboard.
src/components/pastor/PastorDashboard.tsx	focusBezalelDate	Named module function supporting focus bezalel date.
src/components/pastor/PastorDashboard.tsx	exportCurrentCalendarMonth	Named module function supporting export current calendar month.
src/components/pastor/calendar/calendar.email.ts	getMeetingRequestEmail	Retrieves or derives meeting request email.
src/components/pastor/calendar/calendar.email.ts	getMeetingRequesterLocale	Retrieves or derives meeting requester locale.
src/components/pastor/calendar/calendar.forms.ts	createInitialAvailabilityForm	Creates initial availability form and/or its persisted representation.
src/components/pastor/calendar/calendar.forms.ts	createInitialUnavailabilityForm	Creates initial unavailability form and/or its persisted representation.
src/components/pastor/calendar/calendar.forms.ts	buildAvailabilityDates	Builds availability dates from validated or normalized inputs.
src/components/pastor/calendar/calendar.forms.ts	toggleAvailabilityWeekday	Toggles availability weekday state.
src/components/pastor/calendar/calendar.forms.ts	isAvailabilityFormValid	Predicate evaluating whether availability form valid.
src/components/pastor/calendar/calendar.forms.ts	isUnavailabilityFormValid	Predicate evaluating whether unavailability form valid.
src/components/pastor/calendar/calendar.slots.ts	getDateString	Retrieves or derives date string.
src/components/pastor/calendar/calendar.slots.ts	getAvailabilityBlocksForDate	Retrieves or derives availability blocks for date.
src/components/pastor/calendar/calendar.slots.ts	getUnavailabilityBlocksForDate	Retrieves or derives unavailability blocks for date.
src/components/pastor/calendar/calendar.slots.ts	getMeetingsForDate	Retrieves or derives meetings for date.
src/components/pastor/calendar/calendar.slots.ts	getPendingRequestsForDate	Retrieves or derives pending requests for date.
src/components/pastor/calendar/calendar.slots.ts	getAvailabilityRange	Retrieves or derives availability range.

Source module	Function	Reference role
src/components/pastor/calendar/calendar.slots.ts	getUnavailabilityRange	Retrieves or derives unavailability range.
src/components/pastor/calendar/calendar.slots.ts	isPastorSlotInsideAvailability	Predicate evaluating whether pastor slot inside availability.
src/components/pastor/calendar/calendar.slots.ts	getBlockingUnavailabilityForSlot	Retrieves or derives blocking unavailability for slot.
src/components/pastor/calendar/calendar.slots.ts	isPastorSlotBooked	Predicate evaluating whether pastor slot booked.
src/components/pastor/calendar/calendar.slots.ts	getPastorSlotStatus	Retrieves or derives pastor slot status.
src/components/pastor/calendar/calendar.slots.ts	getPastorSlotTranslationKey	Retrieves or derives pastor slot translation key.
src/components/pastor/calendar/calendar.slots.ts	buildSlotBlockHours	Builds slot block hours from validated or normalized inputs.
src/components/pastor/calendar/calendar.utils.ts	timeToHour	Named module function supporting time to hour.
src/components/pastor/calendar/calendar.utils.ts	hourToTime	Named module function supporting hour to time.
src/components/pastor/calendar/calendar.utils.ts	hourToLabel	Named module function supporting hour to label.
src/components/pastor/calendar/calendar.utils.ts	timeToLabel	Named module function supporting time to label.
src/components/pastor/calendar/calendar.utils.ts	timeRangeToLabel	Named module function supporting time range to label.
src/components/pastor/calendar/calendar.utils.ts	buildTimeOptions	Builds time options from validated or normalized inputs.
src/components/pastor/calendar/calendar.utils.ts	slotOverlaps	Named module function supporting slot overlaps.
src/components/pastor/hooks/useAvailability.ts	useAvailability	React hook coordinating availability state, data access, and UI actions.
src/components/pastor/hooks/useAvailability.ts	createAvailability	Creates availability and/or its persisted representation.
src/components/pastor/hooks/useAvailability.ts	deleteAvailability	Deletes availability using the owning feature boundary.
src/components/pastor/hooks/useAvailability.ts	createUnavailability	Creates unavailability and/or its persisted representation.
src/components/pastor/hooks/useAvailability.ts	deleteUnavailability	Deletes unavailability using the owning feature boundary.
src/components/pastor/hooks/useAvailability.ts	resetAvailabilityForm	Named module function supporting reset availability form.
src/components/pastor/hooks/useAvailability.ts	resetUnavailabilityForm	Named module function supporting reset unavailability form.
src/components/pastor/hooks/useAvailability.ts	closeAvailabilityModal	Closes or resets availability modal UI state.
src/components/pastor/hooks/useAvailability.ts	closeUnavailabilityModal	Closes or resets unavailability modal UI state.
src/components/pastor/hooks/useCalendarMonth.ts	useCalendarMonth	React hook coordinating calendar month state, data access, and UI actions.
src/components/pastor/hooks/useCalendarMonth.ts	goToPreviousMonth	Named module function supporting go to previous month.
src/components/pastor/hooks/useCalendarMonth.ts	goToNextMonth	Named module function supporting go to next month.
src/components/pastor/hooks/useCalendarMonth.ts	goToToday	Named module function supporting go to today.
src/components/pastor/hooks/useMeetingRequests.ts	useMeetingRequests	React hook coordinating meeting requests state, data access, and UI actions.
src/components/pastor/hooks/useMeetingRequests.ts	toggleRequests	Toggles requests state.
src/components/pastor/hooks/useMeetings.ts	useMeetings	React hook coordinating meetings state, data access, and UI actions.
src/components/pastor/hooks/useMeetings.ts	createInitialMeeting	Creates initial meeting and/or its persisted representation.
src/components/pastor/hooks/useMeetings.ts	resetMeetingForm	Named module function supporting reset meeting form.
src/components/pastor/hooks/useMeetings.ts	openNewMeeting	Opens or initializes new meeting UI state.
src/components/pastor/hooks/useMeetings.ts	openNewMeetingForDate	Opens or initializes new meeting for date UI state.
src/components/pastor/hooks/useMeetings.ts	openMeetingEditor	Opens or initializes meeting editor UI state.
src/components/pastor/hooks/useMeetings.ts	closeMeetingEditor	Closes or resets meeting editor UI state.
src/components/pastor/hooks/useMeetings.ts	toggleParticipant	Toggles participant state.
src/components/pastor/hooks/useMeetings.ts	handleDeleteMeeting	UI/event handler for delete meeting.
src/components/pastor/hooks/useNextGen.ts	useNextGen	React hook coordinating next gen state, data access, and UI actions.
src/components/pastor/hooks/useParticipants.ts	useParticipants	React hook coordinating participants state, data access, and UI actions.
src/components/pastor/hooks/useParticipants.ts	asRecord	Named module function supporting as record.
src/components/pastor/hooks/useParticipants.ts	normalizeLookupKey	Normalizes lookup key into the module's canonical representation.
src/components/pastor/hooks/useParticipants.ts	unwrapStoredValue	Named module function supporting unwrap stored value.
src/components/pastor/hooks/useParticipants.ts	extractResponseValue	Extracts response value from a broader record.
src/components/pastor/hooks/useParticipants.ts	safeFirebaseKey	Named module function supporting safe firebase key.
src/components/pastor/hooks/useParticipants.ts	getFirstName	Retrieves or derives first name.
src/components/pastor/hooks/useParticipants.ts	getPrimaryGift	Retrieves or derives primary gift.
src/components/pastor/hooks/useParticipants.ts	parseParticipants	Parses and validates participants.
src/components/pastor/hooks/usePastorCalendarData.ts	usePastorCalendarData	React hook coordinating pastor calendar data state, data access, and UI actions.
src/components/pastor/hooks/usePastorCalendarData.ts	handleScroll	UI/event handler for scroll.
src/components/pastor/hooks/usePeopleDevelopment.ts	usePeopleDevelopment	React hook coordinating people development state, data access, and UI actions.
src/components/pastor/hooks/usePeopleDevelopment.ts	closeAssignmentsPopup	Closes or resets assignments popup UI state.
src/components/pastor/hooks/usePeopleDevelopment.ts	closePeopleNotePopup	Closes or resets people note popup UI state.
src/components/pastor/hooks/usePeopleDevelopment.ts	resetCombinedAssignmentFileInput	Named module function supporting reset combined assignment file input.
src/components/pastor/hooks/usePeopleDevelopment.ts	clearCombinedAssignmentFile	Named module function supporting clear combined assignment file.
src/components/pastor/hooks/usePeopleDevelopment.ts	selectAllCombinedAssignmentGroups	Named module function supporting select all combined assignment groups.

Source module	Function	Reference role
src/components/pastor/hooks/usePeopleDevelopment.ts	clearCombinedAssignmentGroups	Named module function supporting clear combined assignment groups.
src/components/pastor/hooks/usePeopleDevelopment.ts	postCombinedAssignment	Named module function supporting post combined assignment.
src/components/pastor/hooks/usePeopleDevelopmentMeetingSchedules.ts	usePeopleDevelopmentMeetingSchedules	React hook coordinating people development meeting schedules state, data access, and UI actions.
src/components/pastor/hooks/usePeopleDevelopmentMeetingSchedules.ts	createDefaultDraft	Creates default draft and/or its persisted representation.
src/components/pastor/hooks/usePeopleDevelopmentMeetingSchedules.ts	startCreatingPeopleDevelopmentMeetingSchedule	Starts creating people development meeting schedule workflow.
src/components/pastor/hooks/usePeopleDevelopmentMeetingSchedules.ts	cancelEditingPeopleDevelopmentMeetingSchedule	Named module function supporting cancel editing people development meeting schedule.
src/components/pastor/hooks/usePeopleDevelopmentMeetingSchedules.ts	saveCurrentPeopleDevelopmentMeetingSchedule	Saves current people development meeting schedule state or persistence changes.
src/components/pastor/meeting-requests/MeetingRequestsSection.tsx	MeetingRequestsSection	Named module function supporting meeting requests section.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	normalizeMeetingRequesterLocale	Normalizes meeting requester locale into the module's canonical representation.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	getMeetingRequestId	Retrieves or derives meeting request id.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	getMeetingRequestDerivedDetails	Retrieves or derives meeting request derived details.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	getPendingMeetingRequests	Retrieves or derives pending meeting requests.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	findMeetingRequest	Finds meeting request from normalized or persisted data.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	buildMeetingFromRequest	Builds meeting from request from validated or normalized inputs.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	requestHasUsableEmail	Named module function supporting request has usable email.
src/components/pastor/meeting-requests/meetingRequests.utils.ts	getMeetingRequestCount	Retrieves or derives meeting request count.
src/components/pastor/nextgen/NextGenQaSessionsAdmin.tsx	NextGenQaSessionsAdmin	Named module function supporting next gen qa sessions admin.
src/components/pastor/nextgen/NextGenQaSessionsAdmin.tsx	createSession	Creates session and/or its persisted representation.
src/components/pastor/nextgen/NextGenQaSessionsAdmin.tsx	setStatus	Named module function supporting set status.
src/components/pastor/nextgen/NextGenQaSessionsAdmin.tsx	reviewParticipant	Named module function supporting review participant.
src/components/pastor/nextgen/NextGenQaSessionsAdmin.tsx	setDiscussionSelection	Named module function supporting set discussion selection.
src/components/pastor/nextgen/NextGenQuestionsSection.tsx	NextGenQuestionsSection	Named module function supporting next gen questions section.
src/components/pastor/nextgen/NextGenQuestionsSection.tsx	getQuestionStatusLabel	Retrieves or derives question status label.
src/components/pastor/nextgen/NextGenQuestionsSection.tsx	getSelectionButtonLabel	Retrieves or derives selection button label.
src/components/pastor/nextgen/NextGenRegistrationsSection.tsx	NextGenRegistrationsSection	Named module function supporting next gen registrations section.
src/components/pastor/nextgen/NextGenRegistrationsSection.tsx	getStatusLabel	Retrieves or derives status label.
src/components/pastor/nextgen/NextGenRegistrationsSection.tsx	getStatusClasses	Retrieves or derives status classes.
src/components/pastor/nextgen/NextGenRegistrationsSection.tsx	getCreatedLabel	Retrieves or derives created label.
src/components/pastor/nextgen/NextGenSurveyResultsSection.tsx	NextGenSurveyResultsSection	Named module function supporting next gen survey results section.
src/components/pastor/nextgen/NextGenSurveyResultsSection.tsx	getBinaryQuestionText	Retrieves or derives binary question text.
src/components/pastor/nextgen/NextGenSurveyResultsSection.tsx	getBinaryOptionText	Retrieves or derives binary option text.
src/components/pastor/nextgen/NextGenSurveyResultsSection.tsx	getRatingQuestionText	Retrieves or derives rating question text.
src/components/pastor/nextgen/nextgen.actions.ts	updateNextGenRegistrationStatus	Updates next gen registration status while applying module validation and state rules.
src/components/pastor/nextgen/nextgen.actions.ts	updateNextGenQuestionSelection	Updates next gen question selection while applying module validation and state rules.
src/components/pastor/nextgen/nextgen.firebase.ts	subscribeToNextGenQuestions	Subscribes to to next gen questions updates and exposes normalized changes.
src/components/pastor/nextgen/nextgen.firebase.ts	subscribeToNextGenRegistrations	Subscribes to to next gen registrations updates and exposes normalized changes.
src/components/pastor/nextgen/nextgen.firebase.ts	subscribeToNextGenSurveyResults	Subscribes to to next gen survey results updates and exposes normalized changes.
src/components/pastor/nextgen/nextgen.firebase.ts	updateNextGenRegistrationRecord	Updates next gen registration record while applying module validation and state rules.
src/components/pastor/nextgen/nextgen.firebase.ts	createNextGenRegistrationReviewLog	Creates next gen registration review log and/or its persisted representation.
src/components/pastor/nextgen/nextgen.firebase.ts	updateNextGenQuestionRecord	Updates next gen question record while applying module validation and state rules.
src/components/pastor/nextgen/nextgen.utils.ts	normalizeNextGenRegistrationStatus	Normalizes next gen registration status into the module's canonical representation.
src/components/pastor/nextgen/nextgen.utils.ts	normalizeNextGenRegistration	Normalizes next gen registration into the module's canonical representation.
src/components/pastor/nextgen/nextgen.utils.ts	normalizeDuplicateIdentityValue	Normalizes duplicate identity value into the module's canonical representation.
src/components/pastor/nextgen/nextgen.utils.ts	normalizeNextGenQuestion	Normalizes next gen question into the module's canonical representation.
src/components/pastor/nextgen/nextgen.utils.ts	sortNextGenQuestions	Named module function supporting sort next gen questions.
src/components/pastor/nextgen/nextgen.utils.ts	isNextGenQuestionSelected	Predicate evaluating whether next gen question selected.
src/components/pastor/nextgen/nextgen.utils.ts	getSelectedNextGenQuestions	Retrieves or derives selected next gen questions.
src/components/pastor/nextgen/nextgen.utils.ts	getSelectableNextGenQuestions	Retrieves or derives selectable next gen questions.
src/components/pastor/nextgen/nextgen.utils.ts	sortNextGenRegistrations	Named module function supporting sort next gen registrations.
src/components/pastor/nextgen/nextgen.utils.ts	buildNextGenRegistrationDuplicateCounts	Builds next gen registration duplicate counts from validated or normalized inputs.
src/components/pastor/nextgen/nextgen.utils.ts	nextGenRegistrationHasDuplicate	Named module function supporting next gen registration has duplicate.
src/components/pastor/nextgen/nextgen.utils.ts	getDuplicateNextGenRegistrations	Retrieves or derives duplicate next gen registrations.
src/components/pastor/nextgen/nextgen.utils.ts	getNextGenRegistrationsByStatus	Retrieves or derives next gen registrations by status.
src/components/pastor/nextgen/nextgen.utils.ts	filterAndSortNextGenRegistrations	Named module function supporting filter and sort next gen registrations.
src/components/pastor/nextgen/nextgen.utils.ts	createEmptyNextGenSurveyResults	Creates empty next gen survey results and/or its persisted representation.

Source module	Function	Reference role
src/components/pastor/nextgen/nextgen.utils.ts	normalizeSurveyBinaryAnswer	Normalizes survey binary answer into the module's canonical representation.
src/components/pastor/nextgen/nextgen.utils.ts	normalizeSurveyRating	Normalizes survey rating into the module's canonical representation.
src/components/pastor/nextgen/nextgen.utils.ts	buildNextGenSurveyAggregateResults	Builds next gen survey aggregate results from validated or normalized inputs.
src/components/pastor/nextgen/nextgen.utils.ts	getSurveyPercentage	Retrieves or derives survey percentage.
src/components/pastor/nextgen/nextgen.utils.ts	formatSurveyPercentage	Formats survey percentage for display or transport.
src/components/pastor/nextgen/nextgen.utils.ts	normalizeNumber	Normalizes number into the module's canonical representation.
src/components/pastor/nextgen/nextgen.utils.ts	asRecord	Named module function supporting as record.
src/components/pastor/people-development/PeopleAssignmentsCalendarModal.tsx	PeopleAssignmentsCalendarModal	Named module function supporting people assignments calendar modal.
src/components/pastor/people-development/PeopleAssignmentsCalendarModal.tsx	downloadBase64Attachment	Downloads or prepares base64 attachment for the user.
src/components/pastor/people-development/PeopleDevelopmentCombinedPostPanel.tsx	PeopleDevelopmentCombinedPostPanel	Named module function supporting people development combined post panel.
src/components/pastor/people-development/PeopleDevelopmentGroupPanel.tsx	PeopleDevelopmentGroupPanel	Named module function supporting people development group panel.
src/components/pastor/people-development/PeopleDevelopmentGroupPanel.tsx	getPostCountLabel	Retrieves or derives post count label.
src/components/pastor/people-development/PeopleDevelopmentGroupPanel.tsx	getMemberCountLabel	Retrieves or derives member count label.
src/components/pastor/people-development/PeopleDevelopmentMeetingSchedulesSection.tsx	PeopleDevelopmentMeetingSchedulesSection	Named module function supporting people development meeting schedules section.
src/components/pastor/people-development/PeopleDevelopmentMeetingSchedulesSection.tsx	formatOccurrenceDate	Formats occurrence date for display or transport.
src/components/pastor/people-development/PeopleDevelopmentMeetingsCalendar.tsx	PeopleDevelopmentMeetingsCalendar	Named module function supporting people development meetings calendar.
src/components/pastor/people-development/PeopleDevelopmentMeetingsCalendar.tsx	buildCalendarDays	Builds calendar days from validated or normalized inputs.
src/components/pastor/people-development/PeopleDevelopmentSection.tsx	PeopleDevelopmentSection	Named module function supporting people development section.
src/components/pastor/people-development/PeopleDevelopmentSection.tsx	getParticipantDisplayName	Retrieves or derives participant display name.
src/components/pastor/people-development/PeopleDevelopmentSection.tsx	getPersonalNoteCountLabel	Retrieves or derives personal note count label.
src/components/pastor/people-development/PeopleDevelopmentSection.tsx	closeGroupPopup	Closes or resets group popup UI state.
src/components/pastor/people-development/PeoplePersonalNoteModal.tsx	PeoplePersonalNoteModal	Named module function supporting people personal note modal.
src/components/pastor/people-development/PeoplePersonalNoteModal.tsx	getNoteTypeLabel	Retrieves or derives note type label.
src/components/pastor/people-development/peopleDevelopment.actions.ts	assignPersonToPeopleDevelopmentGroup	Assigns person to people development group using the domain's update rules.
src/components/pastor/people-development/peopleDevelopment.actions.ts	savePeoplePersonalNote	Saves people personal note state or persistence changes.
src/components/pastor/people-development/peopleDevelopment.actions.ts	postPeopleDevelopmentAssignment	Named module function supporting post people development assignment.
src/components/pastor/people-development/peopleDevelopment.actions.ts	removePeopleDevelopmentAssignment	Removes people development assignment from UI state or persistent storage.
src/components/pastor/people-development/peopleDevelopment.actions.ts	removePeoplePersonalNote	Removes people personal note from UI state or persistent storage.
src/components/pastor/people-development/peopleDevelopment.actions.ts	savePeopleDevelopmentMeetingSchedule	Saves people development meeting schedule state or persistence changes.
src/components/pastor/people-development/peopleDevelopment.actions.ts	removePeopleDevelopmentMeetingSchedule	Removes people development meeting schedule from UI state or persistent storage.
src/components/pastor/people-development/peopleDevelopment.actions.ts	setPeopleDevelopmentMeetingScheduleActive	Named module function supporting set people development meeting schedule active.
src/components/pastor/people-development/peopleDevelopment.actions.ts	getLocalDateKey	Retrieves or derives local date key.
src/components/pastor/people-development/peopleDevelopment.actions.ts	normalizeSourceKeys	Normalizes source keys into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.actions.ts	validatePerson	Validates person against feature rules.
src/components/pastor/people-development/peopleDevelopment.actions.ts	validatePeopleDevelopmentMeetingScheduleDraft	Validates people development meeting schedule draft against feature rules.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeopleDevelopmentGroupLabel	Retrieves or derives people development group label.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getParticipantPeopleDevelopmentGroup	Retrieves or derives participant people development group.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeopleDevelopmentGroupParticipants	Retrieves or derives people development group participants.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeopleDevelopmentGroupAssignments	Retrieves or derives people development group assignments.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeopleAssignmentDateKey	Retrieves or derives people assignment date key.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeopleAssignmentsInMonth	Retrieves or derives people assignments in month.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	groupPeopleAssignmentsByDate	Named module function supporting group people assignments by date.

Source module	Function	Reference role
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeoplePersonalNotesForParticipant	Retrieves or derives people personal notes for participant.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	searchPeopleDevelopmentParticipants	Named module function supporting search people development participants.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeopleDevelopmentEmailRecipients	Retrieves or derives people development email recipients.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getPeopleDevelopmentMeetingSchedulesForGroup	Retrieves or derives people development meeting schedules for group.
src/components/pastor/people-development/peopleDevelopment.selectors.ts	getActivePeopleDevelopmentMeetingSchedules	Retrieves or derives active people development meeting schedules.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeopleDevelopmentGroup	Normalizes people development group into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeopleDevelopmentGroups	Normalizes people development groups into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeoplePersonalNoteType	Normalizes people personal note type into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	extractPeopleDevelopmentGroup	Extracts people development group from a broader record.
src/components/pastor/people-development/peopleDevelopment.utils.ts	getPeopleDevelopmentStaticGroupLabel	Retrieves or derives people development static group label.
src/components/pastor/people-development/peopleDevelopment.utils.ts	formatFileSize	Formats file size for display or transport.
src/components/pastor/people-development/peopleDevelopment.utils.ts	readFileAsBase64	Named module function supporting read file as base64.
src/components/pastor/people-development/peopleDevelopment.utils.ts	isUsableEmail	Predicate evaluating whether usable email.
src/components/pastor/people-development/peopleDevelopment.utils.ts	truncateEmailText	Named module function supporting truncate email text.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeopleDevelopmentMeetingAudience	Normalizes people development meeting audience into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeopleDevelopmentMeetingOrdinal	Normalizes people development meeting ordinal into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeopleDevelopmentMeetingWeekday	Normalizes people development meeting weekday into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeopleDevelopmentMeetingTime	Normalizes people development meeting time into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	normalizePeopleDevelopmentDateKey	Normalizes people development date key into the module's canonical representation.
src/components/pastor/people-development/peopleDevelopment.utils.ts	getPeopleDevelopmentLocalDateKey	Retrieves or derives people development local date key.
src/components/pastor/people-development/peopleDevelopment.utils.ts	getPeopleDevelopmentMeetingScheduleTitle	Retrieves or derives people development meeting schedule title.
src/components/pastor/people-development/peopleDevelopment.utils.ts	getPeopleDevelopmentMeetingRecurrenceLabel	Retrieves or derives people development meeting recurrence label.
src/components/pastor/people-development/peopleDevelopment.utils.ts	formatPeopleDevelopmentMeetingTime	Formats people development meeting time for display or transport.
src/components/pastor/people-development/peopleDevelopment.utils.ts	getMonthlyOrdinalWeekdayDate	Retrieves or derives monthly ordinal weekday date.
src/components/pastor/people-development/peopleDevelopment.utils.ts	isPeopleDevelopmentMeetingOccurrenceInRange	Predicate evaluating whether people development meeting occurrence in range.
src/components/pastor/people-development/peopleDevelopment.utils.ts	getPeopleDevelopmentMeetingOccurrencesForMonth	Retrieves or derives people development meeting occurrences for month.
src/components/pastor/people-development/peopleDevelopment.utils.ts	getNextPeopleDevelopmentMeetingOccurrence	Retrieves or derives next people development meeting occurrence.
src/components/pastor/people-development/peopleDevelopment.utils.ts	addMeetingMinutes	Named module function supporting add meeting minutes.
src/components/pastor/people-development/peopleDevelopmentEmail.ts	buildPeopleDevelopmentAssignmentNotificationEmailHtml	Builds people development assignment notification email html from validated or normalized inputs.
src/components/pastor/people-development/peopleDevelopmentEmail.ts	escapeHtml	Named module function supporting escape html.
src/components/people-notes/AddDevelopmentItemForm.tsx	AddDevelopmentItemForm	Named module function supporting add development item form.
src/components/people-notes/AddPersonPanel.tsx	AddPersonPanel	Named module function supporting add person panel.
src/components/people-notes/DevelopmentItemCard.tsx	DevelopmentItemCard	Named module function supporting development item card.
src/components/people-notes/DevelopmentItemSection.tsx	DevelopmentItemSection	Named module function supporting development item section.
src/components/people-notes/PeopleList.tsx	PeopleList	Named module function supporting people list.
src/components/people-notes/PeopleNotesHeader.tsx	PeopleNotesHeader	Named module function supporting people notes header.
src/components/people-notes/SelectedPersonDetails.tsx	SelectedPersonDetails	Named module function supporting selected person details.
src/components/people-notes/SelectedPersonHeader.tsx	SelectedPersonHeader	Named module function supporting selected person header.
src/components/people-notes/peopleNotes.utils.ts	todayDateString	Named module function supporting today date string.
src/components/people-notes/peopleNotes.utils.ts	getErrorMessage	Retrieves or derives error message.
src/components/people-notes/peopleNotes.utils.ts	formatDateLabel	Formats date label for display or transport.
src/components/people-notes/peopleNotes.utils.ts	formatDateTimeLabel	Formats date time label for display or transport.
src/components/people-notes/usePeopleNotes.ts	usePeopleNotes	React hook coordinating people notes state, data access, and UI actions.
src/components/people-notes/usePeopleNotesData.ts	usePeopleNotesData	React hook coordinating people notes data state, data access, and UI actions.
src/components/people-notes/usePeopleNotesMutations.ts	usePeopleNotesMutations	React hook coordinating people notes mutations state, data access, and UI actions.
src/components/people-notes/usePeopleNotesMutations.ts	handleAddPerson	UI/event handler for add person.
src/components/people-notes/usePeopleNotesMutations.ts	handleAddItem	UI/event handler for add item.
src/components/people-notes/usePeopleNotesMutations.ts	handleAddComment	UI/event handler for add comment.
src/components/people-notes/usePeopleNotesMutations.ts	handleUpdateFollowUpDate	UI/event handler for update follow up date.

Source module	Function	Reference role
src/components/people-notes/usePeopleNotesMutations.ts	handleDeletePerson	UI/event handler for delete person.
src/components/people-notes/usePeopleNotesMutations.ts	handleDeleteItem	UI/event handler for delete item.
src/components/people-notes/usePeopleNotesMutations.ts	handleDeleteComment	UI/event handler for delete comment.
src/components/tutorial-builder/TutorialBuilderSection.tsx	TutorialBuilderSection	Named module function supporting tutorial builder section.
src/components/tutorial-builder/TutorialBuilderSection.tsx	openCreate	Opens or initializes create UI state.
src/components/tutorial-builder/TutorialBuilderSection.tsx	openEdit	Opens or initializes edit UI state.
src/components/tutorial-builder/TutorialBuilderSection.tsx	handleSave	UI/event handler for save.
src/components/tutorial-builder/TutorialBuilderSection.tsx	handleDelete	UI/event handler for delete.
src/components/tutorial-builder/TutorialBuilderSection.tsx	handleDuplicate	UI/event handler for duplicate.
src/components/tutorial-builder/TutorialBuilderSection.tsx	handlePublishToggle	UI/event handler for publish toggle.
src/components/tutorial-builder/TutorialBuilderSection.tsx	handlePreview	UI/event handler for preview.
src/components/tutorial-builder/TutorialContext.tsx	TutorialProvider	Named module function supporting tutorial provider.
src/components/tutorial-builder/TutorialContext.tsx	useTutorials	React hook coordinating tutorials state, data access, and UI actions.
src/components/tutorial-builder/TutorialEditorModal.tsx	TutorialEditorModal	Named module function supporting tutorial editor modal.
src/components/tutorial-builder/TutorialEditorModal.tsx	copyFor	Named module function supporting copy for.
src/components/tutorial-builder/TutorialEditorModal.tsx	getInsertIndexAfterCurrentStep	Retrieves or derives insert index after current step.
src/components/tutorial-builder/TutorialEditorModal.tsx	beginAddStep	Named module function supporting begin add step.
src/components/tutorial-builder/TutorialEditorModal.tsx	addInformationStep	Named module function supporting add information step.
src/components/tutorial-builder/TutorialEditorModal.tsx	deleteStep	Deletes step using the owning feature boundary.
src/components/tutorial-builder/TutorialEditorModal.tsx	duplicateStep	Named module function supporting duplicate step.
src/components/tutorial-builder/TutorialEditorModal.tsx	moveStep	Named module function supporting move step.
src/components/tutorial-builder/TutorialEditorModal.tsx	handleTargetSelect	UI/event handler for target select.
src/components/tutorial-builder/TutorialEditorModal.tsx	validate	Validates against feature rules.
src/components/tutorial-builder/TutorialEditorModal.tsx	handleSave	UI/event handler for save.
src/components/tutorial-builder/TutorialEditorModal.tsx	handlePreview	UI/event handler for preview.
src/components/tutorial-builder/TutorialLibrary.tsx	TutorialLibrary	Named module function supporting tutorial library.
src/components/tutorial-builder/TutorialPlayer.tsx	TutorialPlayer	Named module function supporting tutorial player.
src/components/tutorial-builder/TutorialPlayer.tsx	TutorialStepContent	Named module function supporting tutorial step content.
src/components/tutorial-builder/TutorialPlayer.tsx	clearSimulatedTarget	Named module function supporting clear simulated target.
src/components/tutorial-builder/TutorialSimulationLayer.tsx	TutorialSimulationLayer	Named module function supporting tutorial simulation layer.
src/components/tutorial-builder/TutorialTargetPicker.tsx	TutorialTargetPicker	Named module function supporting tutorial target picker.
src/components/tutorial-builder/TutorialTargetPicker.tsx	handlePointerMove	UI/event handler for pointer move.
src/components/tutorial-builder/TutorialTargetPicker.tsx	handleClick	UI/event handler for click.
src/components/tutorial-builder/TutorialTargetPicker.tsx	handleKeyDown	UI/event handler for key down.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	subscribeToTutorials	Subscribes to to tutorials updates and exposes normalized changes.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	saveTutorial	Saves tutorial state or persistence changes.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	deleteTutorial	Deletes tutorial using the owning feature boundary.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	setTutorialPublished	Named module function supporting set tutorial published.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	duplicateTutorial	Named module function supporting duplicate tutorial.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	subscribeToTutorialProgress	Subscribes to to tutorial progress updates and exposes normalized changes.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	markTutorialStarted	Named module function supporting mark tutorial started.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	updateTutorialProgress	Updates tutorial progress while applying module validation and state rules.
src/components/tutorial-builder/tutorialBuilder.firebase.ts	markTutorialCompleted	Named module function supporting mark tutorial completed.
src/components/tutorial-builder/tutorialBuilder.utils.ts	createTutorialStep	Creates tutorial step and/or its persisted representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	createTutorialDraft	Creates tutorial draft and/or its persisted representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	tutorialToDraft	Named module function supporting tutorial to draft.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizeTutorialStep	Normalizes tutorial step into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizeTutorial	Normalizes tutorial into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizeTutorialSteps	Normalizes tutorial steps into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	validateTutorialDraft	Validates tutorial draft against feature rules.
src/components/tutorial-builder/tutorialBuilder.utils.ts	tutorialMatchesAudience	Named module function supporting tutorial matches audience.
src/components/tutorial-builder/tutorialBuilder.utils.ts	getTutorialElementLabel	Retrieves or derives tutorial element label.
src/components/tutorial-builder/tutorialBuilder.utils.ts	getTutorialSelector	Retrieves or derives tutorial selector.
src/components/tutorial-builder/tutorialBuilder.utils.ts	isTutorialCandidateVisible	Predicate evaluating whether tutorial candidate visible.
src/components/tutorial-builder/tutorialBuilder.utils.ts	scanTutorialTargets	Named module function supporting scan tutorial targets.
src/components/tutorial-builder/tutorialBuilder.utils.ts	waitForTutorialTarget	Named module function supporting wait for tutorial target.

Source module	Function	Reference role
src/components/tutorial-builder/tutorialBuilder.utils.ts	getCurrentRoute	Retrieves or derives current route.
src/components/tutorial-builder/tutorialBuilder.utils.ts	getTutorialCategoryMap	Retrieves or derives tutorial category map.
src/components/tutorial-builder/tutorialBuilder.utils.ts	asRecord	Named module function supporting as record.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizeNumber	Normalizes number into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizeBoolean	Normalizes boolean into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizeAudience	Normalizes audience into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizeAction	Normalizes action into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	normalizePlacement	Normalizes placement into the module's canonical representation.
src/components/tutorial-builder/tutorialBuilder.utils.ts	escapeAttributeValue	Named module function supporting escape attribute value.
src/components/tutorial-builder/tutorialBuilder.utils.ts	getElementPositionSelector	Retrieves or derives element position selector.
src/firebase.ts	signInWithGoogle	Named module function supporting sign in with google.
src/firebase.ts	signUpWithEmail	Named module function supporting sign up with email.
src/firebase.ts	signInWithEmail	Named module function supporting sign in with email.
src/hooks/usePastorAccess.ts	usePastorAccess	React hook coordinating pastor access state, data access, and UI actions.
src/i18n/index.tsx	I18nProvider	Named module function supporting i18n provider.
src/i18n/index.tsx	useI18n	React hook coordinating i18n state, data access, and UI actions.
src/lib/firebase.ts	handleFirestoreError	UI/event handler for firestore error.
src/pages/AboutUs.tsx	AboutUs	Named module function supporting about us.
src/pages/BookingCalendar.tsx	BookingCalendar	Named module function supporting booking calendar.
src/pages/BookingCalendar.tsx	timeToHour	Named module function supporting time to hour.
src/pages/BookingCalendar.tsx	hourToLabel	Named module function supporting hour to label.
src/pages/BookingCalendar.tsx	hourToTime	Named module function supporting hour to time.
src/pages/BookingCalendar.tsx	handleDayClick	UI/event handler for day click.
src/pages/BookingCalendar.tsx	handleSlotClick	UI/event handler for slot click.
src/pages/BookingCalendar.tsx	handlePopupSlotClick	UI/event handler for popup slot click.
src/pages/BookingCalendar.tsx	handleSubmit	UI/event handler for submit.
src/pages/BookingCalendar.tsx	focusBezaelDate	Named module function supporting focus bezael date.
src/pages/CongregationGroupNotes.tsx	CongregationGroupNotes	Named module function supporting congregation group notes.
src/pages/GuidePage.tsx	GuidePage	Named module function supporting guide page.
src/pages/LandingPage.tsx	LandingPage	Named module function supporting landing page.
src/pages/NextGenActivities.tsx	NextGenActivities	Named module function supporting next gen activities.
src/pages/NextGenActivities.tsx	PortalButton	Named module function supporting portal button.
src/pages/PeopleNotesPage.tsx	PeopleNotesPage	Named module function supporting people notes page.
src/pages/PrivacyPolicy.tsx	PrivacyPolicy	Named module function supporting privacy policy.
src/pages/TermsOfService.tsx	TermsOfService	Named module function supporting terms of service.
src/services/administrator.test.ts	apiResponse	Named module function supporting api response.
src/services/administrator.ts	getAdministratorSession	Retrieves or derives administrator session.
src/services/administrator.ts	saveAdministratorAuthority	Saves administrator authority state or persistence changes.
src/services/administrator.ts	suspendAdministrator	Named module function supporting suspend administrator.
src/services/administrator.ts	getArchiveFolders	Retrieves or derives archive folders.
src/services/administrator.ts	createArchiveFolder	Creates archive folder and/or its persisted representation.
src/services/administrator.ts	deleteArchiveFolder	Deletes archive folder using the owning feature boundary.
src/services/administrator.ts	getArchiveFiles	Retrieves or derives archive files.
src/services/administrator.ts	uploadArchiveFile	Uploads archive file through the configured storage workflow.
src/services/administrator.ts	completeArchiveFileUpload	Named module function supporting complete archive file upload.
src/services/administrator.ts	getArchiveFileDownloadUrl	Retrieves or derives archive file download url.
src/services/administrator.ts	deleteArchiveFile	Deletes archive file using the owning feature boundary.
src/services/administrator.ts	send	Sends through the configured communication or API boundary.
src/services/assessment.ts	getAssessmentFormStates	Retrieves or derives assessment form states.
src/services/assessment.ts	submitAssessment	Named module function supporting submit assessment.
src/services/assessment.ts	submitDirectAssessmentSignup	Named module function supporting submit direct assessment signup.
src/services/assessment.ts	getAssessmentResponses	Retrieves or derives assessment responses.
src/services/assessment.ts	updateAssessmentLinkage	Updates assessment linkage while applying module validation and state rules.
src/services/assessment.ts	deleteAssessmentResponse	Deletes assessment response using the owning feature boundary.
src/services/assessment.ts	sendAssessmentIdentifierEmail	Sends assessment identifier email through the configured communication or API boundary.
src/services/assessment.ts	updateAssessmentFormState	Updates assessment form state while applying module validation and state rules.

Source module	Function	Reference role
src/services/assessment.ts	send	Sends through the configured communication or API boundary.
src/services/bezalel.ts	chatWithPastorBezalel	Named module function supporting chat with pastor bezalel.
src/services/bezalel.ts	chatWithBookingBezalel	Named module function supporting chat with booking bezalel.
src/services/bezalel.ts	send	Sends through the configured communication or API boundary.
src/services/booking.ts	getPublicBookingSchedule	Retrieves or derives public booking schedule.
src/services/booking.ts	createPublicBooking	Creates public booking and/or its persisted representation.
src/services/congregationGroupNotes.ts	getCongregationGroupAccess	Retrieves or derives congregation group access.
src/services/gmail.ts	handleOAuthCallback	UI/event handler for oauth callback.
src/services/gmail.ts	startGoogleAuth	Starts google auth workflow.
src/services/gmail.ts	getStoredTokens	Retrieves or derives stored tokens.
src/services/gmail.ts	storeTokens	Named module function supporting store tokens.
src/services/gmail.ts	clearTokens	Named module function supporting clear tokens.
src/services/gmail.ts	sendGmailEmail	Sends gmail email through the configured communication or API boundary.
src/services/gmail.ts	createCalendarMeetLink	Creates calendar meet link and/or its persisted representation.
src/services/gmail.ts	generatePlaceholderLink	Generates placeholder link from controlled inputs.
src/services/gmail.ts	sendEmailViaEmailJS	Sends email via email js through the configured communication or API boundary.
src/services/gmail.ts	sendMeetingConfirmationViaEmailJS	Sends meeting confirmation via email js through the configured communication or API boundary.
src/services/gmail.ts	b64EncodeUnicode	Named module function supporting b64 encode unicode.
src/services/gmail.ts	resolveEmailJSTemplateId	Resolves email jstemplate id from available inputs.
src/services/meetingInvitations.ts	sendMeetingInvitationsViaBackend	Sends meeting invitations via backend through the configured communication or API boundary.
src/services/meetingInvitations.ts	send	Sends through the configured communication or API boundary.
src/services/nextGenMissionMap.ts	getNextGenMissionMap	Retrieves or derives next gen mission map.
src/services/nextGenMissionMap.ts	requestMissionMap	Named module function supporting request mission map.
src/services/nextGenPortal.ts	uploadNextGenFile	Uploads next gen file through the configured storage workflow.
src/services/nextGenPortal.ts	nextGenResponseError	Named module function supporting next gen response error.
src/services/nextGenPortal.ts	getNextGenSessions	Retrieves or derives next gen sessions.
src/services/nextGenPortal.ts	getNextGenSession	Retrieves or derives next gen session.
src/services/nextGenPortal.ts	submitNextGenQuestion	Named module function supporting submit next gen question.
src/services/nextGenPortal.ts	submitNextGenVote	Named module function supporting submit next gen vote.
src/services/nextGenPortal.ts	getPastorNextGenSessions	Retrieves or derives pastor next gen sessions.
src/services/nextGenPortal.ts	createPastorNextGenSession	Creates pastor next gen session and/or its persisted representation.
src/services/nextGenPortal.ts	updatePastorNextGenSession	Updates pastor next gen session while applying module validation and state rules.
src/services/nextGenPortal.ts	updateNextGenQuestionDiscussionSelection	Updates next gen question discussion selection while applying module validation and state rules.
src/services/nextGenPortal.ts	getPastorNextGenSession	Retrieves or derives pastor next gen session.
src/services/nextGenPortal.ts	updateNextGenParticipantStatus	Updates next gen participant status while applying module validation and state rules.
src/services/nextGenPortal.ts	getNextGenFolders	Retrieves or derives next gen folders.
src/services/nextGenPortal.ts	createNextGenFolder	Creates next gen folder and/or its persisted representation.
src/services/nextGenPortal.ts	deleteNextGenFolder	Deletes next gen folder using the owning feature boundary.
src/services/nextGenPortal.ts	getNextGenFiles	Retrieves or derives next gen files.
src/services/nextGenPortal.ts	completeNextGenFile	Named module function supporting complete next gen file.
src/services/nextGenPortal.ts	getNextGenDownloadUrl	Retrieves or derives next gen download url.
src/services/nextGenPortal.ts	deleteNextGenFile	Deletes next gen file using the owning feature boundary.
src/services/nextGenPortal.ts	send	Sends through the configured communication or API boundary.
src/services/pastorAuthorization.ts	getPastorSession	Retrieves or derives pastor session.
src/services/pastorAuthorization.ts	requestPastorSession	Named module function supporting request pastor session.
src/services/pastorCalendar.ts	getPastorCalendarSnapshot	Retrieves or derives pastor calendar snapshot.
src/services/pastorCalendar.ts	createPastorMeeting	Creates pastor meeting and/or its persisted representation.
src/services/pastorCalendar.ts	updatePastorMeeting	Updates pastor meeting while applying module validation and state rules.
src/services/pastorCalendar.ts	deletePastorMeeting	Deletes pastor meeting using the owning feature boundary.
src/services/pastorCalendar.ts	createPastorCalendarBlock	Creates pastor calendar block and/or its persisted representation.
src/services/pastorCalendar.ts	updatePastorCalendarBlock	Updates pastor calendar block while applying module validation and state rules.
src/services/pastorCalendar.ts	deletePastorCalendarBlock	Deletes pastor calendar block using the owning feature boundary.
src/services/pastorCalendar.ts	decidePastorMeetingRequest	Named module function supporting decide pastor meeting request.

Source module	Function	Reference role
src/services/pastorCalendar.ts	downloadPastorCalendarExport	Downloads or prepares pastor calendar export for the user.
src/services/pastorCalendar.ts	pastorAuthorizationHeader	Named module function supporting pastor authorization header.
src/services/pastorCalendar.ts	send	Sends through the configured communication or API boundary.
src/services/peopleDevelopment.test.ts	successfulResponse	Named module function supporting successful response.
src/services/peopleDevelopment.ts	subscribeToPeopleDevelopmentMembers	Subscribes to to people development members updates and exposes normalized changes.
src/services/peopleDevelopment.ts	subscribeToPeopleDevelopmentAssignments	Subscribes to to people development assignments updates and exposes normalized changes.
src/services/peopleDevelopment.ts	subscribeToPeoplePersonalNotes	Subscribes to to people personal notes updates and exposes normalized changes.
src/services/peopleDevelopment.ts	subscribeToPeopleDevelopmentMeetingSchedules	Subscribes to to people development meeting schedules updates and exposes normalized changes.
src/services/peopleDevelopment.ts	assignPeopleDevelopmentMember	Assigns people development member using the domain's update rules.
src/services/peopleDevelopment.ts	createPeopleDevelopmentAssignment	Creates people development assignment and/or its persisted representation.
src/services/peopleDevelopment.ts	deletePeopleDevelopmentAssignment	Deletes people development assignment using the owning feature boundary.
src/services/peopleDevelopment.ts	replacePeopleDevelopmentAssignmentAttachments	Named module function supporting replace people development assignment attachments.
src/services/peopleDevelopment.ts	createPeoplePersonalNote	Creates people personal note and/or its persisted representation.
src/services/peopleDevelopment.ts	deletePeoplePersonalNote	Deletes people personal note using the owning feature boundary.
src/services/peopleDevelopment.ts	createPeopleDevelopmentMeetingSchedule	Creates people development meeting schedule and/or its persisted representation.
src/services/peopleDevelopment.ts	updatePeopleDevelopmentMeetingSchedule	Updates people development meeting schedule while applying module validation and state rules.
src/services/peopleDevelopment.ts	deletePeopleDevelopmentMeetingSchedule	Deletes people development meeting schedule using the owning feature boundary.
src/services/peopleDevelopment.ts	refreshSnapshot	Refreshes snapshot from its authoritative source.
src/services/peopleDevelopment.ts	subscribe	Subscribes to updates and exposes normalized changes.
src/services/peopleDevelopment.ts	send	Sends through the configured communication or API boundary.
src/services/peopleDevelopmentNotifications.ts	sendPeopleDevelopmentNotificationViaBackend	Sends people development notification via backend through the configured communication or API boundary.
src/services/peopleDevelopmentNotifications.ts	createFailedResult	Creates failed result and/or its persisted representation.
src/services/peopleNotes.ts	getPeopleNotes	Retrieves or derives people notes.
src/services/peopleNotes.ts	createPerson	Creates person and/or its persisted representation.
src/services/peopleNotes.ts	createDevelopmentItem	Creates development item and/or its persisted representation.
src/services/peopleNotes.ts	createDevelopmentComment	Creates development comment and/or its persisted representation.
src/services/peopleNotes.ts	updateDevelopmentFollowUpDate	Updates development follow up date while applying module validation and state rules.
src/services/peopleNotes.ts	deletePerson	Deletes person using the owning feature boundary.
src/services/peopleNotes.ts	deleteDevelopmentItem	Deletes development item using the owning feature boundary.
src/services/peopleNotes.ts	deleteDevelopmentComment	Deletes development comment using the owning feature boundary.
src/services/peopleNotes.ts	encodeId	Named module function supporting encode id.
src/services/peopleNotes.ts	send	Sends through the configured communication or API boundary.
backend/src/admin/adminArchives.ts	listArchiveFolders	Lists archive folders from the relevant store.
backend/src/admin/adminArchives.ts	createArchiveFolder	Creates archive folder and/or its persisted representation.
backend/src/admin/adminArchives.ts	deleteArchiveFolder	Deletes archive folder using the owning feature boundary.
backend/src/admin/adminArchives.ts	listArchiveFiles	Lists archive files from the relevant store.
backend/src/admin/adminArchives.ts	getArchiveFile	Retrieves or derives archive file.
backend/src/admin/adminArchives.ts	createPendingArchiveFile	Creates pending archive file and/or its persisted representation.
backend/src/admin/adminArchives.ts	completeArchiveFile	Named module function supporting complete archive file.
backend/src/admin/adminArchives.ts	deleteArchiveFileMetadata	Deletes archive file metadata using the owning feature boundary.
backend/src/admin/adminArchives.ts	createArchiveObjectKey	Creates archive object key and/or its persisted representation.
backend/src/admin/adminArchives.ts	hasInvalidArchiveNameCharacter	Predicate evaluating whether invalid archive name character is present or valid.
backend/src/admin/adminArchives.ts	normalizeArchiveFolders	Normalizes archive folders into the module's canonical representation.
backend/src/admin/adminArchives.ts	normalizeArchiveFiles	Normalizes archive files into the module's canonical representation.
backend/src/admin/adminArchives.ts	normalizeArchiveFolder	Normalizes archive folder into the module's canonical representation.
backend/src/admin/adminArchives.ts	normalizeArchiveFile	Normalizes archive file into the module's canonical representation.
backend/src/admin/adminArchives.ts	normalizedContentType	Normalizes d content type into the module's canonical representation.
backend/src/admin/adminArchives.ts	finiteNumber	Named module function supporting finite number.
backend/src/admin/adminAuthorization.ts	getAdminAccount	Retrieves or derives admin account.
backend/src/admin/adminAuthorization.ts	requireAdminAuthority	Enforces the admin authority authorization or validation boundary.
backend/src/admin/adminAuthorization.ts	normalizeAdminAccount	Normalizes admin account into the module's canonical representation.
backend/src/admin/adminAuthorization.ts	numberOrZero	Named module function supporting number or zero.
backend/src/admin/adminAuthorization.ts	optionalNumber	Named module function supporting optional number.

Source module	Function	Reference role
backend/src/assessment/assessment.service.ts	assessmentFormIds	Named module function supporting assessment form ids.
backend/src/assessment/assessment.service.ts	normalizeFormState	Normalizes form state into the module's canonical representation.
backend/src/assessment/assessment.service.ts	buildAssessmentRecord	Builds assessment record from validated or normalized inputs.
backend/src/assessment/assessment.service.ts	buildDirectSignupRecord	Builds direct signup record from validated or normalized inputs.
backend/src/assessment/assessment.service.ts	normalizeAssessmentResponses	Normalizes assessment responses into the module's canonical representation.
backend/src/assessment/assessment.service.ts	extractAssessmentResponseDetails	Extracts assessment response details from a broader record.
backend/src/assessment/assessment.service.ts	validateAndNormalizeAnswers	Validates and normalize answers against feature rules.
backend/src/assessment/assessment.service.ts	buildStoredFields	Builds stored fields from validated or normalized inputs.
backend/src/assessment/assessment.service.ts	rank	Ranks according to feature scoring rules.
backend/src/assessment/assessment.service.ts	responseBelongsToForm	Named module function supporting response belongs to form.
backend/src/assessment/assessment.service.ts	extractValue	Extracts value from a broader record.
backend/src/assessment/assessment.service.ts	easternTime	Named module function supporting eastern time.
backend/src/assessment/assessment.service.ts	ratingFields	Named module function supporting rating fields.
backend/src/assessment/assessment.service.ts	localizedResults	Named module function supporting localized results.
backend/src/bezalel/bezalel.ai.ts	translateQaTheme	Named module function supporting translate qa theme.
backend/src/bezalel/bezalel.ai.ts	reviewNextGenQuestion	Named module function supporting review next gen question.
backend/src/bezalel/bezalel.ai.ts	runPastorAgent	Executes the pastor agent workflow.
backend/src/bezalel/bezalel.ai.ts	runPublicBookingAgent	Executes the public booking agent workflow.
backend/src/bezalel/bezalel.ai.ts	objectSchema	Named module function supporting object schema.
backend/src/booking/booking.service.ts	buildPublicBookingSchedule	Builds public booking schedule from validated or normalized inputs.
backend/src/booking/booking.service.ts	isBookingSlotAvailable	Predicate evaluating whether booking slot available.
backend/src/booking/booking.service.ts	hasCalendarConflict	Predicate evaluating whether calendar conflict is present or valid.
backend/src/booking/booking.service.ts	normalizeRecurringGroupMeetings	Normalizes recurring group meetings into the module's canonical representation.
backend/src/booking/booking.service.ts	normalizeReservations	Normalizes reservations into the module's canonical representation.
backend/src/booking/booking.service.ts	omitKey	Named module function supporting omit key.
backend/src/booking/booking.service.ts	eachDate	Named module function supporting each date.
backend/src/booking/booking.service.ts	addMinutes	Named module function supporting add minutes.
backend/src/booking/booking.service.ts	normalizeBlocks	Normalizes blocks into the module's canonical representation.
backend/src/booking/booking.service.ts	withoutStatus	Named module function supporting without status.
backend/src/booking/booking.service.ts	stringValue	Named module function supporting string value.
backend/src/booking/booking.service.ts	isWithinRange	Predicate evaluating whether within range.
backend/src/booking/booking.service.ts	overlaps	Named module function supporting overlaps.
backend/src/booking/booking.service.ts	sortAndDedupe	Named module function supporting sort and dedupe.
backend/src/calendar/calendar.ics.ts	buildCalendarIcs	Builds calendar ics from validated or normalized inputs.
backend/src/calendar/calendar.ics.ts	formatUtc	Formats utc for display or transport.
backend/src/calendar/calendar.ics.ts	escapeIcs	Named module function supporting escape ics.
backend/src/calendar/calendar.time.ts	calendarTemporalFields	Named module function supporting calendar temporal fields.
backend/src/calendar/calendar.time.ts	zonedDateTimeToEpoch	Named module function supporting zoned date time to epoch.
backend/src/calendar/calendar.time.ts	calendarReservationKey	Named module function supporting calendar reservation key.
backend/src/calendar/calendar.time.ts	timeZoneOffsetMilliseconds	Named module function supporting time zone offset milliseconds.
backend/src/emails/assessment.email.ts	buildAssessmentEmail	Builds assessment email from validated or normalized inputs.
backend/src/emails/assessment.email.ts	buildIdentifierEmail	Builds identifier email from validated or normalized inputs.
backend/src/emails/assessment.email.ts	escapeHtml	Named module function supporting escape html.
backend/src/emails/bookingRequest.email.ts	buildBookingRequestEmail	Builds booking request email from validated or normalized inputs.
backend/src/emails/bookingRequest.email.ts	escapeHtml	Named module function supporting escape html.
backend/src/emails/calendarNotification.email.ts	buildCalendarNotificationEmail	Builds calendar notification email from validated or normalized inputs.
backend/src/emails/calendarNotification.email.ts	englishCopy	Named module function supporting english copy.
backend/src/emails/calendarNotification.email.ts	arabicCopy	Named module function supporting arabic copy.
backend/src/emails/calendarNotification.email.ts	escapeHtml	Named module function supporting escape html.
backend/src/emails/meetingInvitation.email.ts	buildMeetingInvitationEmail	Builds meeting invitation email from validated or normalized inputs.
backend/src/emails/meetingInvitation.email.ts	escapeHtml	Named module function supporting escape html.
backend/src/emails/meetingInvitation.email.ts	parseMeetingDate	Parses and validates meeting date.
backend/src/emails/meetingInvitation.email.ts	formatMeetingDate	Formats meeting date for display or transport.
backend/src/emails/meetingInvitation.email.ts	formatMeetingTime	Formats meeting time for display or transport.
backend/src/emails/meetingInvitation.email.ts	buildEnglishEmail	Builds english email from validated or normalized inputs.
backend/src/emails/meetingInvitation.email.ts	buildArabicEmail	Builds arabic email from validated or normalized inputs.

Source module	Function	Reference role
backend/src/emails/peopleDevelopmentNotification.email.ts	buildPeopleDevelopmentNotificationEmail	Builds people development notification email from validated or normalized inputs.
backend/src/emails/peopleDevelopmentNotification.email.ts	escapeHtml	Named module function supporting escape html.
backend/src/emails/peopleDevelopmentNotification.email.ts	truncateText	Named module function supporting truncate text.
backend/src/emails/peopleDevelopmentNotification.email.ts	formatFileSize	Formats file size for display or transport.
backend/src/index.ts	createApp	Creates app and/or its persisted representation.
backend/src/nextgen/nextGenFiles.ts	listNextGenFolders	Lists next gen folders from the relevant store.
backend/src/nextgen/nextGenFiles.ts	createNextGenFolder	Creates next gen folder and/or its persisted representation.
backend/src/nextgen/nextGenFiles.ts	deleteNextGenFolder	Deletes next gen folder using the owning feature boundary.
backend/src/nextgen/nextGenFiles.ts	listNextGenFiles	Lists next gen files from the relevant store.
backend/src/nextgen/nextGenFiles.ts	getNextGenFile	Retrieves or derives next gen file.
backend/src/nextgen/nextGenFiles.ts	createPendingNextGenFile	Creates pending next gen file and/or its persisted representation.
backend/src/nextgen/nextGenFiles.ts	completeNextGenFile	Named module function supporting complete next gen file.
backend/src/nextgen/nextGenFiles.ts	deleteNextGenFileRecord	Deletes next gen file record using the owning feature boundary.
backend/src/nextgen/nextGenFiles.ts	createNextGenObjectKey	Creates next gen object key and/or its persisted representation.
backend/src/nextgen/nextGenFiles.ts	normalizeFolders	Normalizes folders into the module's canonical representation.
backend/src/nextgen/nextGenFiles.ts	normalizeFiles	Normalizes files into the module's canonical representation.
backend/src/nextgen/nextGenFiles.ts	normalizeFile	Normalizes file into the module's canonical representation.
backend/src/nextgen/nextGenFiles.ts	normalizeContentType	Normalizes content type into the module's canonical representation.
backend/src/nextgen/nextGenFiles.ts	asRecord	Named module function supporting as record.
backend/src/nextgen/nextGenFiles.ts	stringValue	Named module function supporting string value.
backend/src/nextgen/nextGenFiles.ts	numberValue	Named module function supporting number value.
backend/src/nextgen/nextGenPortal.ts	listQaSessions	Lists qa sessions from the relevant store.
backend/src/nextgen/nextGenPortal.ts	ensureLegacyQaSession	Named module function supporting ensure legacy qa session.
backend/src/nextgen/nextGenPortal.ts	getQaSession	Retrieves or derives qa session.
backend/src/nextgen/nextGenPortal.ts	listQaQuestions	Lists qa questions from the relevant store.
backend/src/nextgen/nextGenPortal.ts	createQaSession	Creates qa session and/or its persisted representation.
backend/src/nextgen/nextGenPortal.ts	createQaQuestion	Creates qa question and/or its persisted representation.
backend/src/nextgen/nextGenPortal.ts	recordQaVote	Named module function supporting record qa vote.
backend/src/nextgen/nextGenPortal.ts	getMemberSessionView	Retrieves or derives member session view.
backend/src/nextgen/nextGenPortal.ts	updateQuestionDiscussionSelection	Updates question discussion selection while applying module validation and state rules.
backend/src/nextgen/nextGenPortal.ts	getPastorSessionView	Retrieves or derives pastor session view.
backend/src/nextgen/nextGenPortal.ts	updateParticipantStatus	Updates participant status while applying module validation and state rules.
backend/src/nextgen/nextGenPortal.ts	createEmailVoteKey	Creates email vote key and/or its persisted representation.
backend/src/nextgen/nextGenPortal.ts	normalizeSessions	Normalizes sessions into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	normalizeSession	Normalizes session into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	normalizeVote	Normalizes vote into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	backfillLegacyDiscussionSelection	Named module function supporting backfill legacy discussion selection.
backend/src/nextgen/nextGenPortal.ts	normalizeQuestions	Normalizes questions into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	normalizeQuestion	Normalizes question into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	normalizeParticipants	Normalizes participants into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	normalizeParticipant	Normalizes participant into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	asRecord	Named module function supporting as record.
backend/src/nextgen/nextGenPortal.ts	stringValue	Named module function supporting string value.
backend/src/nextgen/nextGenPortal.ts	numberValue	Named module function supporting number value.
backend/src/nextgen/nextGenPortal.ts	patchIfMissing	Named module function supporting patch if missing.
backend/src/nextgen/nextGenPortal.ts	normalizeTheme	Normalizes theme into the module's canonical representation.
backend/src/nextgen/nextGenPortal.ts	score	Computes or returns score for the domain.
backend/src/pastorCalendar/pastorCalendar.normalize.ts	normalizePastorCalendarSnapshot	Normalizes pastor calendar snapshot into the module's canonical representation.
backend/src/pastorCalendar/pastorCalendar.normalize.ts	normalizeCollection	Normalizes collection into the module's canonical representation.
backend/src/pastorCalendar/pastorCalendar.normalize.ts	compareByDate	Named module function supporting compare by date.
backend/src/pastorCalendar/pastorCalendar.normalize.ts	numberValue	Named module function supporting number value.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	asRecord	Named module function supporting as record.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	normalizeGroup	Normalizes group into the module's canonical representation.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	normalizeMembers	Normalizes members into the module's canonical representation.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	normalizeAssignments	Normalizes assignments into the module's canonical representation.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	normalizePersonalNotes	Normalizes personal notes into the module's canonical representation.

Source module	Function	Reference role
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	normalizeSchedules	Normalizes schedules into the module's canonical representation.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	safeFirebaseKey	Named module function supporting safe firebase key.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	extractResponseValue	Extracts response value from a broader record.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	isGroupId	Predicate evaluating whether group id.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	numberValue	Named module function supporting number value.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	normalizeGroups	Normalizes groups into the module's canonical representation.
backend/src/peopleDevelopment/peopleDevelopment.normalize.ts	normalizeAttachments	Normalizes attachments into the module's canonical representation.
backend/src/peopleNotes/peopleNotes.normalize.ts	normalizePeopleNotes	Normalizes people notes into the module's canonical representation.
backend/src/peopleNotes/peopleNotes.normalize.ts	asRecord	Named module function supporting as record.
backend/src/peopleNotes/peopleNotes.normalize.ts	objectEntries	Named module function supporting object entries.
backend/src/peopleNotes/peopleNotes.normalize.ts	asString	Named module function supporting as string.
backend/src/peopleNotes/peopleNotes.normalize.ts	asNumber	Named module function supporting as number.
backend/src/routes/admin.routes.ts	createAdminRoutes	Creates admin routes and/or its persisted representation.
backend/src/routes/admin.routes.ts	archiveStorage	Named module function supporting archive storage.
backend/src/routes/admin.routes.ts	publicArchiveFile	Named module function supporting public archive file.
backend/src/routes/admin.routes.ts	archiveFileNotFound	Named module function supporting archive file not found.
backend/src/routes/admin.routes.ts	withArchiveAuthority	Named module function supporting with archive authority.
backend/src/routes/admin.routes.ts	withChief	Named module function supporting with chief.
backend/src/routes/admin.routes.ts	withDatabase	Named module function supporting with database.
backend/src/routes/admin.routes.ts	normalizeAccounts	Normalizes accounts into the module's canonical representation.
backend/src/routes/admin.routes.ts	readJson	Named module function supporting read json.
backend/src/routes/admin.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/admin.routes.ts	notFound	Named module function supporting not found.
backend/src/routes/admin.routes.ts	archiveError	Named module function supporting archive error.
backend/src/routes/admin.routes.ts	archiveOperationError	Named module function supporting archive operation error.
backend/src/routes/assessment.routes.ts	createAssessmentRoutes	Creates assessment routes and/or its persisted representation.
backend/src/routes/assessment.routes.ts	sendAssessmentNotifications	Sends assessment notifications through the configured communication or API boundary.
backend/src/routes/assessment.routes.ts	withAssessmentAdmin	Named module function supporting with assessment admin.
backend/src/routes/assessment.routes.ts	withDatabase	Named module function supporting with database.
backend/src/routes/assessment.routes.ts	readJson	Named module function supporting read json.
backend/src/routes/assessment.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/assessment.routes.ts	isEmail	Predicate evaluating whether email.
backend/src/routes/auth.routes.ts	createAuthRoutes	Creates auth routes and/or its persisted representation.
backend/src/routes/bezalel.routes.ts	createBezalelRoutes	Creates bezalel routes and/or its persisted representation.
backend/src/routes/bezalel.routes.ts	pastorCalendarContext	Named module function supporting pastor calendar context.
backend/src/routes/bezalel.routes.ts	publicScheduleContext	Named module function supporting public schedule context.
backend/src/routes/bezalel.routes.ts	safeRecords	Named module function supporting safe records.
backend/src/routes/bezalel.routes.ts	validatePastorAction	Validates pastor action against feature rules.
backend/src/routes/bezalel.routes.ts	validatePublicBooking	Validates public booking against feature rules.
backend/src/routes/bezalel.routes.ts	requiredDatabaseUrl	Enforces the database url authorization or validation boundary.
backend/src/routes/bezalel.routes.ts	agentError	Named module function supporting agent error.
backend/src/routes/bezalel.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/bezalel.routes.ts	readJson	Named module function supporting read json.
backend/src/routes/bezalel.routes.ts	dateInToronto	Named module function supporting date in toronto.
backend/src/routes/bezalel.routes.ts	addDays	Named module function supporting add days.
backend/src/routes/bezalel.routes.ts	createBookingRoutes	Creates booking routes and/or its persisted representation.
backend/src/routes/booking.routes.ts	notifyPastor	Named module function supporting notify pastor.
backend/src/routes/booking.routes.ts	withDatabase	Named module function supporting with database.
backend/src/routes/booking.routes.ts	readJsonBody	Named module function supporting read json body.
backend/src/routes/booking.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/booking.routes.ts	dateInToronto	Named module function supporting date in toronto.
backend/src/routes/meetingInvitations.routes.ts	createMeetingInvitationRoutes	Creates meeting invitation routes and/or its persisted representation.
backend/src/routes/nextGenMissionMap.routes.ts	createNextGenMissionMapRoutes	Creates next gen mission map routes and/or its persisted representation.
backend/src/routes/nextGenMissionMap.routes.ts	mapConfigurationError	Named module function supporting map configuration error.
backend/src/routes/nextGenPortal.routes.ts	createNextGenPortalRoutes	Creates next gen portal routes and/or its persisted representation.
backend/src/routes/nextGenPortal.routes.ts	withMemberSession	Named module function supporting with member session.

Source module	Function	Reference role
backend/src/routes/nextGenPortal.routes.ts	withQaManager	Named module function supporting with qa manager.
backend/src/routes/nextGenPortal.routes.ts	withFile	Named module function supporting with file.
backend/src/routes/nextGenPortal.routes.ts	withDatabase	Named module function supporting with database.
backend/src/routes/nextGenPortal.routes.ts	storage	Named module function supporting storage.
backend/src/routes/nextGenPortal.routes.ts	publicFile	Named module function supporting public file.
backend/src/routes/nextGenPortal.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/nextGenPortal.routes.ts	notFound	Named module function supporting not found.
backend/src/routes/nextGenPortal.routes.ts	readJson	Named module function supporting read json.
backend/src/routes/nextGenPortal.routes.ts	portalError	Named module function supporting portal error.
backend/src/routes/nextGenPortal.routes.ts	geminiPortalError	Named module function supporting gemini portal error.
backend/src/routes/nextGenPortal.routes.ts	fileError	Named module function supporting file error.
backend/src/routes/nextGenPortal.routes.ts	storageOrFileError	Named module function supporting storage or file error.
backend/src/routes/pastorCalendar.routes.ts	createPastorCalendarRoutes	Creates pastor calendar routes and/or its persisted representation.
backend/src/routes/pastorCalendar.routes.ts	addBlockRoutes	Named module function supporting add block routes.
backend/src/routes/pastorCalendar.routes.ts	notifyRequester	Named module function supporting notify requester.
backend/src/routes/pastorCalendar.routes.ts	withDatabase	Named module function supporting with database.
backend/src/routes/pastorCalendar.routes.ts	readJsonBody	Named module function supporting read json body.
backend/src/routes/pastorCalendar.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/pastorCalendar.routes.ts	parseId	Parses and validates id.
backend/src/routes/pastorCalendar.routes.ts	safeId	Named module function supporting safe id.
backend/src/routes/pastorCalendar.routes.ts	invalidIdError	Named module function supporting invalid id error.
backend/src/routes/pastorCalendar.routes.ts	notFound	Named module function supporting not found.
backend/src/routes/pastorCalendar.routes.ts	calendarConflict	Named module function supporting calendar conflict.
backend/src/routes/pastorCalendar.routes.ts	normalizeCollectionWithIds	Normalizes collection with ids into the module's canonical representation.
backend/src/routes/pastorCalendar.routes.ts	calendarCollectionEvents	Named module function supporting calendar collection events.
backend/src/routes/pastorCalendar.routes.ts	actor	Named module function supporting actor.
backend/src/routes/pastorCalendar.routes.ts	stringValue	Named module function supporting string value.
backend/src/routes/pastorCalendar.routes.ts	localeValue	Named module function supporting locale value.
backend/src/routes/pastorCalendar.routes.ts	prefixChanges	Named module function supporting prefix changes.
backend/src/routes/peopleDevelopment.routes.ts	createPeopleDevelopmentRoutes	Creates people development routes and/or its persisted representation.
backend/src/routes/peopleDevelopment.routes.ts	findPortalMember	Finds portal member from normalized or persisted data.
backend/src/routes/peopleDevelopment.routes.ts	deleteById	Deletes by id using the owning feature boundary.
backend/src/routes/peopleDevelopment.routes.ts	readJson	Named module function supporting read json.
backend/src/routes/peopleDevelopment.routes.ts	parseId	Parses and validates id.
backend/src/routes/peopleDevelopment.routes.ts	invalidId	Named module function supporting invalid id.
backend/src/routes/peopleDevelopment.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/peopleDevelopment.routes.ts	torontoDateKey	Named module function supporting toronto date key.
backend/src/routes/peopleDevelopment.routes.ts	getPortalClientKey	Retrieves or derives portal client key.
backend/src/routes/peopleDevelopment.routes.ts	createPortalRateLimiter	Creates portal rate limiter and/or its persisted representation.
backend/src/routes/peopleDevelopment.routes.ts	withDatabase	Named module function supporting with database.
backend/src/routes/peopleNotes.routes.ts	createPeopleNotesRoutes	Creates people notes routes and/or its persisted representation.
backend/src/routes/peopleNotes.routes.ts	readJsonBody	Named module function supporting read json body.
backend/src/routes/peopleNotes.routes.ts	parseId	Parses and validates id.
backend/src/routes/peopleNotes.routes.ts	getActor	Retrieves or derives actor.
backend/src/routes/peopleNotes.routes.ts	parsePushResult	Parses and validates push result.
backend/src/routes/peopleNotes.routes.ts	validationError	Named module function supporting validation error.
backend/src/routes/peopleNotes.routes.ts	invalidIdError	Named module function supporting invalid id error.
backend/src/routes/peopleNotes.routes.ts	withDatabase	Named module function supporting with database.
backend/src/schemas/booking.schema.ts	timeToMinutes	Named module function supporting time to minutes.
backend/src/schemas/booking.schema.ts	isCalendarDate	Predicate evaluating whether calendar date.
backend/src/schemas/meetingInvitation.schema.ts	isValidCalendarDate	Predicate evaluating whether valid calendar date.
backend/src/schemas/meetingInvitation.schema.ts	timeToMinutes	Named module function supporting time to minutes.
backend/src/schemas/pastorCalendar.schema.ts	optionalText	Named module function supporting optional text.
backend/src/security/firebaseAuth.ts	createFirebaseAuthMiddleware	Creates firebase auth middleware and/or its persisted representation.
backend/src/security/firebaseAuth.ts	optionalString	Named module function supporting optional string.
backend/src/security/firebaseAuth.ts	authenticationError	Named module function supporting authentication error.

Source module	Function	Reference role
backend/src/security/firebaseServiceAccount.ts	getFirebaseServiceAccountAccessToken	Retrieves or derives firebase service account access token.
backend/src/security/firebaseServiceAccount.ts	createServiceAccountAssertion	Creates service account assertion and/or its persisted representation.
backend/src/security/firebaseServiceAccount.ts	importPrivateKey	Named module function supporting import private key.
backend/src/security/firebaseServiceAccount.ts	encodeJson	Named module function supporting encode json.
backend/src/security/firebaseServiceAccount.ts	encodeBase64Url	Named module function supporting encode base64 url.
backend/src/security/pastorAuthorization.ts	isPastorUser	Predicate evaluating whether pastor user.
backend/src/security/pastorAuthorization.ts	requirePastorAccess	Enforces the pastor access authorization or validation boundary.
backend/src/services/backblazeArchive.service.ts	createBackblazeArchiveStorage	Creates backblaze archive storage and/or its persisted representation.
backend/src/services/backblazeArchive.service.ts	delay	Named module function supporting delay.
backend/src/services/backblazeArchive.service.ts	normalizedEndpoint	Normalizes d endpoint into the module's canonical representation.
backend/src/services/backblazeArchive.service.ts	requiredValue	Enforces the d value authorization or validation boundary.
backend/src/services/backblazeArchive.service.ts	storageResponseError	Named module function supporting storage response error.
backend/src/services/backblazeArchive.service.ts	objectUrl	Named module function supporting object url.
backend/src/services/brevo.service.ts	sendBrevoEmail	Sends brevo email through the configured communication or API boundary.
backend/src/services/brevo.service.ts	sendBrevoBccEmail	Sends brevo bcc email through the configured communication or API boundary.
backend/src/services/brevo.service.ts	sendBrevoRequest	Sends brevo request through the configured communication or API boundary.
backend/src/services/firebaseRealtimeDatabase.service.ts	createFirebaseRealtimeDatabaseClient	Creates firebase realtime database client and/or its persisted representation.
backend/src/services/firebaseRealtimeDatabase.service.ts	createFirebaseAdminRealtimeDatabaseClient	Creates firebase admin realtime database client and/or its persisted representation.
backend/src/services/firebaseRealtimeDatabase.service.ts	putIfAbsent	Named module function supporting put if absent.
backend/src/services/firebaseRealtimeDatabase.service.ts	normalizeDatabaseUrl	Normalizes database url into the module's canonical representation.
backend/src/services/gemini.service.ts	generateGeminiJson	Generates gemini json from controlled inputs.

Appendix E

Automated Test Catalog

The supplied repository contains 167 named test cases matched from frontend and backend test files. The list below preserves test intent so maintainers can see which contracts already have regression coverage.

Test file	Test case / asserted behavior
backend/test/admin.spec.ts	requires Firebase authentication for the administrator session
backend/test/admin.spec.ts	registers a new non-chief account as pending
backend/test/admin.spec.ts	initializes the exact Pastor as chief when no chief exists
backend/test/admin.spec.ts	denies authority allocation to a non-chief administrator
backend/test/admin.spec.ts	denies LInC Archives to an administrator without archive authority
backend/test/admin.spec.ts	lists normalized archive folders for an allocated administrator
backend/test/admin.spec.ts	creates a nested archive folder with server-owned metadata
backend/test/admin.spec.ts	refuses to delete an archive folder that contains nested folders
backend/test/admin.spec.ts	lists persistent archive files for an allocated administrator
backend/test/admin.spec.ts	prepares and records a private archive upload
backend/test/admin.spec.ts	verifies an uploaded object before marking the archive file ready
backend/test/admin.spec.ts	signs downloads and deletes private objects through the archive service
backend/test/assessment.spec.ts	returns public form states without authentication
backend/test/assessment.spec.ts	stores a validated assessment with server calculations and timestamps
backend/test/assessment.spec.ts	rejects unknown answers and invalid ratings
backend/test/assessment.spec.ts	refuses submissions when an administrator disabled the form
backend/test/assessment.spec.ts	keeps a stored assessment successful when Brevo fails
backend/test/assessment.spec.ts	stores a public direct signup with backend-controlled fields
backend/test/assessment.spec.ts	requires Firebase authentication for User Linkage
backend/test/assessment.spec.ts	denies an active administrator without Manage Assessment Forms allocation
backend/test/assessment.spec.ts	allows an allocated administrator to list normalized User Linkage responses
backend/test/assessment.spec.ts	lets an allocated administrator update linkage and form availability
backend/test/auth.spec.ts	rejects a request without a Bearer token
backend/test/auth.spec.ts	rejects a token that fails verification
backend/test/auth.spec.ts	returns the compatible session response for a Pastor
backend/test/auth.spec.ts	returns an authenticated but unauthorized session for a non-Pastor
backend/test/auth.spec.ts	forbids the Pastor endpoint for a non-Pastor
backend/test/auth.spec.ts	allows the exact Pastor email on the protected endpoint
backend/test/auth.spec.ts	returns 503 when Firebase authentication is not configured
backend/test/auth.spec.ts	allows the Authorization header in CORS preflight
backend/test/backblazeArchive.service.spec.ts	creates a five-minute signed upload URL scoped to the archive object
backend/test/backblazeArchive.service.spec.ts	signs private downloads with an attachment filename override
backend/test/backblazeArchive.service.spec.ts	retrieves a newly uploaded object when the first HEAD request returns 404
backend/test/bezalel.spec.ts	requires Pastor authentication before exposing the private calendar agent
backend/test/bezalel.spec.ts	provides a sanitized calendar to the Pastor agent and returns one validated action
backend/test/bezalel.spec.ts	lets the public booking agent inspect only public availability
backend/test/booking.spec.ts	returns only privacy-safe schedule ranges without authentication
backend/test/booking.spec.ts	rejects an excessive schedule range before database access
backend/test/booking.spec.ts	stores an anonymous request with server-controlled fields
backend/test/booking.spec.ts	rejects client-controlled status and timestamps
backend/test/booking.spec.ts	rejects a request that overlaps an existing pending request
backend/test/booking.spec.ts	rejects past dates before obtaining database access
backend/test/booking.spec.ts	keeps a stored request successful when notification delivery fails
backend/test/booking.spec.ts	returns 503 when server credentials are absent
backend/test/brevo.service.spec.ts	sends one provider request with hidden recipients in BCC

Test file	Test case / asserted behavior
backend/test/brevo.service.spec.ts	rejects an empty BCC recipient list before calling Brevo
backend/test/calendar.spec.ts	stores Toronto wall times with the correct daylight-saving offset
backend/test/calendar.spec.ts	expands recurring People Development meetings and blocks overlapping bookings
backend/test/calendar.spec.ts	builds an importable calendar using UTC timestamps
backend/test/gemini.service.spec.ts	uses the current stable Flash model by default
backend/test/index.spec.ts	returns Hello Hono from the root route
backend/test/index.spec.ts	exposes the deployed NextGen contract health endpoint
backend/test/index.spec.ts	returns a meaningful JSON error for an unknown authenticated NextGen endpoint
backend/test/index.spec.ts	identifies a Firebase service-account authentication failure
backend/test/index.spec.ts	identifies the Firebase Database HTTP status without exposing credentials
backend/test/index.spec.ts	returns closed QA sessions to members as visible records while keeping drafts private
backend/test/index.spec.ts	lets an authenticated member create a question with fixed voting choices
backend/test/index.spec.ts	lets an authenticated member change an existing vote through the HTTP contract
backend/test/index.spec.ts	lets an allocated administrator select a member question for discussion
backend/test/index.spec.ts	allows an administrator with NextGen QA authority to use the management endpoint
backend/test/index.spec.ts	returns a meaningful authorization error to an administrator without NextGen QA authority
backend/test/index.spec.ts	rejects an invalid email test request body
backend/test/index.spec.ts	handles a successful Brevo sandbox request
backend/test/index.spec.ts	sends a People Development BCC request without an authorization header
backend/test/index.spec.ts	requires Pastor authentication before sending meeting invitations
backend/test/nextGenBezalel.spec.ts	enforces two submitted questions per account before calling Gemini
backend/test/nextGenBezalel.spec.ts	rejects an off-theme question without writing it
backend/test/nextGenBezalel.spec.ts	requires a theme and stores Bezalel translations when a Pastor creates a session
backend/test/nextGenMissionMap.spec.ts	requires a Firebase Bearer token
backend/test/nextGenMissionMap.spec.ts	accepts another authenticated Firebase email
backend/test/nextGenMissionMap.spec.ts	accepts any Firebase sign-in provider
backend/test/nextGenMissionMap.spec.ts	returns validated map data for the exact login
backend/test/nextGenMissionMap.spec.ts	fails closed when the secret map data is invalid
backend/test/nextGenPortal.spec.ts	stores one current vote per email key and allows changing that vote
backend/test/nextGenPortal.spec.ts	creates member questions with fixed upvote and downvote choices
backend/test/nextGenPortal.spec.ts	migrates legacy questions once into a closed QA Session 1
backend/test/nextGenPortal.spec.ts	keeps a closed session read-only for voting
backend/test/nextGenPortal.spec.ts	returns voter identities separately from aggregate results and excludes unverified votes
backend/test/nextGenPortal.spec.ts	filters personal upvotes and ranks by verified net votes without returning scores
backend/test/nextGenPortal.spec.ts	lets management select and unselect a member question for discussion
backend/test/nextGenPortal.spec.ts	places every NextGen object below the isolated nextgen prefix
backend/test/pastorCalendar.spec.ts	requires authentication before calendar database access
backend/test/pastorCalendar.spec.ts	rejects a non-Pastor before calendar database access
backend/test/pastorCalendar.spec.ts	returns one normalized no-store calendar snapshot
backend/test/pastorCalendar.spec.ts	creates a meeting with backend-controlled audit fields
backend/test/pastorCalendar.spec.ts	exports confirmed, pending, and recurring calendar items as ICS
backend/test/pastorCalendar.spec.ts	updates a meeting and its source request in one root patch
backend/test/pastorCalendar.spec.ts	moves the atomic reservation when an accepted meeting is rescheduled
backend/test/pastorCalendar.spec.ts	creates availability with server-controlled timestamps
backend/test/pastorCalendar.spec.ts	accepts a request with an atomic meeting/request update
backend/test/pastorCalendar.spec.ts	rejects a decision for an already decided request
backend/test/pastorCalendar.spec.ts	requires authentication for meeting invitations
backend/test/peopleDevelopment.spec.ts	rejects invalid portal identifiers before database access
backend/test/peopleDevelopment.spec.ts	returns only the matched group assignments and active schedules
backend/test/peopleDevelopment.spec.ts	returns no group data for a recognized unassigned member
backend/test/peopleDevelopment.spec.ts	rate limits public identifier attempts before database access
backend/test/peopleDevelopment.spec.ts	requires Firebase authentication for pastor routes
backend/test/peopleDevelopment.spec.ts	rejects authenticated non-pastors before database access
backend/test/peopleDevelopment.spec.ts	rejects arbitrary member source paths

Test file	Test case / asserted behavior
backend/test/peopleDevelopment.spec.ts	assigns a member using backend-controlled narrow updates
backend/test/peopleDevelopment.spec.ts	creates assignments with server timestamps and generated IDs
backend/test/peopleNotes.spec.ts	requires a Firebase Bearer token
backend/test/peopleNotes.spec.ts	rejects an authenticated non-Pastor before Firebase access
backend/test/peopleNotes.spec.ts	lists and normalizes People Notes with no-store caching
backend/test/peopleNotes.spec.ts	rejects unknown person fields without calling Firebase
backend/test/peopleNotes.spec.ts	creates a person with backend-controlled audit fields
backend/test/peopleNotes.spec.ts	creates a development item and updates the parent timestamp
backend/test/peopleNotes.spec.ts	adds a comment and updates item and person timestamps
backend/test/peopleNotes.spec.ts	updates a follow-up date atomically with timestamps
backend/test/peopleNotes.spec.ts	deletes a person
backend/test/peopleNotes.spec.ts	deletes an item with an atomic null patch
backend/test/peopleNotes.spec.ts	deletes a comment with an atomic null patch
backend/test/peopleNotes.spec.ts	returns 503 when the database URL is missing
backend/test/peopleNotes.spec.ts	maps Firebase Rules denial to HTTP 403
backend/test/peopleNotes.spec.ts	maps Firebase network failures to HTTP 502
backend/test/peopleNotes.spec.ts	allows People Notes PATCH and DELETE CORS preflight
src/components/admin/admin.utils.test.ts	keeps archive access disabled for legacy authority records
src/components/admin/admin.utils.test.ts	recognizes an explicit LInC Archives allocation
src/components/admin/admin.utils.test.ts	recognizes an explicit NextGen QA allocation
src/components/admin/archives/archives.utils.test.ts	builds and alphabetizes a nested folder tree
src/components/admin/archives/archives.utils.test.ts	builds breadcrumbs from the archive root to the selected folder
src/components/admin/archives/archives.utils.test.ts	filters folders and stored files to the current location
src/components/admin/archives/archives.utils.test.ts	formats file sizes for archive rows
src/components/admin/attendance/attendance.utils.test.ts	normalizes person records and legacy raw Base64 photos
src/components/admin/attendance/attendance.utils.test.ts	adds attendance dates once and keeps them sorted
src/components/admin/attendance/attendance.utils.test.ts	parses stored attendance strings without blank entries
src/components/admin/attendance/attendance.utils.test.ts	builds a six-week calendar and marks Sundays
src/components/admin/attendance/attendance.utils.test.ts	finds the first Sunday in May and builds inclusive Sunday ranges
src/components/assessment/assessment.calculations.test.ts	accepts real email addresses and rejects placeholders
src/components/assessment/assessment.calculations.test.ts	creates defaults and enforces required answers
src/components/assessment/assessment.calculations.test.ts	scores groups, ranks them, and builds localized result copy
src/components/assessment/assessment.calculations.test.ts	traces a derived calculation back to its source section
src/components/brand/LincLogo.test.tsx	provides accessible alternative text when used as meaningful branding
src/components/brand/LincLogo.test.tsx	is decorative by default and supports responsive sizing
src/components/congregation-group-notes/congregationGroupNotes.utils.test.ts	finds identifiers inside nested legacy form responses
src/components/congregation-group-notes/congregationGroupNotes.utils.test.ts	normalizes supported group aliases
src/components/congregation-group-notes/congregationGroupNotes.utils.test.ts	normalizes a shared assignment and its PDF attachment
src/components/congregation-group-notes/congregationGroupNotes.utils.test.ts	rejects assignments without a valid group or content
src/components/congregation-group-notes/congregationGroupNotes.utils.test.ts	formats attachment sizes and returns readable Arabic group labels
src/components/landing/landingNavigation.test.tsx	shows the five agreed top-level destinations
src/components/landing/landingNavigation.test.tsx	keeps assessment, booking, and group notes inside the featured program
src/components/landing/landingNavigation.test.tsx	restores all five destinations in the floating navigation
src/components/nextgen/activities/NextGenActivityMenu.test.tsx	shows completion status and reports the selected activity

Test file	Test case / asserted behavior
src/components/nextgen/activities/nextGenActivities.utils.test.ts	normalizes identifiers to four uppercase alphanumeric characters
src/components/nextgen/activities/nextGenActivities.utils.test.ts	validates usable email addresses
src/components/nextgen/activities/nextGenActivities.utils.test.ts	normalizes a user record and defaults unknown status to pending
src/components/nextgen/activities/nextGenActivities.utils.test.ts	collects voter identifiers from legacy and current storage shapes
src/components/nextgen/activities/nextGenActivities.utils.test.ts	derives net votes when the stored value is absent
src/components/nextgen/activities/nextGenActivities.utils.test.ts	filters reviewed sessions and sorts remaining questions by vote priority
src/components/nextgen/activities/nextGenActivities.utils.test.ts	counts completed binary and rating survey answers
src/components/nextgen/portal/NextGenQaSessions.test.tsx	lists each open session as its own form
src/components/nextgen/portal/NextGenQaSessions.test.tsx	shows a closed session as disabled instead of hiding it
src/components/nextgen/portal/NextGenQaSessions.test.tsx	retries a transient backend failure without reloading the page
src/components/nextgen/portal/NextGenQaSessions.test.tsx	keeps voted questions visible, reflects discussion selection, and allows changing a vote
src/components/nextgen/portal/NextGenQaSessions.test.tsx	lets members submit questions and requests server-side filtered views
src/components/people-notes/peopleNotes.utils.test.ts	creates a valid ISO date default
src/components/people-notes/peopleNotes.utils.test.ts	extracts readable error messages
src/components/people-notes/peopleNotes.utils.test.ts	formats stored dates for display
src/components/people-notes/peopleNotes.utils.test.ts	uses readable localized fallbacks for missing dates
src/services/administrator.test.ts	uploads directly to a signed URL and completes the authenticated archive record
src/services/administrator.test.ts	preserves a successfully uploaded object when completion verification is delayed
src/services/congregationGroupNotes.test.ts	sends only the submitted identifier to the Hono portal
src/services/nextGenPortal.test.ts	identifies an undeployed backend endpoint
src/services/nextGenPortal.test.ts	preserves a meaningful backend JSON error with its HTTP status
src/services/nextGenPortal.test.ts	provides useful authentication and authorization errors
src/services/peopleDevelopment.test.ts	sends only pastor-entered personal-note fields and leaves audit fields to Hono
src/services/peopleDevelopment.test.ts	strips schedule audit fields from create and update requests

Appendix F

Residual Direct-Firebase Client Catalog

The following current frontend files import `firebase/database` directly and therefore remain relevant to Firebase Security Rules and client-side authorization analysis:

- `src/components/admin/AdministratorPanel.tsx`
- `src/components/admin/attendance/useAttendanceManagement.ts`
- `src/components/landing/CommunityCarousel.tsx`
- `src/components/nextgen/activities/nextGenActivities.firebase.ts`
- `src/components/pastor/hooks/useParticipants.ts`
- `src/components/pastor/nextgen/nextgen.firebase.ts`
- `src/components/tutorial-builder/tutorialBuilder.firebase.ts`
- `src/firebase.ts` (Firebase initialization/export)

This list is an important migration checklist: a domain is not fully behind the backend while its sensitive persistence still depends on direct browser database access.

Maintenance Checklist

1. Update this document whenever a route is added/removed, a database root changes, a capability is added, an external provider changes, or an authorization boundary changes.
2. Keep backend README authorization statements synchronized with route middleware.
3. Treat `documentation/database/*.mmd` as versioned schema documentation and update it with migrations.
4. Add new public/backend endpoints to the API matrix and test catalog in the same release.
5. Record new environment variables in both deployment configuration and this catalog; never put secret values in documentation.
6. When a direct-Firebase domain migrates to Hono, remove it from the residual-client appendix only after source imports and runtime flow are both gone.
7. Remove legacy modules only after verifying no route, barrel export, migration routine, or production data still depends on them.

Note – Release 1.4 Baseline

This document is deliberately long because the repository is now a real multi-domain product. The goal is not to maximize pages; it is to prevent a future engineer from needing to rediscover feature behavior, security boundaries, data paths, storage providers, and source ownership by reverse-engineering the project again.